

Antitrust Methods: Research and Practice
Empirical, Qualitative, and Litigation Tools

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Antitrust Methods: Research and Practice

Empirical, Qualitative, and Litigation Tools

Welcome. This book translates lecture notes and practitioner experience into a reproducible playbook for antitrust and regulation. It pairs empirical methods with qualitative evidence, case law, and agency practice.

- Audience: upper-undergraduates, early graduate students, and practitioners.
- Approach: code-forward (R), with “Method,” “Qualitative,” “Case,” and “Debate” callouts in each chapter.
- Jurisdictions: US focus with comparative notes for EU/UK/Japan/China where relevant.
- Reproducibility: keys live in `.Renviron`; data are cached in `data/raw` and `data/derived`; figures regenerate from code.

Use the navigation to jump into chapters or start with the Preface.

Preface and How to Use This Book

Audience and approach

- Upper-undergraduates, early graduate students, and practitioners needing applied antitrust methods.
- Each chapter pairs empirical tools with qualitative evidence guidance and litigation context.
- Code boxes use R; figures are reproducible from public or licensed data.

How to read

- Skim the case boxes first to anchor theory.
- Run the R chunks; every figure is meant to be regenerated locally.
- Method boxes flag assumptions, diagnostics, and common failure modes.
- Debate callouts mark unsettled areas where multiple approaches coexist.

Data and reproducibility

- Keys live in `.Renviron`; data are cached in `data/raw` and `data/derived`.
- Render with `quarto render`; set `execute: freeze: auto` to cache slow chunks.
- Use `renv` to lock package versions for teaching and litigation consistency.

Evidence triad

- Empirics: estimation and inference with transparent code.
- Documentary: facts from discovery and public sources woven into case boxes.
- Expert judgment: where and how to exercise professional judgment, including survey design and qualitative interviews.

Citations and jurisdictions

- Cite claims and data with [`@key`] inline; include papers, books, agency actions, and case law (US focus with EU/UK/Japan/China notes where relevant).
- All references live in `references/references.bib`; add `note = {Last accessed: YYYY-MM-DD}` for web/agency sources.
- When a claim hinges on jurisdictional nuance, flag it explicitly (e.g., `{US}`, `{EU}`) and note divergences in standards or burdens.

Orientation: Antitrust and Regulation

Learning goals

This opening chapter orients you to the institutional settings where economists, researchers, and lawyers work side by side. By the end you should be able to articulate how competition authorities define their objectives, how they translate statutes or guidelines into investigative hypotheses, and how evidentiary burdens shift between agencies and courts. You will also learn how to map economic questions—market definition, market power, theory of harm, efficiencies, and remedy selection—onto the specific legal elements regulators must prove or rebut.

An equally important goal is judgment: when should you push for additional data work, and when is documentary, testimonial, or industry expertise the better marginal investment? Because this course serves practitioners in South Africa who often face multi-jurisdictional matters, we will continually compare US-led doctrines with approaches in the EU, UK, Asia, and regional African regulators. Expect a mix of narrative history, process walkthroughs, and applied examples from roughly 2005–2024.

Core narrative

Institutions and mandates

Modern antitrust work is anchored by a handful of institutions with overlapping mandates. In the United States the Department of Justice (DOJ) Antitrust Division and the Federal Trade Commission (FTC) share merger and conduct jurisdiction, aided by state attorneys general and sector regulators such as the Federal Communications Commission or Surface Transportation Board. Post-2000 history matters: the Clinton-era Microsoft litigation cemented exclusionary conduct frameworks, the 2010 Horizontal Merger Guidelines elevated unilateral effects, and the 2023 Merger Guidelines re-centered structural presump-

tions and platform dynamics. Across the Atlantic, the European Commission’s Directorate-General for Competition (DG COMP) pairs investigations with inquisitorial decision making, while national authorities like the UK Competition and Markets Authority (CMA) play an increasingly global role (e.g., blocking Microsoft/Activision until remedies evolved). Asian agencies—including Japan’s JFTC, Korea’s KFTC, and China’s SAMR—and South Africa’s Competition Commission and Tribunal have expanded resources to scrutinize digital markets, food supply chains, and cartels affecting the Southern African Development Community (SADC). Understanding each body’s statutory tests, procedural timelines, and evidentiary cultures is essential to crafting persuasive analyses.

Workflow from intake to remedies

Regardless of jurisdiction, investigations follow a recognizable arc. Matters typically begin with a complaint, leniency application, or merger notification. Case teams triage with “hot docs” and quick descriptive analytics, draft a plan that allocates legal and economic workstreams, and establish preservation plus data-request protocols. The second phase is scoping: enumerating data systems, prioritizing custodians, negotiating production formats, and deciding whether to run early econometric screens—such as price-cost margins or diversion ratios—before collecting more costly qualitative evidence. In phase three the team synthesizes theory and evidence into statements of issues, white papers, or ultimately pleadings and testimony. Finally, remedies discussions integrate forward-looking modeling with institutional constraints: can the Competition Tribunal in Pretoria supervise behavioral commitments as effectively as US courts, or is divestiture the only reliable fix? Always document these choices so that tribunals (and future teams) can reconstruct the evidentiary chain.

Scoping data needs early

Practitioners treat data, documents, and interviews as complements rather than substitutes. Intake meetings should generate inventories of transactional databases, billing systems, CRM extracts, internal forecasts, board presentations, and benchmark studies. The task is to identify what can be measured quickly (e.g., monthly price series, bidding outcomes, churn metrics) and what requires longitudinal assembly (e.g., claims-level healthcare data or network telemetry). Teams should simultaneously plan qualitative work: establish search terms for electronically stored information (ESI), design interview protocols, and identify third parties—customers, suppliers, former employees—whose perspectives sharpen or rebut economic priors. Missing this early sequencing is costly; see the Competition Commission South Africa’s review of Netcare/Community Hospital Group, where late-stage data disputes shortened the time available for substantive modeling.

Illustrative vignette: Search dominance diagnostics

The 2020–2023 DOJ challenge to Google Search illustrates how process and substance intertwine. Economists combined browser default share calculations, auction revenue data, and margin analyses with interview testimony from handset makers and browser developers. Those fact narratives clarified why certain regression designs mattered (e.g., estimating counterfactual query volume absent default contracts). Compare that to the Competition Commission South Africa’s investigation into digital platforms and app stores, where documentary evidence about self-preferencing and ad-tech fee structures filled gaps in transaction data. The lesson: quantitative rigor carries the day only when tied to institutional detail about distribution agreements, switching costs, and remedy administrability.

i Method box: Integrating empirics here

Early-stage screening rarely needs sophisticated econometrics. Build dashboards that pull together market shares, HHI trends, entry/exit indicators, and switching matrices from customer-level data. Combine transactional datasets with public sources (e.g., SEC filings, the FTC’s merger statistics, Stats SA tariff books) to benchmark plausible margins. When data are thin, like in new fintech or biotech markets pivot, quickly to qualitative or expert evidence rather than forcing underpowered regressions. Document these decisions in a running technical memo so litigators can later explain why the team prioritized one technique over another.

i Qualitative evidence

Treat qualitative work as disciplined research, not anecdote hunting. Begin with an interview guide that connects each theory of harm to specific questions, and pair every interview with contemporaneous notes plus sourcing metadata (custodian, date, privilege status). For document review, align search strings with economic hypotheses—for example, pairing “capacity discipline” with “shutdown” when probing fertilizer cartel allegations in the US Midwest and South Africa’s Sasol case. Create chronologies that marry documentary excerpts with data milestones; these become invaluable when explaining to a tribunal why a particular conduct pattern aligns with measured price changes or customer churn.

i Case box: Quick tour

Recent matters illustrate the diversity of evidentiary mixes. The US v. Microsoft legacy is still instructive for tying and exclusion, but newer platform cases—US v. Google Search and the FTC’s Meta/Within suit—

highlight how product design data and third-party developer testimony reinforce each other. Airline collaborations such as DOJ v. American Airlines/JetBlue (NEA) show how scheduling data, revenue management simulations, and traveler surveys combine to test unilateral and coordinated effects. Global cartel work remains vibrant: the EU Trucks cartel decisions, the KFTC's memory chip probes, and South Africa's construction and bread cartels each turned on leniency statements corroborated by bidding records and cost analyses. Labor markets are now front-page antitrust: US wage-fixing cases in healthcare and the 2023 consent orders in the poultry industry relied on HR databases and interview evidence from recruiters.

i Comparative note

While the US emphasizes adversarial hearings and judicial precedent, DG COMP and the CMA operate administrative processes where agencies both investigate and decide. This affects research timing: EU cases often allow broader use of compelled internal data but require detailed written submissions early, whereas South African proceedings feature public-interest considerations that demand additional qualitative evidence (jobs, ownership, regional development). Asian agencies have diverse discovery rules—SAMR can request granular consumer data with little advance notice—so plan modular analyses that adapt as production obligations change. Throughout the book, sidebar notes will flag where legal standards (dominance vs. monopoly), burdens of proof, or remedy preferences diverge.

Looking ahead

Orientation work is only useful if it flows directly into research design. Your next task is to convert the institutional map above into concrete analytical workplans: build a case chronology template, sketch a data inventory that spans US, EU, and SADC sources, and identify at least one flagship dataset (e.g., claims-level healthcare data or national tender records) that will power the quantitative chapters. Keep a running list of figures—HHI trends, entry timelines, or platform governance maps—that we will revisit later so the narrative remains visually anchored and cumulative.

Visualizations

Agency and case timeline

This timeline pairs statutory milestones with landmark enforcement actions so readers can anchor later quantitative work in institutional change. Extend or localize the tibble with jurisdiction-specific events (e.g., Competition Amendment Acts, CMA guidelines, SAMR platform rules) before teaching or consulting engagements.

```
library(dplyr)
library(ggplot2)
library(scales)

milestones <- tibble::tribble(
  ~year, ~event, ~type,
  1890, "Sherman Act", "Statute",
  1914, "Clayton & FTC Acts", "Statute",
  1982, "DOJ Merger Guidelines", "Guidance",
  1998, "US v. Microsoft", "Case",
  2013, "US v. Apple eBooks", "Case",
  2019, "Competition Amendment\nAct (SA)", "Statute",
  2020, "EC Google Shopping", "Case",
  2023, "Joint Merger Guidelines", "Guidance"
) |>
  dplyr::mutate(
    y = dplyr::if_else(dplyr::row_number() %% 2 == 0, 1.2, -1.2),
    vjust = dplyr::if_else(y > 0, -0.5, 1.5)
  )

ggplot(milestones, aes(x = year, y = y, color = type)) +
  geom_hline(yintercept = 0, color = "gray70", linewidth = 0.8) +
  geom_segment(aes(xend = year, yend = 0), linewidth = 0.8, alpha
    ↵ = 0.7) +
  geom_point(size = 4) +
  geom_text(aes(label = event, vjust = vjust),
    size = 3.5, show.legend = FALSE, lineheight = 0.9) +
  scale_y_continuous(limits = c(-2, 2)) +
  scale_x_continuous(labels = scales::comma, breaks = seq(1880,
    ↵ 2030, 20)) +
  scale_color_manual(values = c("Statute" = "#D55E00",
    "Guidance" = "#0072B2",
    "Case" = "#009E73")) +

labs(
  title = "Key Antitrust Milestones",
  subtitle = "Blend statutory, guidance, and litigation anchors
    ↵ to frame workflows",
```

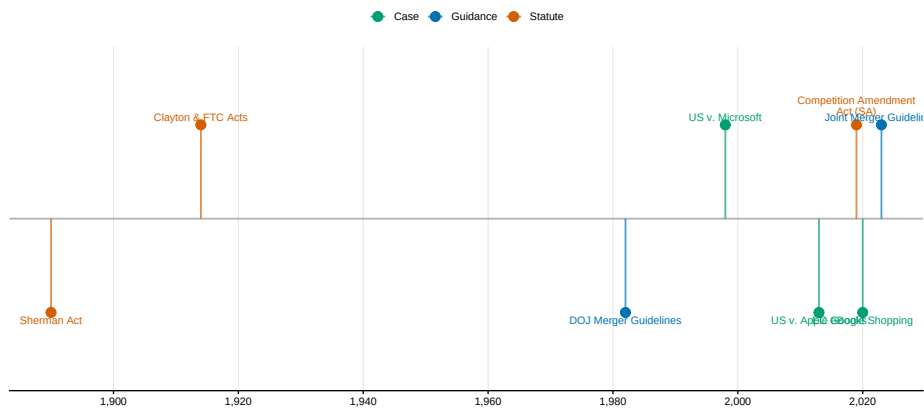
```

x = NULL, y = NULL, color = NULL
) +
theme_classic(base_size = 13) +
theme(
  axis.text.y = element_blank(),
  axis.ticks.y = element_blank(),
  axis.line.y = element_blank(),
  panel.grid.major.x = element_line(color = "gray90", linewidth
    ↪ = 0.3),
  legend.position = "top",
  plot.title = element_text(face = "bold")
)

```

Key Antitrust Milestones

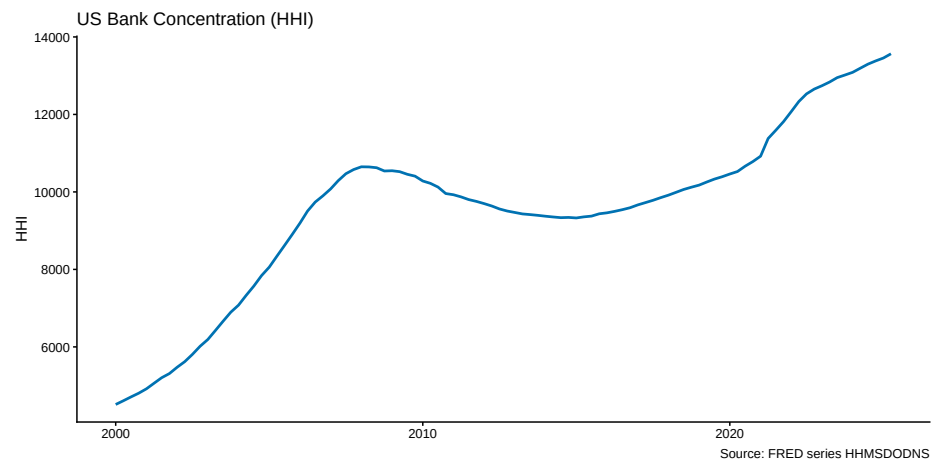
Blend statutory, guidance, and litigation anchors to frame workflows



HHI trend example

Track concentration trends using FRED data as a baseline for cross-jurisdictional comparisons.

10 Orientation: Antitrust and Regulation



Research Design: Quantitative and Qualitative

Learning goals

The goal of this chapter is to help you translate messy case facts into evidence that withstands scrutiny from opposing experts, agency staff, and judges. You will learn how to articulate research questions that map directly to legal theories of harm and to choose empirical strategies—diff-in-diff, event study, instrumental variables (IV), regression discontinuity (RDD), matching, or structural models—that are feasible with available data. Equally important, you will practice designing qualitative instruments (surveys, interviews, document coding schemes) that align with those empirical tests so each stream of evidence reinforces the others.

Because antitrust matters often unfold across jurisdictions, we emphasize how to document design choices so they travel: a memorandum prepared for the US DOJ should be intelligible to DG COMP economists or the South African Competition Tribunal. That means explicit statements about identifying assumptions, robustness diagnostics, and data provenance. Throughout the chapter we draw on recent matters such as US hospital merger retrospectives, EU telecom remedies, and South African ride-hailing and grocery market inquiries.

Workflow

Scoping memo

Every matter should begin with a scoping memo linking the narrative theory of harm to measurable outcomes. Specify the primary question (“Did the 2018 hospital merger in Texas raise commercial insurance prices?”), outline plausible channels (unilateral effects vs. coordination), and list essential datasets

with owners, legal process required, and likely cleaning steps. Capture timing constraints—the DOJ’s Second Request clock, the CMA’s Phase I deadlines, or the Competition Commission South Africa’s 60-business-day merger review period—and highlight non-negotiable assumptions (e.g., availability of claims-level data or ability to survey procurement managers). Include citations to precedent such as merger retrospectives (O. Ashenfelter and Hosken 2010) or class-certification common-impact decisions (Dickey and Rubinfeld 2014) so legal teams can anticipate how courts reacted to similar designs.

Data pipeline architecture

Treat the data pipeline as infrastructure. Raw productions, public datasets, and hand-entered chronologies should land in **data/raw**, with documented schemas and hashing to ensure integrity. Cleaning scripts housed in **scripts/** or **R/** should emit analytic files to **data/derived**, complete with README files explaining variable creation, filtering rules, and version numbers. Cloud-based teams should use reproducible environments (Renv, Conda, Docker) so every regression or visualization is rerunnable months later when litigation heats up. For South African matters, plan for hybrid data sources—local procurement records may arrive as PDFs, while US financials might come via SEC APIs—so build ingestion scripts that normalize currencies, indexation, and time zones.

Pre-analysis and litigation deliverables

Before estimating anything, write a pre-analysis plan that documents main outcomes, treatment definitions, control groups, identifying assumptions, and robustness checks. This discipline is not just for academics: DOJ and FTC staff frequently ask for pre-specified models, DG COMP expects methodological appendices, and South African tribunals often require plain-language summaries usable in simultaneous interpretation. Include sensitivity tests (e.g., alternative market definitions, exclusion of outlier distributors, placebo periods) and specify the qualitative evidence that will contextualize results. As you progress, convert technical memos into courtroom-ready outputs—slide decks, white papers, expert declarations—without sacrificing reproducibility.

The Modern Causal Toolkit: The Credibility Revolution

Modern antitrust economics increasingly relies on the “credibility revolution” in applied microeconomics—a shift from complex structural adjustments to transparent research designs that mimic randomized experiments. As championed by Angrist and Pischke (2010) and popularized in *Causal Inference: The Mixtape* (Cunningham, 2021), the goal is to identify a “natural experiment” where treatment (e.g., a merger, a cartel breakdown, or a regulation) is as good as randomly assigned.

The “Design-Based” Philosophy

Instead of asking “what controls do I need to fix my model?”, design-based thinking asks “where does the variation come from?” If you cannot draw a Directed Acyclic Graph (DAG) showing how the treatment was assigned independent of the outcome, no amount of regression control will save the analysis. This “reduced form” approach contrasts with the “structural” methods used in merger simulation (see Chapter 4), which rely on theoretical models to estimate “deep parameters”.

This approach prioritizes: 1. **Clean Identification:** Finding shocks (institutional details, policy boundaries, timing quirks) that separate treated and control groups. 2. **Transparency:** Presenting raw data visuals (like the event study plots below) before showing regression tables. 3. **Falsification:** Rigorously testing “placebos” (e.g., testing for effects before the merger happened, showing no effects in a time period where the treatment is not expected to have an effect) to prove the design is valid.

Core Methods (“The Mixtape”)

While the toolkit is vast, antitrust practitioners lean heavily on a few workhorses: - **Difference-in-Differences (DiD):** The gold standard for retrospective merger analysis. - **Synthetic Control:** Crucial for “N=1” cases (e.g., a single national merger) where no single control unit exists. - **Instrumental Variables (IV):** Used when prices are endogenous; we look for “shifters” (like tax changes or weather) that move supply but not demand. - **Regression Discontinuity (RDD):** Exploits arbitrary cutoffs (e.g., population thresholds for regulation) to compare similar firms on either side of the line.

The Evolution of DiD

A critical development in recent years is the realization that the standard Two-Way Fixed Effects (TWFE) estimator can be biased when treatment timing varies (staggered adoption). If you are analyzing a rollup strategy where a firm buys competitors in 2018, 2019, and 2020, standard regressions might compare early-treated units to late-treated units in ways that invert the sign of the effect.

Modern estimators like Callaway & Sant’Anna or Sun & Abraham (implemented in R packages `did` and `fixest`) explicitly handle this heterogeneity. In litigation, relying on “old” TWFE without robustness checks is now a vulnerability.

Getting Up to Speed

For a practical guide to these methods, we rely on “**Causal Inference: The Mixtape**” (Cunningham, 2021). It bridges the gap between textbook theory and coding reality. - **Start here:** mixtape.scunning.com

(Chapters on DAGs and DiD). - **Context:** To understand how these methods displaced earlier approaches, see Angrist & Pischke (2010), “The Credibility Revolution in Empirical Economics” (AEA link).

i Method box: Causal tools

Diff-in-diff and event studies: Always test for pre-trend equivalence using graphical diagnostics and formal tests. Dynamic specifications (leads/lags) are persuasive in telecom or energy cases where policy shocks phase in. The DOJ used this approach in Spirit/JetBlue analyses, while South African regulators applied similar diagnostics to evaluate grocery supplier rebates.

Instrumental variables: When supply or demand shocks are endogenous—think of hospital mergers with network design responses—search for plausibly exogenous instruments, such as regulatory bed caps or travel-time thresholds. Document relevance and exclusion explicitly; DG COMP is unforgiving when those steps are skipped.

Panel estimators: Fixed-effects (FE) or two-way FE models remain workhorses but require caution under staggered adoption. Use estimators like `did::att_gt` or `fixest`’s Sun-Abraham implementation, and explain weighting schemes in your declarations.

Synthetic control / matrix completion: For markets with single treated units (e.g., the CMA’s analysis of a UK airport slot divestiture), synthetic control and ridge-regularized matrix completion provide transparent counterfactuals. Include donor-pool rationale and placebo reassignments.

Inference: With few clusters (common in mining or port cases), complement cluster-robust standard errors with wild bootstrap or randomization inference to avoid overconfident p-values.

i Qualitative methods box

Surveys and diversion studies: Define the decision-maker (consumer vs. procurement lead), sampling frame, and recall period before crafting questions. The FTC’s Staples/Essendant review and the Competition Commission South Africa’s ride-hailing inquiry both relied on carefully screened respondents to estimate diversion ratios. Pre-test instruments to detect anchoring or order effects, and document weighting schemes.

Structured case studies: Build matrices that compare conduct across firms, time, and geographies—e.g., analyzing fertilizer cartels in the US, EU, and South Africa to see whether “industry meetings” align with price spikes. Use the same template so qualitative coders can draw parallels to econometric findings.

Document coding: Create controlled vocabularies for internal emails, board decks, and pricing memos. Tag each excerpt with custodian, date, and issue code; this facilitates linking to regression covariates (e.g., a “capacity discipline” tag aligned with utilization metrics).

Expert elicitation: When engaging industry experts (engineers, procurement veterans), capture their priors, data references, and compensation structures. Tribunals increasingly ask for transparency, especially in South African public-interest hearings.

i Debate

Three debates recur in nearly every case. First, **stated vs. revealed preferences**: surveys can illuminate multi-sided platforms where transaction data are limited (app stores, adtech), but agencies often prefer revealed behavior. One compromise is to calibrate survey responses against observable churn or clickstream metrics, as seen in the DOJ’s ad-tech inquiries. Second, **parallel trends under innovation**: in markets with rapid product cycles (semiconductors, fintech), clean control groups may not exist. Explain how you use feature-adjusted indices, synthetic controls, or machine-learning prognostics to approximate counterfactuals, and disclose residual risk. Third, **placebo tests**: courts increasingly expect falsification exercises. In wage-fixing litigation (e.g., poultry processors, US tech no-poach cases), teams ran placebo periods and alternative employee groups to show effects were specific to collusive periods rather than macro shocks.

Visualizations

Event study around a merger announcement

This figure uses publicly available equity data to demonstrate how to structure an event-study diagnostic for merger retrospectives or policy shocks. Replace tickers and the `event_date` to align with a specific matter; the scaffold keeps everything tidy for quick reporting.

```
library(tidyquant)
library(dplyr)
library(ggplot2)

event_date <- as.Date("2013-02-14") # American Airlines/US
  ↳ Airways announcement
tickers <- c("AAL", "DAL", "LUV", "SPY") # SPY proxy for market
  ↳ return
```

```

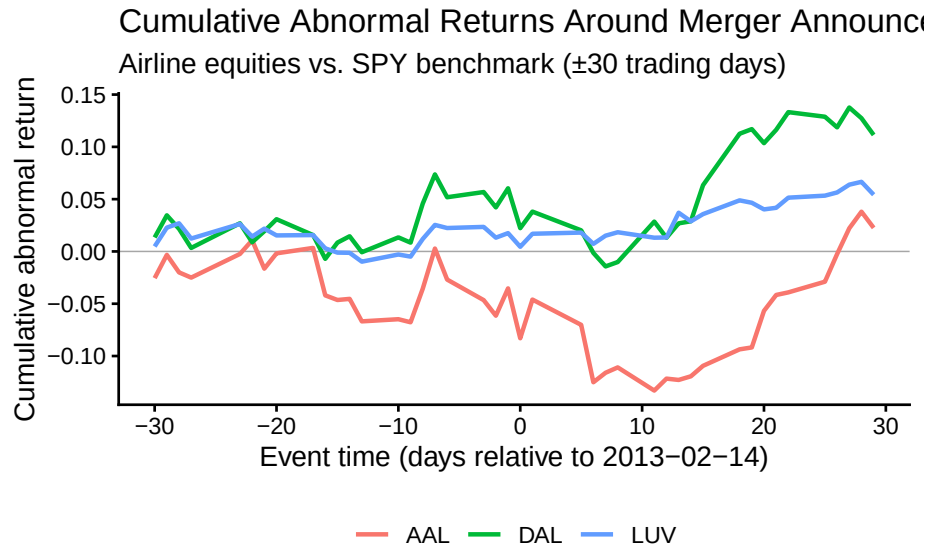
prices <- tq_get(
  tickers,
  from = event_date - lubridate::days(180),
  to   = event_date + lubridate::days(180)
) |>
  dplyr::group_by(symbol) |>
  dplyr::arrange(date) |>
  dplyr::mutate(ret = log(adjusted) - log(dplyr::lag(adjusted)))
↪ |>
  dplyr::ungroup()

market <- prices |>
  dplyr::filter(symbol == "SPY") |>
  dplyr::select(date, mkt_ret = ret)

cars <- prices |>
  dplyr::filter(symbol != "SPY") |>
  dplyr::left_join(market, by = "date") |>
  dplyr::mutate(
    abnormal_ret = ret - mkt_ret,
    rel_day = as.integer(date - event_date)
  ) |>
  dplyr::filter(rel_day >= -30, rel_day <= 30,
  ↪ !is.na(abnormal_ret)) |>
  dplyr::group_by(symbol) |>
  dplyr::arrange(rel_day) |>
  dplyr::mutate(car = cumsum(abnormal_ret))

ggplot(cars, aes(x = rel_day, y = car, color = symbol)) +
  geom_hline(yintercept = 0, linewidth = 0.3, color = "gray65") +
  geom_line(linewidth = 0.9) +
  labs(
    title = "Cumulative Abnormal Returns Around Merger
    ↪ Announcement",
    subtitle = "Airline equities vs. SPY benchmark (±30 trading
    ↪ days)",
    x = "Event time (days relative to 2013-02-14)",
    y = "Cumulative abnormal return",
    color = NULL
  ) +
  theme_classic(base_size = 12) +
  theme(legend.position = "bottom")

```



Looking ahead

Document every decision from scoping memo to estimation so later chapters can layer on industry-specific models. Begin compiling a reusable appendix that includes sample interview protocols, diversion survey templates, and code snippets for data validation. We will lean on these assets when we turn to market definition, industrial-organization toolkits, and merger simulation, so note any gaps (e.g., need a telecom-grade cost index or South African procurement benchmark) while the evidence record is still flexible.

Code box: diff-in-diff scaffold

```
library(fixest)
library(did)

# expected columns:
# - id: firm, product, or region
# - time: monthly/quarterly period (Date or numeric)
# - treat: 0/1 indicator
# - outcome: price, margin, volume, quality score, etc.
# - controls: cost indices, demand shifters, policy dummies

# panel_data <- panel_data %>%
#   filter(between(time, as.Date("2016-01-01"),
#     ↪ as.Date("2023-12-31"))) %>%
#   mutate(post = if_else(time >= treat_date[id], 1, 0))
```

```
# att_gt <- att_gt(yname = "outcome", tname = "time", idname =  
  ↪ "id",  
#                               gname = "treat_time", data = panel_data, panel  
  ↪ = TRUE,  
#                               xformula = ~ controls, clustervars = "id")  
# summary(att_gt)  
  
# Sun & Abraham style with fixest, clustered at market level:  
# feols(outcome ~ i(time, treat, ref = REF_PERIOD) + controls |  
  ↪ id + time,  
#       data = panel_data, cluster = ~market_id)
```

Market Definition and Competitive Landscape

Learning goals

Market definition is no longer a rote SSNIP ritual, but agencies and courts still expect a disciplined articulation of product and geographic boundaries before debating competitive effects. This chapter shows you how to frame those boundaries with data and qualitative evidence, evaluate SSNIP/SSNDQ logic, and communicate results across DOJ/FTC, DG COMP, CMA, SAMR, and Competition Tribunal South Africa processes.

By the end of the chapter you should be able to:

- Frame relevant product and geographic markets using demand evidence, diversion ratios, and qualitative industry facts.
- Apply and critique SSNIP/SSNDQ reasoning, critical-loss vs. actual-loss tests, and switching-matrix screens.
- Integrate survey evidence, internal documents, and econometrics into narratives tuned to the burden of proof in each jurisdiction.
- Reference key precedents (e.g., (U.S. Department of Justice and Federal Trade Commission 2010), (European Commission 2024), (Competition and Markets Authority 2021)) and literature when defending methodological choices.

Why market definition still matters

Even in unilateral-effects cases where agencies focus on margins, diversion, and price effects directly, courts often ask, “what is the market?” The answer shapes presumptions (HHI thresholds, dominance), jurisdictional hooks (public-interest test vs. consumer welfare), and remedy feasibility. Practitioners should treat market definition as an evidence-integration exercise: describe customer substitution paths, quantify them when possible, and tie those findings to the legal standard at issue (e.g., “reasonable interchangeability” in US case law vs. “suffi-

ciently interchangeable” in EC practice). Keep records of how you screened candidate markets so future teams can expand or narrow scope without re-litigating data prep.

“The ‘relevant market’ is not an intrinsic feature of the economy, but a tool for analyzing competitive effects... It is defined by the constraints that prevent a hypothetical monopolist from raising prices.”
— *United States v. Google LLC*, Memorandum Opinion (D.D.C. 2024)

Core tools and workflow

Modern market-definition work typically cycles through four toolkits:

1. **Descriptive analytics.** Price/volume trends, seasonality, and policy shocks that reveal when products or regions move together. Use `ggplot2` quick looks before heavy modeling.
2. **Demand elasticities & diversion.** Log-log regressions, AIDS-lite models, discrete choice, or churn-based diversion ratios that quantify substitutability. Always document instruments, fixed effects, or controls used.
3. **Critical-loss vs. actual-loss.** Compare hypothetical post-SSNIP losses to observed switching, stating margin assumptions and capacity constraints. Highlight when brand repositioning or supply limits break the classic calculations.
4. **Geographic screens.** Shipment flows, travel-time analyses, or catchment-area heatmaps built from loyalty data, mobile location pings, or Stats SA transport datasets. Pair with qualitative evidence on delivery commitments or regulatory boundaries.

The workflow mirrors Chapter 02’s research design steps: scoping memo → data inventory → qualitative plan → estimation → memo/slide deck. Revisit `chapters/13-empirical-appendix.qmd` for templates.

Core tools

Use these specific diagnostics to anchor your analysis:

- **Price/quantity dashboards.** Track cointegration, variance ratios, and shock responses to see which products move together.
- **Elasticities/diversion.** Estimate log-log or `fixest` panel regressions; or use choice data (`mlogit`, `apollo`) when products are differentiated.
- **Critical vs. actual loss.** Compare theoretical thresholds to observed share loss; revisit margin assumptions for multi-product firms.
- **Switching matrices, flows, and gravity models.** Summarize customer-level churn or shipment adjacency to motivate geographic splits.

Critical-loss worksheet

```
library(dplyr)
library(ggplot2)

critical_loss <- function(margin, price_increase) {
  price_increase / (price_increase + margin)
}

margin <- 0.25
price_increase <- 0.05
cl_threshold <- critical_loss(margin, price_increase)

observed <- tibble::tribble(
  ~segment, ~observed_share_loss,
  "Commercial group", 0.08,
  "Government", 0.03,
  "SMB", 0.12
) |>
  mutate(
    critical_loss = cl_threshold,
    passes_test = observed_share_loss > critical_loss
  )

observed
```

```
# A tibble: 3 x 4
  segment      observed_share_loss critical_loss passes_test
  <chr>                <dbl>         <dbl> <lgl>
1 Commercial group      0.08          0.167 FALSE
2 Government            0.03          0.167 FALSE
3 SMB                   0.12          0.167 FALSE
```

Critical-loss vs. actual-loss curve

This visualization compares the theoretical critical loss threshold to observed share losses across different margin assumptions. It helps communicate whether a candidate market “passes” the SSNIP test.

```
library(dplyr)
library(ggplot2)
library(patchwork)

# Generate critical loss curves for different price increases
margins <- seq(0.05, 0.60, by = 0.01)
price_increases <- c(0.05, 0.10, 0.15)
```

```

cl_curves <- expand.grid(
  margin = margins,
  price_increase = price_increases
) |>
  mutate(
    critical_loss = price_increase / (price_increase + margin),
    price_increase_label = paste0(price_increase * 100, "%
    ↪ SSNIP")
  )

# Observed actual loss from customer switching data
↪ (illustrative)
# In practice, these come from loyalty data, surveys, or
↪ econometric diversion estimates
actual_loss_examples <- tibble::tribble(
  ~margin, ~actual_loss, ~scenario,
  0.20, 0.12, "Observed switching (loyalty data)",
  0.30, 0.08, "Survey-based diversion",
  0.40, 0.06, "Econometric estimate"
)

# Main plot: critical loss curves
p1 <- ggplot(cl_curves, aes(x = margin, y = critical_loss, color
↪ = price_increase_label)) +
  geom_line(linewidth = 1.2) +
  geom_point(data = actual_loss_examples,
    aes(x = margin, y = actual_loss, shape = scenario),
    color = "black", size = 3, inherit.aes = FALSE) +
  scale_x_continuous(labels = scales::percent_format()) +
  scale_y_continuous(labels = scales::percent_format(), limits =
↪ c(0, 0.30)) +
  scale_color_manual(values = c("#0072B2", "#D55E00", "#009E73"))
↪ +
  labs(
    title = "Critical Loss vs. Actual Loss Test",
    subtitle = "Points above the curve suggest the candidate
    ↪ market fails SSNIP (too much switching)",
    x = "Margin (Price - MC) / Price",
    y = "Share loss threshold",
    color = "Price increase",
    shape = "Actual loss from data"
  ) +
  theme_classic(base_size = 12) +
  theme(
    legend.position = "bottom",
    legend.box = "vertical",

```

```

    plot.title.position = "plot"
  ) +
  guides(color = guide_legend(order = 1), shape =
    ↪ guide_legend(order = 2))

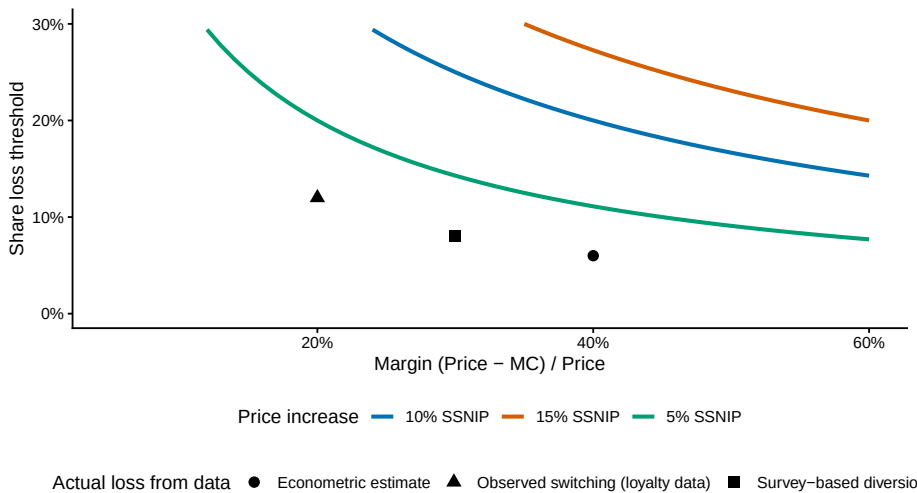
# Companion table showing specific calculations
cl_table <- tibble::tribble(
  ~margin, ~price_increase, ~critical_loss, ~actual_loss,
  ↪ ~interpretation,
  0.25, 0.05, 0.167, 0.08, "Market likely too narrow (actual <
  ↪ critical)",
  0.25, 0.10, 0.286, 0.12, "Market likely too narrow (actual <
  ↪ critical)",
  0.40, 0.05, 0.111, 0.15, "Market likely too broad (actual >
  ↪ critical)"
) |>
mutate(
  across(c(margin, price_increase, critical_loss, actual_loss),
    ~scales::percent(., accuracy = 0.1))
)

p1

```

Critical Loss vs. Actual Loss Test

Points above the curve suggest the candidate market fails SSNIP (too much switching)



```
cat("\nCritical loss decision table:\n")
```

Critical loss decision table:

```
print(cl_table, n = Inf)
```

```
# A tibble: 3 x 5
```

	margin	price_increase	critical_loss	actual_loss	interpretation
	<chr>	<chr>	<chr>	<chr>	<chr>
1	25.0%	5.0%	16.7%	8.0%	Market likely too narrow (act~
2	25.0%	10.0%	28.6%	12.0%	Market likely too narrow (act~
3	40.0%	5.0%	11.1%	15.0%	Market likely too broad (actu~

How to use this chart: For a given margin estimate (horizontal axis), find the critical loss threshold (the curve). If your observed or estimated actual loss (from switching data, surveys, or diversion ratios) falls *above* the curve, customers switch too readily for a hypothetical monopolist to profitably raise prices by 5-10%, suggesting the candidate market is too narrow and should be expanded. Conversely, if actual loss is *below* the curve, the market definition may be defensible.

Swap `margin` and `price_increase` with matter-specific values. For theoretical derivations, see Katz and Shapiro (2003). For alternatives to traditional market definition, see Farrell and Shapiro (2010a) and Schmalensee (2009). For quantitative techniques, consult Davis and Garcés (2010).

Route-share heatmap: NYC airline concentration

This heatmap visualizes carrier concentration on specific routes, helping identify whether route-by-route markets are appropriate or whether broader origin-destination-pairs should be considered. High concentration (dark colors) on specific routes may signal competitive concerns.

```
library(nycflights13)
library(dplyr)
library(tidyr)
library(ggplot2)

# Calculate route-level shares and HHI
route_shares <- flights |>
  filter(origin %in% c("JFK", "LGA", "EWR")) |>
  mutate(route = paste(origin, dest, sep = "-")) |>
  count(route, carrier, name = "flights") |>
  group_by(route) |>
  mutate(
    share = flights / sum(flights),
    total_flights = sum(flights)
  ) |>
  ungroup()

# Calculate HHI for each route
```

```

route_hhi <- route_shares |>
  group_by(route) |>
  summarize(
    hhi = sum((share * 100)^2),
    total_flights = first(total_flights),
    n_carriers = n()
  ) |>
  arrange(desc(hhi))

# Select top 30 routes by volume for visualization clarity
top_routes <- route_shares |>
  group_by(route) |>
  summarize(total = sum(flights)) |>
  slice_max(total, n = 30) |>
  pull(route)

# Filter to top routes and add HHI
route_shares_top <- route_shares |>
  filter(route %in% top_routes) |>
  left_join(route_hhi |> select(route, hhi), by = "route") |>
  mutate(
    route_labeled = paste0(route, " (HHI: ", round(hhi), ")")
  ) |>
  arrange(desc(hhi))

# Create ordered factor for proper sorting
route_shares_top <- route_shares_top |>
  mutate(route_labeled = factor(route_labeled,
                                levels =
                                  ↪ unique(route_labeled[order(hhi,
                                  ↪ decreasing = TRUE)])))

# Heatmap
p_heatmap <- ggplot(route_shares_top, aes(x = carrier, y =
  ↪ route_labeled, fill = share)) +
  geom_tile(color = "white", linewidth = 0.5) +
  geom_text(aes(label = ifelse(share >= 0.05,
  ↪ scales::percent(share, accuracy = 1), "")),
    size = 2.5, color = "white") +
  scale_fill_viridis_c(
    labels = scales::percent_format(),
    option = "plasma",
    begin = 0.1,
    end = 0.9
  ) +
  labs(

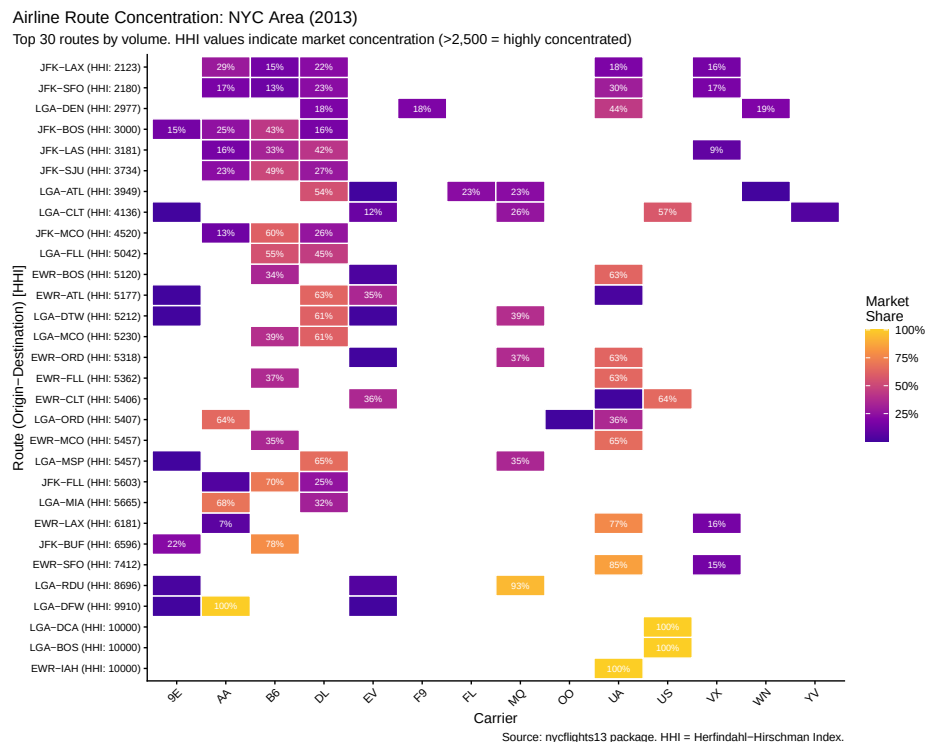
```

```

title = "Airline Route Concentration: NYC Area (2013)",
subtitle = "Top 30 routes by volume. HHI values indicate
  ↳ market concentration (>2,500 = highly concentrated)",
x = "Carrier",
y = "Route (Origin-Destination) [HHI]",
fill = "Market\\nShare",
caption = "Source: nycflights13 package. HHI =
  ↳ Herfindahl-Hirschman Index."
) +
theme_classic(base_size = 11) +
theme(
  axis.text.y = element_text(size = 8),
  axis.text.x = element_text(angle = 45, hjust = 1),
  plot.title.position = "plot",
  legend.position = "right",
  panel.grid = element_blank()
)

```

p_heatmap



```

# Summary table of most concentrated routes
cat("\nMost concentrated routes (Top 10 by HHI):\n")

```

Most concentrated routes (Top 10 by HHI):

```
route_hhi |>
  slice_max(hhi, n = 10) |>
  mutate(
    hhi = round(hhi),
    concentration = case_when(
      hhi >= 2500 ~ "Highly concentrated",
      hhi >= 1500 ~ "Moderately concentrated",
      TRUE ~ "Unconcentrated"
    )
  ) |>
  select(route, hhi, n_carriers, total_flights, concentration) |>
  print(n = Inf)
```

A tibble: 96 x 5

	route	hhi	n_carriers	total_flights	concentration
	<chr>	<dbl>	<int>	<int>	<chr>
1	EWB-ALB	10000	1	439	Highly concentrated
2	EWB-ANC	10000	1	8	Highly concentrated
3	EWB-AVL	10000	1	265	Highly concentrated
4	EWB-BQN	10000	1	297	Highly concentrated
5	EWB-BTV	10000	1	931	Highly concentrated
6	EWB-BUF	10000	1	973	Highly concentrated
7	EWB-BZN	10000	1	36	Highly concentrated
8	EWB-CAE	10000	1	104	Highly concentrated
9	EWB-CMH	10000	1	724	Highly concentrated
10	EWB-DAY	10000	1	1134	Highly concentrated
11	EWB-DSM	10000	1	411	Highly concentrated
12	EWB-EGE	10000	1	110	Highly concentrated
13	EWB-GRR	10000	1	719	Highly concentrated
14	EWB-GSO	10000	1	1120	Highly concentrated
15	EWB-GSP	10000	1	745	Highly concentrated
16	EWB-HDN	10000	1	15	Highly concentrated
17	EWB-HNL	10000	1	365	Highly concentrated
18	EWB-HOU	10000	1	976	Highly concentrated
19	EWB-IAH	10000	1	3973	Highly concentrated
20	EWB-JAC	10000	1	23	Highly concentrated
21	EWB-JAX	10000	1	1242	Highly concentrated
22	EWB-LAS	10000	1	2010	Highly concentrated
23	EWB-LGA	10000	1	1	Highly concentrated
24	EWB-MCI	10000	1	1356	Highly concentrated
25	EWB-MDW	10000	1	2043	Highly concentrated
26	EWB-MEM	10000	1	926	Highly concentrated

27	EWR-MHT	10000	1	867	Highly concentrated
28	EWR-MKE	10000	1	1094	Highly concentrated
29	EWR-MSN	10000	1	354	Highly concentrated
30	EWR-MTJ	10000	1	15	Highly concentrated
31	EWR-MYR	10000	1	56	Highly concentrated
32	EWR-OKC	10000	1	346	Highly concentrated
33	EWR-ORF	10000	1	584	Highly concentrated
34	EWR-PDX	10000	1	571	Highly concentrated
35	EWR-PHL	10000	1	49	Highly concentrated
36	EWR-PVD	10000	1	376	Highly concentrated
37	EWR-PWM	10000	1	770	Highly concentrated
38	EWR-RIC	10000	1	1667	Highly concentrated
39	EWR-ROC	10000	1	513	Highly concentrated
40	EWR-SAN	10000	1	1134	Highly concentrated
41	EWR-SAT	10000	1	330	Highly concentrated
42	EWR-SAV	10000	1	736	Highly concentrated
43	EWR-SBN	10000	1	4	Highly concentrated
44	EWR-SLC	10000	1	354	Highly concentrated
45	EWR-SNA	10000	1	825	Highly concentrated
46	EWR-STT	10000	1	189	Highly concentrated
47	EWR-SYR	10000	1	157	Highly concentrated
48	EWR-TUL	10000	1	315	Highly concentrated
49	EWR-TVC	10000	1	24	Highly concentrated
50	EWR-TYS	10000	1	323	Highly concentrated
51	EWR-XNA	10000	1	291	Highly concentrated
52	JFK-ABQ	10000	1	254	Highly concentrated
53	JFK-ACK	10000	1	265	Highly concentrated
54	JFK-BHM	10000	1	1	Highly concentrated
55	JFK-BQN	10000	1	599	Highly concentrated
56	JFK-BTV	10000	1	1364	Highly concentrated
57	JFK-BUR	10000	1	371	Highly concentrated
58	JFK-EGE	10000	1	103	Highly concentrated
59	JFK-HNL	10000	1	342	Highly concentrated
60	JFK-HOU	10000	1	714	Highly concentrated
61	JFK-IAH	10000	1	274	Highly concentrated
62	JFK-JAC	10000	1	2	Highly concentrated
63	JFK-LGB	10000	1	668	Highly concentrated
64	JFK-MCI	10000	1	276	Highly concentrated
65	JFK-MEM	10000	1	1	Highly concentrated
66	JFK-MKE	10000	1	183	Highly concentrated
67	JFK-OAK	10000	1	312	Highly concentrated
68	JFK-PBI	10000	1	1739	Highly concentrated
69	JFK-PSE	10000	1	365	Highly concentrated
70	JFK-PSP	10000	1	19	Highly concentrated
71	JFK-PWM	10000	1	1304	Highly concentrated
72	JFK-RIC	10000	1	249	Highly concentrated

73	JFK-SDF	10000	1	46	Highly concentrated
74	JFK-SMF	10000	1	284	Highly concentrated
75	JFK-SRQ	10000	1	474	Highly concentrated
76	JFK-STL	10000	1	1	Highly concentrated
77	LGA-AVL	10000	1	10	Highly concentrated
78	LGA-BHM	10000	1	296	Highly concentrated
79	LGA-BOS	10000	1	4283	Highly concentrated
80	LGA-BWI	10000	1	15	Highly concentrated
81	LGA-CAK	10000	1	864	Highly concentrated
82	LGA-CHO	10000	1	52	Highly concentrated
83	LGA-CRW	10000	1	138	Highly concentrated
84	LGA-DAY	10000	1	391	Highly concentrated
85	LGA-DCA	10000	1	4716	Highly concentrated
86	LGA-EYW	10000	1	17	Highly concentrated
87	LGA-HOU	10000	1	425	Highly concentrated
88	LGA-IAH	10000	1	2951	Highly concentrated
89	LGA-ILM	10000	1	110	Highly concentrated
90	LGA-LEX	10000	1	1	Highly concentrated
91	LGA-MDW	10000	1	2070	Highly concentrated
92	LGA-MYR	10000	1	3	Highly concentrated
93	LGA-SAV	10000	1	68	Highly concentrated
94	LGA-SBN	10000	1	6	Highly concentrated
95	LGA-SDF	10000	1	217	Highly concentrated
96	LGA-TYS	10000	1	308	Highly concentrated

Interpretation: Routes with HHI > 2,500 (US DOJ/FTC threshold for “highly concentrated”) may warrant closer scrutiny. The visualization shows that many NYC routes are served by only 1-2 carriers with dominant shares, which could support narrow route-level market definitions in merger analysis. In practice, you would supplement this with pricing data, switching patterns, and qualitative evidence about entry barriers.

Replace `nycflights13` with live data (slot allocation, loyalty records, booking data, or Stats SA shipment records) when presenting evidence in actual matters.

Geographic market definition: Shipment flows

Understanding where products physically flow helps define geographic markets. This visualization shows the intensity of shipments between regions, which can reveal natural market boundaries, trade patterns, and whether distant regions constrain local pricing.

```
library(dplyr)
library(ggplot2)
library(tidyr)

# Illustrative shipment flow data
```

```
# In practice, pull from Census Commodity Flow Survey, BEA trade
  ↪ data,
# company shipping records, or Stats SA provincial trade data
set.seed(42)
regions <- c("Northeast", "Southeast", "Midwest", "Southwest",
  ↪ "West")

# Create flow matrix (origin x destination)
# Self-shipments (diagonal) are typically highest
flow_data <- expand_grid(
  origin = regions,
  destination = regions
) |>
  mutate(
    # Self-shipments are high, nearby regions moderate, distant
    ↪ regions low
    base_flow = case_when(
      origin == destination ~ runif(n(), 800, 1200),
      (origin == "Northeast" & destination == "Southeast") |
      (origin == "Southeast" & destination == "Northeast") ~
    ↪ runif(n(), 300, 500),
      (origin == "Midwest" & destination == "Northeast") |
      (origin == "Northeast" & destination == "Midwest") ~
    ↪ runif(n(), 250, 450),
      (origin == "West" & destination == "Southwest") |
      (origin == "Southwest" & destination == "West") ~
    ↪ runif(n(), 350, 550),
      TRUE ~ runif(n(), 50, 200)
    ),
    shipments = round(base_flow)
  ) |>
  select(-base_flow)

# Calculate self-sufficiency ratio (intra-regional shipments /
  ↪ total from that origin)
self_sufficiency <- flow_data |>
  group_by(origin) |>
  mutate(
    total_out = sum(shipments),
    self_share = shipments / total_out
  ) |>
  filter(origin == destination) |>
  select(region = origin, self_sufficiency = self_share)

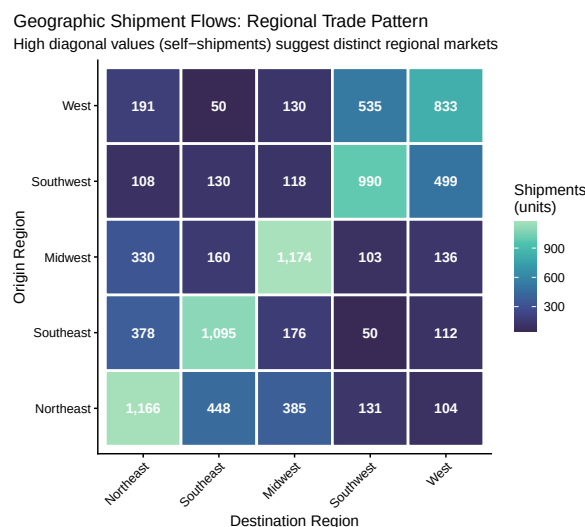
# Heatmap of flows
p_flows <- ggplot(flow_data, aes(x = destination, y = origin,
  ↪ fill = shipments)) +
```

```

geom_tile(color = "white", linewidth = 1) +
geom_text(aes(label = scales::comma(shipments, accuracy = 1)),
          color = "white", size = 4, fontface = "bold") +
scale_fill_viridis_c(
  option = "mako",
  labels = scales::comma,
  begin = 0.2,
  end = 0.9
) +
labs(
  title = "Geographic Shipment Flows: Regional Trade Pattern",
  subtitle = "High diagonal values (self-shipments) suggest
    ↪ distinct regional markets",
  x = "Destination Region",
  y = "Origin Region",
  fill = "Shipments\\n(units)",
  caption = "Illustrative data. In practice, use Census
    ↪ Commodity Flow Survey, BEA data, or proprietary shipping
    ↪ records."
) +
theme_classic(base_size = 12) +
theme(
  plot.title.position = "plot",
  panel.grid = element_blank(),
  axis.text.x = element_text(angle = 45, hjust = 1)
) +
coord_fixed()

p_flows

```



```
# Self-sufficiency table
cat("\nRegional self-sufficiency (intra-regional shipments as %
  ↳ of total):\n")
```

Regional self-sufficiency (intra-regional shipments as % of total):

```
self_sufficiency |>
  mutate(self_sufficiency = scales::percent(self_sufficiency,
    ↳ accuracy = 1)) |>
  arrange(desc(self_sufficiency)) |>
  print(n = Inf)
```

```
# A tibble: 5 x 2
# Groups:   region [5]
  region    self_sufficiency
  <fct>      <chr>
1 Midwest    62%
2 Southeast  60%
3 Southwest  54%
4 Northeast  52%
5 West       48%
```

How to use this analysis: - **High self-sufficiency** (diagonal dominance): If 70-80%+ of shipments from a region stay within that region, it suggests the region may be a distinct geographic market. - **Low cross-flows**: Minimal shipments between distant regions (e.g., Northeast <-> West) suggest they don't constrain each other's pricing. - **Clustering**: Regions with high mutual flows (e.g., Northeast <-> Southeast) might constitute a single broader market.

In practice, combine this with: - **Pricing correlation analysis**: Do prices move together across regions? - **Delivered pricing**: What are freight costs relative to product value? - **Qualitative evidence**: Customer testimony about willingness to source from distant suppliers, delivery time requirements, etc.

Data sources: - US: Census Commodity Flow Survey (5-year intervals), BEA trade flows by state - South Africa: Stats SA provincial trade data, National Treasury procurement records - Company-specific: Shipping manifests, customer address clustering, delivery zone definitions

Diversion ratios from customer switching data

Diversion ratios quantify where customers go when their first choice becomes unavailable or more expensive. They are critical for market definition, UPP calculations, and merger simulation. This example shows how to compute diversion from customer-level switching or choice data.

```
library(dplyr)
library(ggplot2)
library(tidyr)

# Simulated customer switching data
# In practice: loyalty card data, survey "next-best" responses,
# clickstream/shopping cart abandonment, health plan switching
set.seed(123)
n_customers <- 1000

# Customer choices: first choice and second choice (if first
#   ↪ unavailable)
switching_data <- tibble(
  customer_id = 1:n_customers,
  first_choice = sample(c("Product_A", "Product_B", "Product_C",
    ↪ "Product_D", "Outside"),
    n_customers, replace = TRUE,
    prob = c(0.30, 0.25, 0.20, 0.15, 0.10)),
  second_choice = sample(c("Product_A", "Product_B", "Product_C",
    ↪ "Product_D", "Outside"),
    n_customers, replace = TRUE,
    prob = c(0.22, 0.23, 0.22, 0.18, 0.15))
) |>
# Ensure second choice differs from first choice
rowwise() |>
mutate(
  second_choice = if_else(
    second_choice == first_choice,
    sample(setdiff(c("Product_A", "Product_B", "Product_C",
    ↪ "Product_D", "Outside"), first_choice), 1),
```

```

        second_choice
      )
    ) |>
    ungroup()

# Calculate diversion ratios
# Diversion from Product i to Product j =
# (# customers switching from i to j) / (total customers choosing
#   ↪ i first)
diversion_matrix <- switching_data |>
  count(first_choice, second_choice, name = "switchers") |>
  group_by(first_choice) |>
  mutate(
    total_first = sum(switchers),
    diversion_ratio = switchers / total_first
  ) |>
  ungroup() |>
  select(from = first_choice, to = second_choice,
    ↪ diversion_ratio, switchers, total_first)

# Visualize diversion from Product A and Product B (merger
#   ↪ parties)
products_of_interest <- c("Product_A", "Product_B")

p_diversion <- diversion_matrix |>
  filter(from %in% products_of_interest, to != from) |>
  mutate(
    merger_party = to %in% products_of_interest,
    from_label = paste("From", from)
  ) |>
  ggplot(aes(x = reorder(to, diversion_ratio), y =
    ↪ diversion_ratio, fill = merger_party)) +
  geom_col() +
  geom_text(aes(label = scales::percent(diversion_ratio, accuracy
    ↪ = 1)),
    hjust = -0.1, size = 3.5) +
  scale_y_continuous(labels = scales::percent_format(), limits =
    ↪ c(0, 0.35)) +
  scale_fill_manual(
    values = c("FALSE" = "#0072B2", "TRUE" = "#D55E00"),
    labels = c("FALSE" = "Competitor", "TRUE" = "Merger Partner")
  ) +
  coord_flip() +
  facet_wrap(~from_label, ncol = 2) +
  labs(
    title = "Diversion Ratios: Where Do Customers Go?",

```

```

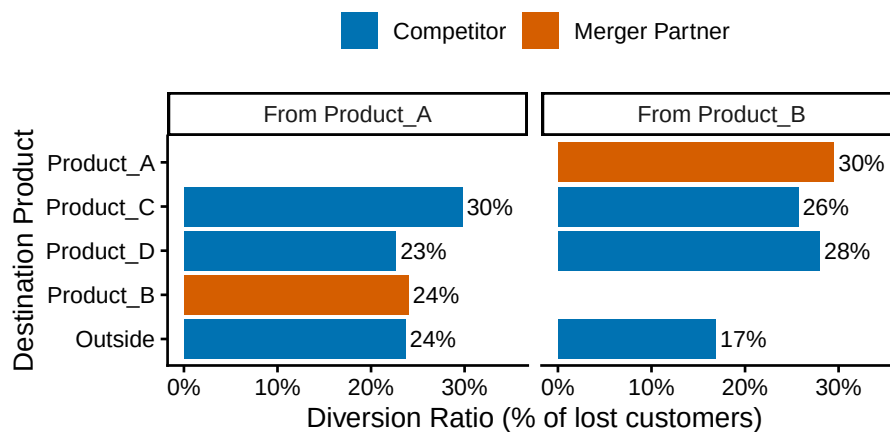
  subtitle = "High diversion to merger partner suggests
    ↳ products compete closely",
  x = "Destination Product",
  y = "Diversion Ratio (% of lost customers)",
  fill = NULL,
  caption = "Source: Simulated switching data. In practice:
    ↳ loyalty panels, surveys, or choice experiments."
) +
  theme_classic(base_size = 12) +
  theme(
    plot.title.position = "plot",
    legend.position = "top"
  )

```

p_diversion

Diversion Ratios: Where Do Customers Go?

High diversion to merger partner suggests products compete closely



ce: Simulated switching data. In practice: loyalty panels, surveys, or choice experiments.

```

# Summary table
cat("\nDiversion from Product A:\n")

```

Diversion from Product A:

```

diversion_matrix |>
  filter(from == "Product_A", to != "Product_A") |>
  arrange(desc(diversion_ratio)) |>
  mutate(
    diversion_ratio = scales::percent(diversion_ratio, accuracy =
      ↳ 0.1),

```

```

interpretation = case_when(
  to == "Product_B" ~ "HIGH - Merger partner (UPP concern)",
  to == "Outside" ~ "Customers leave market",
  TRUE ~ "Substitute competitor"
)
) |>
select(to, diversion_ratio, switchers, interpretation) |>
print(n = Inf)

```

```

# A tibble: 4 x 4
  to      diversion_ratio switchers interpretation
<chr>    <chr>          <int> <chr>
1 Product_C 29.7%                88 Substitute competitor
2 Product_B 24.0%                71 HIGH - Merger partner (UPP concern)
3 Outside   23.6%                70 Customers leave market
4 Product_D 22.6%                67 Substitute competitor

```

Interpretation for merger analysis: - **High diversion between merger parties** (e.g., 25-35% from A→B): Strong evidence products compete closely; supports narrow market definition and raises UPP concerns. - **Low diversion to outside option:** Customers are “captive” to the industry, strengthening market power concerns. - **Symmetric vs. asymmetric diversion:** If A→B is 30% but B→A is only 15%, Product A is a closer substitute for B than vice versa (important for UPP calculations).

Data sources: - **Loyalty/transaction data:** Track customer purchases before/after product unavailability or price changes - **Surveys:** “If Product A were unavailable, what would you buy instead?” (watch for hypothetical bias) - **Natural experiments:** Stockouts, temporary exits, localized price shocks - **Discrete choice experiments:** Show customers product bundles and estimate substitution patterns

i Method box

- **Computing diversion from panel data.** Use customer-level purchase histories to track switches after price changes, stockouts, or exits: `dplyr::group_by(customer) |> arrange(date) |> lag(product)` to identify transitions.
- **Survey-based diversion.** Ask “next best” questions but weight by purchase likelihood; validate against revealed switching where possible. Document screening (exclude inattentive respondents) and show robustness to alternative weighting schemes.
- **Entry/exit event studies.** When competitors enter or exit (Lyft entering a city, clinic closure), run high-frequency event studies on prices/volumes to see whether products constrain each other.

- **Cross-price elasticities with uncertainty.** Pair point estimates with bootstrapped intervals (or Bayesian draws) and present them in `gt` (the R “grammar of tables” package, which produces formatted tables) so tribunals can see both the estimates and their precision.
- **Discrete choice models.** For differentiated products, estimate multinomial logit or nested logit models (`mlogit`, `apollo` packages) to recover full substitution matrices and test market definitions.

i Qualitative evidence

Product positioning. Sales playbooks, win/loss reports, and marketing decks often list “closest competitor” products—quote them and tie to diversion ratios.

Customer evidence. Structured interviews and surveys reveal practical switching hurdles, procurement timelines, and multi-homing behavior. Document sampling rigor following established survey research standards.

RFP and contract language. Bid specs, exclusivity clauses, and default settings (especially on platforms) show whether customers view products as substitutes.

i Case box: Market definition in practice

Hospitals & insurers (US). In *FTC v. Advocate Health Care* (7th Cir. 2017), the court examined whether a hospital merger would harm commercially insured patients in the Chicago North Shore area. The FTC presented patient origin data showing that 80% of patients traveled less than 15 miles, commercial insurer testimony about limited bargaining leverage with “must-have” hospitals, and survey evidence on patient switching. Defendants countered with broader geographic market definitions based on physician referral patterns and specialty care availability. The court sided with the FTC’s narrower market, emphasizing that commercial insurers—not patients—were the direct customers, and their testimony about competitive dynamics was dispositive. **Lesson:** In two-sided markets, identify the relevant customer for market definition purposes and weight their testimony heavily.

Grocery retail (South Africa). The 2015-2019 Grocery Retail Market Inquiry combined loyalty-card transaction data from Pick n Pay and Shoprite covering millions of baskets, mall lease registers showing exclusive clauses, and micro-level CPI data by product and township. Investigators used the loyalty data to map customer catchment areas, finding that 70-80% of shoppers traveled less than 5 kilometers. They simulated a 5% SSNIP on a basket of staples and estimated that only 15-20% of customers would switch to more distant stores, well below the critical

loss threshold for margins of 25-30%. The inquiry documented how long-term exclusive leases between national chains and mall developers foreclosed entry by discounters in townships, effectively segmenting markets by neighborhood. **Lesson:** Granular transaction data can overcome theoretical debates about SSNIP tests; pair with qualitative evidence (lease contracts) to show barriers to expansion.

Tech platforms: Google Search (US/EU). In *US v. Google* (2020) and European Commission decisions, market definition turned on whether general search, specialized search (maps, shopping, travel), and social media search were in the same market. Google argued for a broad “online advertising” or “digital information” market. Agencies presented telemetry showing users rarely substitute between general search and social platforms for commercial queries, default contract terms showing Google paid billions to remain the default on Safari/Chrome (revealing high diversion), and advertiser testimony that search ads have unique intent-based targeting. The narrow “general search” market was upheld. **Lesson:** In zero-price markets, use revealed preference (default payments, multi-homing rates) rather than stated willingness to pay; SSNDQ (Small but Significant Non-transitory Decrease in Quality) can substitute for SSNIP.

Ride-hailing (Asia/Africa). Japan’s JFTC and South Africa’s Competition Commission both investigated ride-hailing platforms. JFTC examined whether taxi services and ride-hailing were in the same market, analyzing driver multi-homing (many drivers used both Uber and local apps), rider app-switching behavior via surveys, and pricing correlation. High multi-homing suggested low switching costs, supporting a broader market. In South Africa, the Commission’s digital platform inquiry examined Uber, Bolt, and local competitors, finding that commission caps and driver exclusivity clauses fragmented the market geographically (Gauteng vs. Western Cape) and by vehicle class (metered taxis vs. e-hailing). **Lesson:** Multi-sided platform markets require separate analysis of each side (riders, drivers) and attention to contractual terms that may segment otherwise integrated markets.

Southern African market evidence

- **Private Healthcare Market Inquiry (Competition Commission, 2014–2019).** Tribunal-appointed panel reviewed six years of patient-level claims covering roughly 70% of the 8.8 million beneficiaries in South Africa’s private schemes, combining them with supplier cost data to test alternative geographic definitions for specialist care. The inquiry found Herfindahl indices above 4,000 in several provinces and documented limited patient switching despite tariff differentials, motivating recommendations on supply-side licensing reform and transparency.
- **Grocery Retail Market Inquiry (2015–2019).** Investigators merged retailer loyalty data, mall lease registers, and micro-CPI data to map catch-

ment areas for supermarkets versus spaza shops. By simulating 5% SSNIP-style shocks with actual basket-level switching elasticities, the Commission showed how long-term exclusive leases between national chains and landlords constrained entry by discounters in townships and secondary towns.

- **Data Services Market Inquiry (2017–2019).** The Competition Commission benchmarked prepaid mobile data prices (30-day 1GB basket) against a peer group of African and BRICS comparators, documenting South African prices that were roughly 20–40% above the median even after controlling for spectrum cost proxies and GDP per capita. Subscriber-level usage data from MTN and Vodacom revealed steep price discrimination by income segment, which fed into the Tribunal-endorsed commitments to cut headline prepaid rates by 30–50% and expand zero-rated educational content.

i Debate: Is market definition still necessary?

A lively debate centers on whether formal market definition remains essential in unilateral effects analysis. **The traditional view** (agencies, most courts) holds that market definition disciplines the analysis: it forces practitioners to articulate which products/regions constrain pricing, establishes evidentiary presumptions (HHI thresholds), and provides a common language across jurisdictions. Without it, arguments risk becoming untethered from competitive reality.

The reform view (some economists, recent US guidelines commentary) argues that diversion ratios, margins, and UPP/GUPPI directly answer the relevant question—“will this merger raise prices?”—without the need for binary in/out market distinctions. Forcing a SSNIP test can be arbitrary (the 5–10% threshold is a convention, not economics), and critical-loss analysis is sensitive to margin measurement. Modern tools (merger simulation, natural experiments) can estimate price effects directly. See Katz & Shapiro (2003), Farrell & Shapiro (2010), and Werden (2012) for the evolution of this debate.

Practical middle ground: Most practitioners define a candidate market to anchor share calculations and HHI, but also present direct evidence (diversion, UPP, simulation) to avoid getting bogged down in SSNIP debates. Courts still expect market definition—especially in monopolization and vertical cases where structural presumptions remain central—so skip it at your peril. Document the market definition exercise even if your ultimate analysis relies on direct effects modeling.

Jurisdictional variation: DG COMP and the CMA increasingly accept “direct effects” arguments in merger reviews but still require formal market definition in abuse-of-dominance cases. The US has moved toward pragmatism: the 2023 Merger Guidelines retain market definition but emphasize it’s not always necessary when direct evidence of competitive effects is available. South African practice follows this middle path:

the Competition Act requires market definition for some statutory tests (dominance, public interest), but practitioners routinely supplement with direct diversion and simulation evidence.

i Citations and comparative note

- Cite market definition sources such as US Merger Guidelines SSNIP discussion (U.S. Department of Justice and Federal Trade Commission 2010) and EC Notice on market definition (European Commission 2024); add CMA cases for UK context.
- When using surveys or switching analyses, reference standards on survey reliability (e.g., reference guides, cases admitting/excluding surveys).
- Flag jurisdictional differences explicitly when SSNDQ or platform-centric approaches apply more than SSNIP (e.g., EU digital contexts).

Looking ahead

Export cleaned switching matrices, elasticity estimates, and qualitative chronologies to the shared appendix so the IO, merger, and cartels chapters can reuse them. Before moving to Chapter 04, confirm that your demand estimates include the product/segment labels used in the IO models and that any survey instruments or interviews are logged in the qualitative template.

Industrial Organization Toolkit

Learning goals

This chapter bridges descriptive evidence and full-blown merger simulations. You will learn when to deploy lightweight demand or supply models, how to explain their assumptions to agencies and courts, and how to integrate qualitative evidence (contracts, governance documents, interviews) so the results feel grounded.

By the end you should be able to:

- Use IO models to link conduct to outcomes: demand estimation, cost modeling, bargaining, and equilibrium effects.
- Choose between structural and reduced-form evidence depending on data, timing, and tribunal expectations.
- Capture platform-specific dynamics (multi-homing, indirect network effects, parity clauses).
- Explain each model's intuition, diagnostics, and jurisdictional track record.

Core components

1. **Demand estimation.** Start with logit/BLP-lite intuition (shares vs. prices and characteristics) (Nevo 2000; Berry, Levinsohn, and Pakes 1995). Add nested logit or random coefficients when differentiation is material, and report diversion matrices.
2. **Supply modeling & pass-through.** Recover marginal costs via first-order conditions (FOCs) or reduced-form models, and estimate pass-through elasticities (see code below).
3. **Bargaining frameworks.** Nash-in-Nash approximations for platform

or healthcare negotiations. For theoretical foundations, see Tirole (1988) and Motta (2004).

4. **Multi-sided interactions.** Incorporate indirect network effects, multi-homing rates, and platform policies (MFNs, parity clauses) as either covariates or structural parameters (Rochet and Tirole 2003).
5. **Diagnostics & validation.** Cross-check simulated price effects against historical shocks, third-party benchmarks (O. Ashenfelter and Hosken 2010), or rival stock reactions.

Structural vs. Reduced Form Approaches

Antitrust economics relies on two distinct methodological traditions. Understanding the difference—and knowing when to use each—is critical for designing an effective expert report.

The Contrast

- **Structural Modeling (Simulation):**
 - **Definition:** Estimates “deep parameters” (primitives like utility weights and marginal costs) that are assumed to remain stable even when the market changes. It imposes a specific theoretical framework (e.g., Bertrand-Nash competition with Logit demand) to link these parameters to outcomes.
 - **Primary Use: Merger Simulation.** Because the post-merger world does not yet exist, we cannot measure it directly. We must *simulate* the new equilibrium by feeding the estimated parameters into a model of the merged firm.
 - **Pros/Cons:** It is the only way to predict the future (prospective analysis), but the results are only as good as the model’s assumptions (functional form, conduct).
- **Reduced Form (Causal Inference):**
 - **Definition:** Estimates the relationship between observable variables (e.g., price and a policy dummy) without recovering the underlying primitives or imposing a full model of competition. It focuses on identifying a causal effect from historical variation.
 - **Primary Use: Retrospectives and Damages.** Used to answer “what happened?” (e.g., “Did the cartel raise prices?” or “Did the 2018 merger harm consumers?”). Common tools include Difference-in-Differences (DiD) and instrumental variables (IV) regressions.
 - **Pros/Cons:** Relies on fewer assumptions about market conduct, but cannot typically predict the effect of a *new* policy or merger (the Lucas Critique).

Complementarity in Practice

In modern practice, these approaches reinforce each other. You might use a **reduced-form** event study to show that a past merger in the same industry raised prices by 5%. You then calibrate a **structural** simulation model to reproduce that 5% increase. Once the model is “validated” by the historical data, you can confidently use it to predict the effects of the *current* merger.

i Method box

UPP/GUPPI. Use diversion ratios and margins to compute Upward Pricing Pressure (UPP) or Gross Upward Pricing Pressure Index (GUPPI). See Farrell & Shapiro (2010) “Antitrust Evaluation of Horizontal Mergers: An Economic Alternative to Market Definition” for the theoretical foundation.

Pass-through regressions. Run `feols(price ~ cost_shifter | product + time)` with clustered standard errors to get empirical pass-through before turning to structural models.

Simulation scaffolds. When time is short, build a logit demand system with constant marginal costs to approximate unilateral effects, then refine later.

i Qualitative evidence

Contracts & governance. MFNs, parity clauses, data-sharing obligations, and service-level agreements indicate how flexible margins are; tie them to modeling assumptions.

Design choices. Product defaults, API throttling, or app-store ranking algorithms signal platform steering and inform multi-sided parameters.

Field interviews. Operations or procurement leads can explain capacity constraints, switching costs, or negotiation cycles—use these insights to set priors for bargaining power or cost recovery.

i Citations and comparative note

- Cite foundational IO models and their antitrust applications (e.g., diversion/UPP notes in merger guidelines (U.S. Department of Justice and Federal Trade Commission 2010; European Commission 2004)).
- Note jurisdictional differences in treatment of MFNs/parity clauses (e.g., EU/UK platform cases vs. US approach).
- Attribute platform multi-sided modeling references (e.g., Rochet-Tirole style papers) and cite any A/B evidence or agency decisions when used.

Demand skeleton: simple logit example

```
library(tidyverse)

products <- tribble(
  ~product, ~price, ~share, ~feature_score,
  "A", 12, 0.25, 0.6,
  "B", 10, 0.20, 0.5,
  "C", 9, 0.15, 0.4,
  "Outside", 0, 0.40, 0
)

logit_data <- products |>
  mutate(
    share_ratio = share / share[product == "Outside"],
    logit_dep = log(share_ratio)
  ) |>
  filter(product != "Outside")

logit_model <- lm(logit_dep ~ price + feature_score, data =
  ↪ logit_data)
coef(logit_model)
```

(Intercept)	price	feature_score
-1.80886494	-0.06453852	3.52220594

Document instruments (cost shifters) if price endogeneity is a concern. Expand to nested logit or BLP as needed.

UPP/GUPPI calculation and sensitivity analysis

Upward Pricing Pressure (UPP) is a quick screen to assess whether a merger creates incentives to raise prices. It combines diversion ratios (where do customers go?) and margins (how profitable are those diverted sales?). High UPP suggests competitive concerns; low or negative UPP (when efficiencies exceed pricing pressure) suggests the merger may be benign.

Basic UPP calculation

```
library(dplyr)

upp <- function(diversion, margin, efficiency = 0) {
  (diversion * margin) - efficiency
}
```

```

upp_results <- tibble::tribble(
  ~pair, ~diversion, ~margin, ~efficiency,
  "Product A -> B", 0.35, 0.45, 0.05,
  "Product B -> A", 0.28, 0.40, 0.02
) |>
mutate(
  upp = upp(diversion, margin, efficiency),
  guppi = diversion * margin, # Gross UPP (before
    ↪ efficiencies)
  interpretation = case_when(
    upp > 0.10 ~ "High pricing pressure",
    upp > 0.05 ~ "Moderate pricing pressure",
    upp > 0 ~ "Low pricing pressure",
    TRUE ~ "No pricing pressure (efficiencies dominate)"
  )
)

upp_results

```

```

# A tibble: 2 x 7
  pair          diversion margin efficiency  upp guppi interpretation
<chr>          <dbl>  <dbl>      <dbl> <dbl> <dbl> <chr>
1 Product A -> B    0.35   0.45        0.05 0.108 0.158 High pricing pressure
2 Product B -> A    0.28   0.4         0.02 0.092 0.112 Moderate pricing press~

```

UPP/GUPPI sensitivity tornado chart

A tornado chart shows how UPP changes as we vary key inputs (diversion, margin, efficiencies) one at a time. This helps communicate uncertainty and identify which parameters matter most for the competitive assessment.

```

library(dplyr)
library(ggplot2)
library(tidyr)

# Base case parameters (Product A -> Product B merger)
base_diversion <- 0.30
base_margin <- 0.40
base_efficiency <- 0.03

base_upp <- (base_diversion * base_margin) - base_efficiency

# Sensitivity ranges: vary each parameter ±30% while holding
  ↪ others constant
sensitivity <- tibble::tribble(
  ~parameter, ~low_value, ~high_value,

```

```

"Diversion ratio",      base_diversion * 0.7, base_diversion *
  ↪ 1.3,
"Margin (B)",          base_margin * 0.7,      base_margin *
  ↪ 1.3,
"Efficiency credit",   base_efficiency * 0.5, base_efficiency *
  ↪ 1.5
) |>
rowwise() |>
mutate(
  upp_low = case_when(
    parameter == "Diversion ratio" ~ (low_value * base_margin)
  ↪ - base_efficiency,
    parameter == "Margin (B)" ~ (base_diversion * low_value) -
  ↪ base_efficiency,
    parameter == "Efficiency credit" ~ (base_diversion *
  ↪ base_margin) - low_value,
    TRUE ~ NA_real_
  ),
  upp_high = case_when(
    parameter == "Diversion ratio" ~ (high_value * base_margin)
  ↪ - base_efficiency,
    parameter == "Margin (B)" ~ (base_diversion * high_value) -
  ↪ base_efficiency,
    parameter == "Efficiency credit" ~ (base_diversion *
  ↪ base_margin) - high_value,
    TRUE ~ NA_real_
  ),
  range = abs(upp_high - upp_low)
) |>
ungroup() |>
arrange(desc(range))

# Create ordered factor for tornado plot
sensitivity <- sensitivity |>
  mutate(parameter = factor(parameter, levels = rev(parameter)))

# Tornado chart
ggplot(sensitivity) +
  geom_segment(aes(x = upp_low, xend = upp_high,
    y = parameter, yend = parameter),
    linewidth = 8, color = "#0072B2", alpha = 0.7) +
  geom_vline(xintercept = base_upp, linetype = "dashed",
    color = "#D55E00", linewidth = 1) +
  geom_vline(xintercept = 0, linetype = "solid",
    color = "black", linewidth = 0.5) +
  annotate("text", x = base_upp, y = 3.5,

```

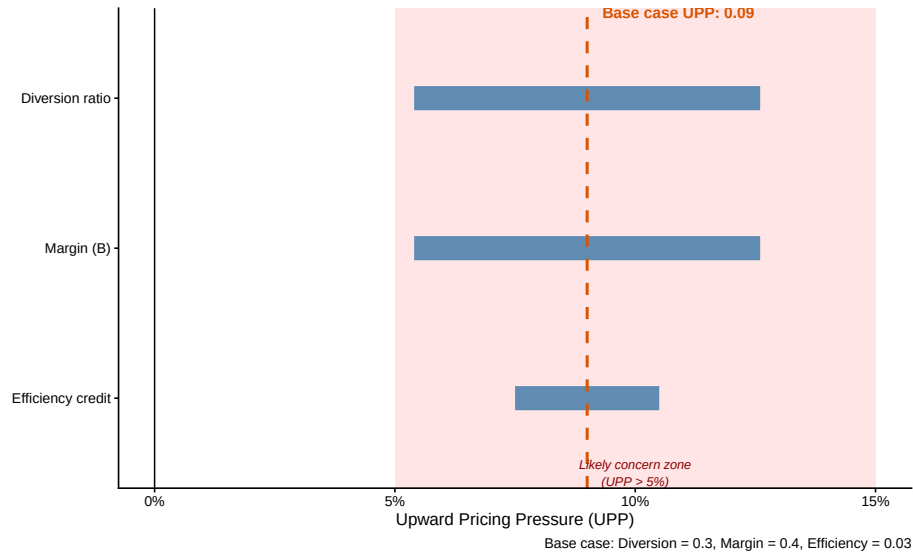
```

    label = paste0("Base case UPP: ", round(base_upp, 3)),
    hjust = -0.1, vjust = -0.5, size = 4, fontface =
      ↪ "bold", color = "#D55E00") +
annotate("rect", xmin = 0.05, xmax = 0.15, ymin = -Inf, ymax =
  ↪ Inf,
      fill = "red", alpha = 0.1) +
annotate("text", x = 0.10, y = 0.5,
      label = "Likely concern zone\n(UPP > 5%)",
      size = 3, color = "darkred", fontface = "italic") +
scale_x_continuous(labels = scales::percent_format(),
      breaks = seq(-0.05, 0.20, by = 0.05)) +
labs(
  title = "UPP Sensitivity Analysis (Tornado Chart)",
  subtitle = "How does UPP change as we vary key inputs? Longer
    ↪ bars = more sensitive parameters",
  x = "Upward Pricing Pressure (UPP)",
  y = NULL,
  caption = paste0("Base case: Diversion = ",
    ↪ round(base_diversion, 2),
      ", Margin = ", round(base_margin, 2),
      ", Efficiency = ", round(base_efficiency, 2))
) +
theme_classic(base_size = 12) +
theme(
  plot.title.position = "plot",
  panel.grid.major.y = element_blank()
)

```

UPP Sensitivity Analysis (Tornado Chart)

How does UPP change as we vary key inputs? Longer bars = more sensitive parameters



```
# Summary table
cat("\nSensitivity analysis summary:\n")
```

Sensitivity analysis summary:

```
sensitivity |>
  mutate(
    across(c(low_value, high_value, upp_low, upp_high),
           ~round(., 4)),
    range = round(range, 4),
    impact = case_when(
      range > 0.04 ~ "HIGH impact on UPP",
      range > 0.02 ~ "MODERATE impact",
      TRUE ~ "LOW impact"
    )
  ) |>
  select(parameter, low_value, high_value, upp_low, upp_high,
         range, impact) |>
  print(n = Inf)
```

A tibble: 3 x 7

	parameter	low_value	high_value	upp_low	upp_high	range	impact
	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1	Diversion ratio	0.21	0.39	0.054	0.126	0.072	HIGH impact on ~
2	Margin (B)	0.28	0.52	0.054	0.126	0.072	HIGH impact on ~

3 Efficiency credit 0.015 0.045 0.105 0.075 0.03 MODERATE impact

How to use this chart: - **Longest bars** = most sensitive parameters. Focus data collection and robustness checks on these. - **Base case** (dashed orange line): Your central UPP estimate. - **Zero line**: $UPP < 0$ suggests efficiencies outweigh pricing pressure. - **Concern threshold** (shaded region): Many agencies use 5% as a rule of thumb; higher UPP warrants closer scrutiny.

Practical insights: - If **diversion** creates the widest swing, invest in better switching data (surveys, natural experiments, loyalty panels). - If **margin** is most sensitive, request detailed cost accounting or validate with third-party benchmarks. - If **efficiencies** have large impact, document synergy claims rigorously (pre/post integration plans, expert affidavits).

GUPPI variant: To calculate GUPPI instead of UPP, simply omit the efficiency credit: $GUPPI = \text{diversion} \times \text{margin}$. Some agencies prefer GUPPI because it isolates the gross pricing pressure before offsetting efficiencies.

Tie the efficiency term to documented synergies or cost savings (procurement economies, network effects, R&D cost sharing). For cross-border mergers, adjust for currency and local cost differences.

Visualizations

Pass-through diagnostic using public price indices

Pass-through analysis estimates how much of a cost change flows through to consumer prices. This is critical for cartel damages (upstream overcharge → downstream harm), merger analysis (will cost savings benefit consumers?), and vertical restraints. This example uses real FRED data for gasoline prices.

```
library(fredr)
library(dplyr)
library(tidyr)
library(ggplot2)
library(patchwork)

fredr_set_key(Sys.getenv("FRED_API_KEY"))

# Producer Price Index: Petroleum refining
ppi <- fredr(
  series_id = "PCU324110324110",
  observation_start = as.Date("2015-01-01")
) |>
  transmute(date, ppi = value)

# Consumer Price Index: Gasoline (all types)
cpi <- fredr(
```

```
series_id = "CUSR0000SETB01",
observation_start = as.Date("2015-01-01")
) |>
  transmute(date, cpi = value)

pass_df <- inner_join(ppi, cpi, by = "date") |>
  arrange(date) |>
  mutate(
    ppi_index = ppi / first(ppi) * 100,
    cpi_index = cpi / first(cpi) * 100,
    # Calculate changes for pass-through regression
    ppi_change = (ppi - lag(ppi)) / lag(ppi),
    cpi_change = (cpi - lag(cpi)) / lag(cpi)
  ) |>
  drop_na()

# Time series plot
pass_long <- pass_df |>
  select(date, Producer = ppi_index, Consumer = cpi_index) |>
  pivot_longer(-date, names_to = "series", values_to = "index")

p1 <- ggplot(pass_long, aes(x = date, y = index, color = series))
  +
  geom_line(linewidth = 1.1) +
  scale_color_manual(values = c("Producer" = "#D55E00",
    "Consumer" = "#0072B2")) +
  labs(
    title = "Producer vs. Consumer Gasoline Price Indices",
    subtitle = "Tracking cost pass-through from refineries to
      retail",
    x = NULL,
    y = "Index (Jan 2015 = 100)",
    color = NULL
  ) +
  theme_classic(base_size = 12) +
  theme(
    legend.position = "bottom",
    plot.title.position = "plot"
  )

# Scatter plot with regression line
pass_model <- lm(cpi_index ~ ppi_index, data = pass_df)
pass_coef <- coef(pass_model)[["ppi_index"]]
pass_r2 <- summary(pass_model)$r.squared

p2 <- ggplot(pass_df, aes(x = ppi_index, y = cpi_index)) +
```

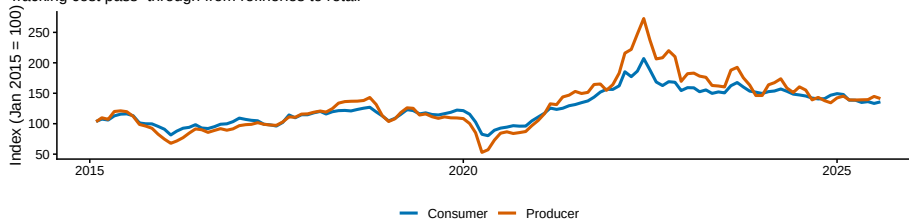
```

geom_point(alpha = 0.5, color = "#0072B2") +
geom_smooth(method = "lm", se = TRUE, color = "#D55E00",
  ↪ linewidth = 1.2) +
annotate("text", x = min(pass_df$ppi_index) + 5, y =
  ↪ max(pass_df$cpi_index) - 5,
  label = paste0("Pass-through rate: ", round(pass_coef,
  ↪ 3),
  "\nR² = ", round(pass_r2, 3)),
  hjust = 0, size = 4, fontface = "bold") +
labs(
  title = "Pass-Through Regression",
  subtitle = "1 unit increase in PPI → ? unit increase in CPI",
  x = "Producer Price Index",
  y = "Consumer Price Index"
) +
theme_classic(base_size = 12) +
theme(plot.title.position = "plot")

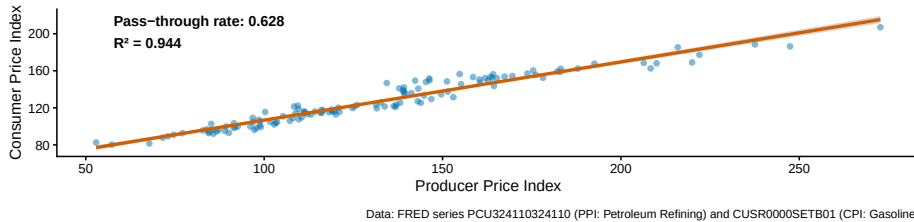
# Combined plot
p1 / p2 + plot_annotation(
  caption = "Data: FRED series PCU324110324110 (PPI: Petroleum
  ↪ Refining) and CUSR0000SETB01 (CPI: Gasoline)"
)

```

Producer vs. Consumer Gasoline Price Indices
Tracking cost pass-through from refineries to retail



Pass-Through Regression
1 unit increase in PPI → ? unit increase in CPI



Data: FRED series PCU324110324110 (PPI: Petroleum Refining) and CUSR0000SETB01 (CPI: Gasoline)

```

# Summary statistics
pass_summary <- tibble::tibble(
  metric = c("Pass-through coefficient", "R-squared",
  ↪ "Observations", "Interpretation"),

```

```
value = c(
  round(pass_coef, 3),
  round(pass_r2, 3),
  nrow(pass_df),
  ifelse(pass_coef < 0.5, "Incomplete pass-through",
    ifelse(pass_coef < 1.0, "Substantial pass-through",
      "Over-shifting (pass-through > 100%)"))
)
)

cat("\nPass-through analysis summary:\n")
```

Pass-through analysis summary:

```
print(pass_summary, n = Inf)
```

A tibble: 4 x 2

metric	value
<chr>	<chr>
1 Pass-through coefficient	0.628
2 R-squared	0.944
3 Observations	127
4 Interpretation	Substantial pass-through

Interpretation: - **Pass-through coefficient:** Measures how much a 1% increase in producer prices flows through to consumer prices. Values < 1 indicate incomplete pass-through (firms absorb some cost increases); values > 1 indicate over-shifting (firms amplify cost increases, possibly due to market power). - **R-squared:** Indicates how well producer prices explain consumer price variation. High R^2 suggests tight cost-price linkage. - **Incomplete pass-through:** Common in competitive markets where firms absorb cost shocks to retain customers. Can also occur with sticky prices or menu costs. - **Over-shifting:** May signal market power (firms use cost increases as “focal points” to raise prices beyond the cost change) or complementarities/network effects.

Applications in antitrust: - **Cartel damages:** If pass-through is 80%, a \$10 upstream overcharge causes an \$8 downstream price increase. - **Merger efficiencies:** If parties claim \$5M in cost savings but pass-through is only 30%, consumers benefit by only \$1.5M. - **Vertical mergers:** Estimate pass-through separately for upstream and downstream stages to predict EDM (Elimination of Double Marginalization) benefits.

Swap series IDs for your industry (e.g., WPUSI012011 for steel PPI, CPIAUCSL for overall CPI) or use firm-specific cost and price data when available.

Bargaining sketches

When platforms or healthcare networks negotiate contracts, full structural models may be infeasible. Use Nash-in-Nash approximations or reduced-form proxies:

- **Regression-based bargaining shares.** Estimate how much of a wholesale cost shock flows into downstream prices for different bargaining parties (e.g., pharmacy benefit managers vs. hospitals).
- **Qualitative overlays.** Pair negotiation timelines, renewal clauses, and concession histories with the empirical results.

Looking ahead

Store demand estimates, diversion matrices, and pass-through/bargaining diagnostics in `data/derived` with READMEs so the mergers and monopolization chapters can drop them into simulations. Before moving to Chapter 05 (Cartels), note which datasets or code paths (e.g., `io-logit`, `upp-gup` templates) should be generalized in `chapters/13-empirical-appendix.qmd`. Update the visualization tracker if you add new figures so the team can coordinate future enhancements.

Cartels and Collusion

Learning goals

Cartel enforcement hinges on weaving together quantitative screens, econometric estimates, and documentary evidence. This chapter walks through that workflow using established methodologies from the OECD and academic literature on cartel detection.

By the end you should be able to:

- Spot suspicious conduct using dashboards, variance/rotation screens, and early break tests.
- Tie raid/leniency events to structural changes in prices, margins, and quantities.
- Estimate overcharge and pass-through with transparent assumptions and uncertainty analysis.
- Triangulate econometrics with leniency statements, chats, and procurement narratives to build court-ready stories.

Workflow roadmap

1. **Timeline & conduct map.** Build a dated chronology from complaints, internal messages, procurement milestones, and raids. Align suspected mechanisms (bid rotation, geographic allocation, price floors) with specific products and periods.
2. **Descriptive diagnostics.** Plot prices, costs, and quantities; look for unnatural stability or compression.
3. **Screens.** Run variance, digit, and rotation screens to triage markets. Treat them as prioritization tools, not dispositive proof.
4. **Event/break tests.** Evaluate how prices or spreads change around raids, leniency announcements, or policy shifts.

5. **Overcharge & pass-through.** Estimate counterfactual prices using panel regressions, yardsticks, or synthetic controls; propagate uncertainty.
6. **Triangulation.** Reconcile quantitative results with leniency statements, buyer interviews, and messaging logs.
7. **Damages & remedies.** Translate overcharge estimates into monetary relief or structural commitments, respecting jurisdictional rules (e.g., Competition Tribunal SA public-interest remedies).

Timeline and conduct map

Document every known communication, meeting, or enforcement action in a single table (date, event, evidence source, hypothesis). Update it as new statements arrive. Every regression or screen should cite the row(s) it tests—this discipline prevents “data dredging” and makes testimony easier.

Descriptives and screens

Start with `ggplot2` dashboards that overlay prices with cost indices, demand proxies, and competitor prices. Flag regimes where prices remain static despite volatile costs or where margins converge across firms.

- **Variance/dispersion screens:** Check price spreads, standard deviations, and coefficients of variation across firms or regions (Abrantes-Metz et al. 2006).
- **Procurement rotation:** Rank bids chronologically to flag turn-taking, convenient price endings, or geographic allocations (Porter and Zona 1993; Conley and Decarolis 2016).
- **Digit/Benford checks:** Use sparingly and only when invoice conventions support the assumptions.
- **Correlation screens:** High correlations in supposedly independent bids can justify deeper probes (Harrington 2008).

Document screen logic following OECD guidance on cartel screens (OECD 2013) and note data limitations (missing bidders, net vs. list prices).

Bid-rotation scaffold

```
library(dplyr)
library(ggplot2)
```

```
rotation <- bids_df |>
  group_by(bidder) |>
  summarise(
    wins = sum(rank == 1),
    avg_rank = mean(rank, na.rm = TRUE),
    rotation_sd = sd(rank, na.rm = TRUE)
  ) |>
  arrange(desc(wins))

ggplot(rotation, aes(x = reorder(bidder, wins), y = wins)) +
  geom_col(fill = "#0072B2") +
  coord_flip() +
  labs(title = "Bid wins by vendor", x = NULL, y = "Wins") +
  theme_classic()
```

Pair charts with tables showing consecutive wins, geographic patterns, or unexplained bid withdrawals.

Bid-rotation network graph

Network graphs help visualize suspected coordination patterns by showing which bidders consistently “yield” to others. This is particularly useful for identifying systematic rotation schemes or hub-and-spoke arrangements.

```
library(dplyr)
library(tidyr)
library(ggplot2)
library(igraph)
library(ggraph)

# Simulated bid data: tender_id, bidder, rank, bid_amount
# In practice, replace with actual procurement data
set.seed(123)
n_tenders <- 50
bidders <- c("Firm A", "Firm B", "Firm C", "Firm D", "Firm E")

bids <- expand_grid(
  tender_id = 1:n_tenders,
  bidder = bidders
) |>
  group_by(tender_id) |>
  mutate(
    # Simulate rotation: certain patterns are more likely
    base_bid = runif(n(), 100, 200),
    rotation_boost = case_when(
      tender_id %% 5 == 0 & bidder == "Firm A" ~ -20,
```

```

    tender_id %% 5 == 1 & bidder == "Firm B" ~ -20,
    tender_id %% 5 == 2 & bidder == "Firm C" ~ -20,
    tender_id %% 5 == 3 & bidder == "Firm D" ~ -20,
    tender_id %% 5 == 4 & bidder == "Firm E" ~ -20,
    TRUE ~ 0
  ),
  bid = base_bid + rotation_boost,
  rank = rank(bid)
) |>
ungroup()

# Identify winners and runner-ups
winners <- bids |>
  filter(rank == 1) |>
  select(tender_id, winner = bidder)

runner_ups <- bids |>
  filter(rank == 2) |>
  select(tender_id, runner_up = bidder)

# Create edge list: who "yielded" to whom (runner-up -> winner)
edges <- inner_join(winners, runner_ups, by = "tender_id") |>
  count(runner_up, winner, name = "weight") |>
  filter(runner_up != winner)

# Create network graph
g <- graph_from_data_frame(edges, directed = TRUE)

# Plot using ggraph
ggraph(g, layout = "fr") +
  geom_edge_link(aes(width = weight, alpha = weight),
    arrow = arrow(length = unit(4, 'mm'), type =
      ↪ "closed"),
    end_cap = circle(3, 'mm'),
    color = "#666666") +
  geom_node_point(size = 12, color = "#0072B2", alpha = 0.8) +
  geom_node_text(
    aes(label = name),
    color = "white",
    fontface = "bold",
    size = 4,
    family = "Arial"
  ) +
  scale_edge_width(range = c(0.5, 3)) +
  scale_edge_alpha(range = c(0.3, 0.9)) +
  labs(

```

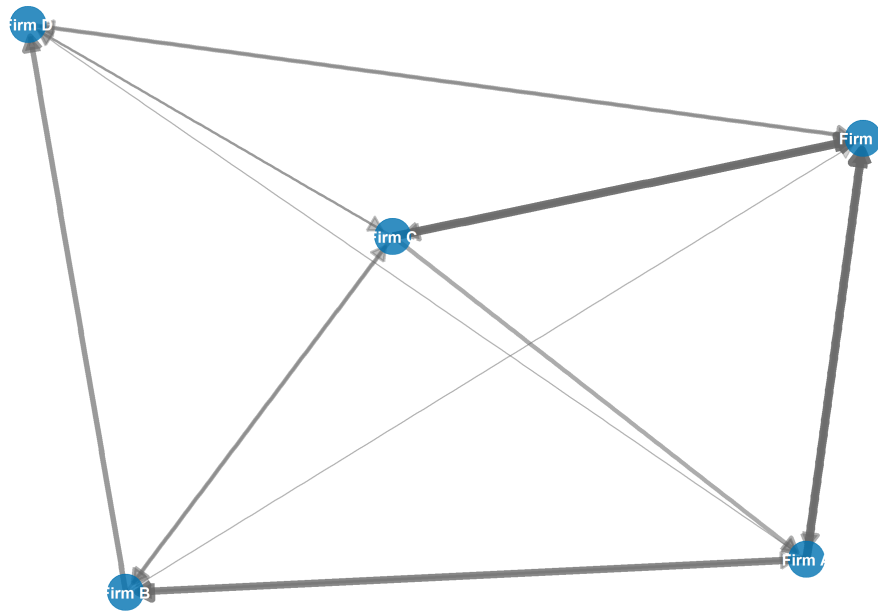
```

title = "Bid Rotation Network Graph",
subtitle = "Arrows show which firms consistently submit
  ↳ 2nd-place 'cover bids' when others win",
caption = "Edge width = frequency of runner-up pattern.
  ↳ Systematic rotation creates hub-spoke or circular
  ↳ patterns."
) +
theme_graph() +
theme(
  text = element_text(family = "Arial"),
  plot.title = element_text(size = 14, face = "bold"),
  plot.subtitle = element_text(size = 11),
  legend.position = "none"
)

```

Bid Rotation Network Graph

Arrows show which firms consistently submit 2nd-place 'cover bids' when others win



Edge width = frequency of runner-up pattern. Systematic rotation creates hub-spoke or circular patterns.

```

# Supplementary table: win rates and rotation metrics
rotation_metrics <- bids |>
  group_by(bidder) |>
  summarise(
    wins = sum(rank == 1),
    second_place = sum(rank == 2),
    avg_rank = mean(rank),

```

```

    rotation_index = wins / n_tenders
  ) |>
  arrange(desc(wins))

cat("\nRotation metrics by bidder:\n")

```

Rotation metrics by bidder:

```
print(rotation_metrics, n = Inf)
```

```

# A tibble: 5 x 5
  bidder  wins second_place avg_rank rotation_index
<fct>  <int>      <int>    <dbl>      <dbl>
1 Firm E    16         9     2.66        0.32
2 Firm A     9        12     3.02        0.18
3 Firm B     9         9     3.12        0.18
4 Firm D     9         6     3.08        0.18
5 Firm C     7        14     3.12        0.14

```

How to interpret this graph: - **Hub-spoke patterns:** If one firm is at the center with many arrows pointing to it, that firm may be the “ringleader” coordinating bids. - **Circular patterns:** Arrows forming a circle ($A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$) suggest systematic rotation where firms take turns winning. - **Asymmetric flows:** If Firm A frequently comes second when Firm B wins, but not vice versa, this may indicate a hierarchical arrangement. - **Edge weights:** Thicker arrows indicate more frequent patterns, which are more suspicious than occasional occurrences.

Practical applications: - Combine with document evidence (emails, meeting logs) to validate rotation schemes. - Compare network patterns across product lines or regions to identify geographic allocation. - Use as demonstrative exhibit in tribunal presentations or expert reports.

Replace simulated data with actual procurement records (World Bank, TED, US DOT, Stats SA infrastructure tenders). Document bid amounts, dates, and project characteristics in `data/raw/procurement/` with provenance notes.

Event and break tests

When raids or leniency filings occur, run event studies on prices, spreads, or quantities. Check both cartel participants and outsiders; in public procurement you can also evaluate engineers’ estimates or cost indices.

Structural break tests (Bai–Perron, CUSUM) highlight regime shifts. Always incorporate costs so you do not misattribute cost-driven changes to conduct.

Present break plots next to relevant timeline entries (e.g., WhatsApp thread confirming a January 2019 meeting).

i Code box: raid event study scaffold

```
# df must include date, price, cost, treated (0/1), firm,
  ↳ product
# library(fixest)
# df <- df |> mutate(post = date >= as.Date("YYYY-MM-DD"))
# feols(log(price) ~ post * treated + cost + month(date) |
  ↳ firm + product + year(date),
#       cluster = ~firm, data = df)
```

Overcharge and pass-through

Pick a counterfactual that matches data availability:

- **Before/after:** Include product, customer, and time fixed effects plus cost controls.
- **Difference-in-differences:** Compare cartel markets to unaffected regions or product classes; test pre-trends.
- **Yardsticks:** Use similar markets (neighboring countries, regulated benchmarks) when local data are thin.
- **Synthetic controls:** Helpful for single-market cartels; combine unaffected markets into a donor pool.

Robustness: alternative functional forms (log vs. level), trimming outliers, placebo periods, heteroskedasticity checks, and randomization inference when N is small. For pass-through, regress downstream prices on upstream costs with lags to quantify harm on intermediaries vs. end-users.

Leniency, documents, and triangulation

Leniency statements, chats, and board minutes pin down conduct mechanisms (rotation order, price floors, trigger strategies). Use them to:

- Set regression windows and sample selections.
- Validate that estimated start/stop dates match qualitative evidence.
- Explain residuals—if econometrics show little effect where documents admit coordination, revisit product mapping or data coverage.

- Clarify what is fact (documented) vs. inference (econometric patterns), citing (OECD 2015) or agency policies.

i Method box: econometric toolkit

Event studies: Evaluate raid/leniency impact on prices or spreads.

Overcharge regressions: `feols(log(price) ~ cartel_period + cost + demand | product + customer + time)` with clustered SEs and placebo checks.

Pass-through: Dynamic regressions or VECMs linking upstream and downstream prices to trace harm along the chain.

i Method box: screens & variance

Distribution screens: Variance compression, digit spikes, Benford where appropriate.

Rotation checks: Consecutive wins, geographic allocation, or sequential ordering.

Cost-pass-through diagnostics: Compare pass-through before/during suspected collusion; suppressed pass-through can corroborate coordination.

i Qualitative evidence

Treat qualitative materials as structured data:

- **Bid sheets & messaging apps:** Extract explicit allocation rules or fallback prices.
- **Buyer interviews:** Document procurement timelines, awareness of cover bids, or complaints.
- **Expert narratives:** Craft timelines tying every empirical result to specific documents.

i Case box: Illustrative matters

Apple eBooks (US). Hub-and-spoke coordination with agency-model contracts and SSNDQ logic.

Lysine & DRAM (US/EU). Classic dawn-raid analyses (see Connor, 2007, “Price-Fixing Overcharges,” American Antitrust Institute Working Paper; Levenstein & Suslow, 2006, “What Determines Cartel Success?,” *Journal of Economic Literature* 44(1): 43-95).

Construction bid-rigging (EU, SA, JFTC). Rotation matrices and dawn raids aligned with procurement archives.

Digital advertising (SAMR). Algorithms and platform policies as qualitative evidence.

i Case box: South African enforcement highlights

- **Bread cartel (Tribunal case 15/CR/Feb07).** Stats SA CPI data plus mill-level costs revealed synchronized 30–40 cent jumps; overcharge estimated at 7–10% with R250 million penalties and consumer relief funds.
- **Construction fast-track settlement (2013).** Self-reported matrices across 300 tenders showed rotation; event windows around raids captured 8–12% price drops relative to engineers' estimates.
- **Fertilizer/ammonia (Sasol/Yara/Omnia).** Export-parity benchmarks and plant-level variable-cost data showed 25–30% overcharges despite spare capacity, leading to penalties and divestitures.

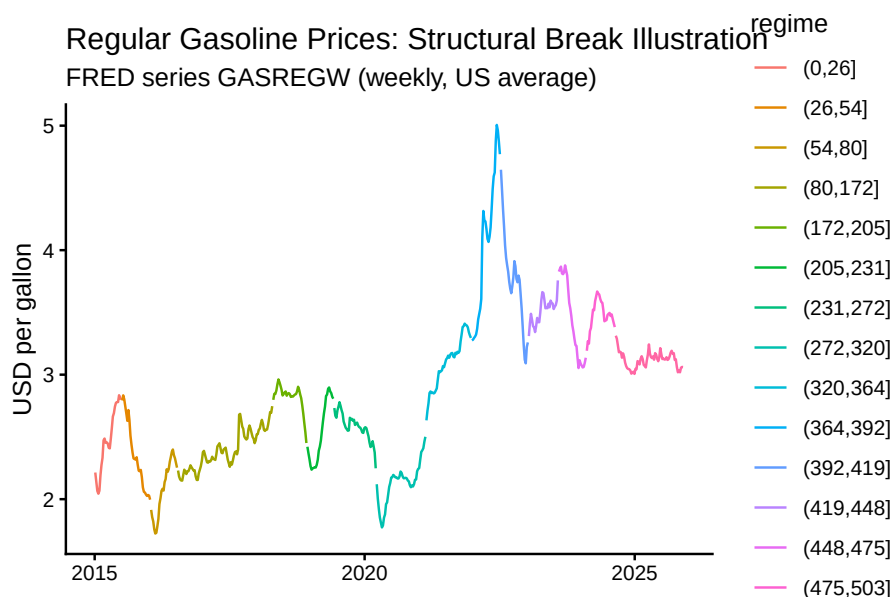
Code box: structural break illustration

```
library(fredr)
library(dplyr)
library(strucchange)
library(ggplot2)

fredr_set_key(Sys.getenv("FRED_API_KEY"))
gas <- fredr(series_id = "GASREGW", observation_start =
  ↪ as.Date("2015-01-01")) |>
  rename(price = value) |>
  mutate(week = row_number())

bp <- breakpoints(price ~ week, data = gas, h = 26)
gas$regime <- factor(cut(gas$week, c(0, bp$breakpoints, Inf)))

ggplot(gas, aes(date, price, color = regime)) +
  geom_line() +
  labs(title = "Regular Gasoline Prices: Structural Break
  ↪ Illustration",
       subtitle = "FRED series GASREGW (weekly, US average)",
       x = NULL, y = "USD per gallon") +
  theme_classic()
```



Replace the public FRED data with product-level transactions to present in litigation.

i Citations and comparative note

- Cite OECD screen guidance, FTC/DOJ econometrics speeches, and EC/JFTC leniency notices when defending methodology.
- Reference cases (lysine, DRAM, auto parts, LIBOR) with docket numbers.
- Flag jurisdictional differences: e.g., SAMR often requires data-driven screens, while US courts scrutinize documentary corroboration.

Looking ahead

Archive every screen, regression, and figure in `data/derived` with READMEs so the damages and litigation chapters can reuse them. Push generic templates (rotation indices, break plots, overcharge tables) into `chapters/13-empirical-appendix.qmd`. When you shift to the mergers chapter, carry over demand estimates, diversion ratios, or pass-through results—post-cartel mergers frequently face heightened scrutiny, and reusing these diagnostics speeds up the next phase.

Mergers

Learning goals

Merger reviews synthesize everything covered so far: market definition, IO modeling, qualitative evidence, and remedies. This chapter provides a practical workflow for assessing unilateral, coordinated, vertical, and public-interest effects across US, EU/UK, and South African jurisdictions, drawing on agency guidance (U.S. Department of Justice and Federal Trade Commission 2023; European Commission 2004; Competition and Markets Authority 2021).

By the end you should be able to:

- Assess unilateral and coordinated effects with descriptive and structural tools (UPP/GUPPI, logit simulations, retrospectives).
- Evaluate vertical/mixed theories (elimination of double marginalization (EDM) vs. foreclosure) and quantify vUPP.
- Integrate efficiencies and public-interest claims with evidence.
- Propose remedies tied explicitly to diagnosed harms.

Core topics

- Market structure pre/post; diversion from shares and switching data.
- UPP/GUPPI screens; thresholds and caveats.
- Merger simulation: differentiated products logit/nested logit; calibrating margins and ownership.
- Coordinated effects: maverick analysis, transparency, capacity and bidding dynamics.
- Retrospectives: diff-in-diff/event studies with rivals and controls.
- Horizontal, vertical, and mixed mergers; define “upstream” and “downstream” relative to the theory of harm and adapt analysis to industry specifics.
- Workflow: (1) transaction and market overview; (2) shares/diversion/HHI; (3) UPP/GUPPI and margins; (4) unilateral and coordinated effects (structural or reduced form); (5) vertical/mixed theories (EDM vs. foreclosure); (6) efficiencies evidence; (7) remedies and retrospectives.

Transaction overview and market structure

- Summarize products, geographies, overlaps, and timing; map to candidate relevant markets.
- Compute shares/HHI and diversion using shares, switching, or survey-based measures. Flag mavericks and fringe.
- Upstream/downstream definitions depend on the theory of harm: be explicit about the vertical chain and platform sides.
- Ground definitions in agency guidance (US Merger Guidelines (U.S. Department of Justice and Federal Trade Commission 2023, 2010), EC/CMA guidance (European Commission 2004; Competition and Markets Authority 2021)) and adjust for platform/digital contexts.

Practical tips: keep a data inventory (chapter 13 template) noting which datasets inform shares (transaction data, Nielsen panels, loyalty cards). Align product labels with the IO models you plan to run later so you avoid remapping midstream.

Unilateral effects: UPP/GUPPI and sim

- Use UPP/GUPPI as a screen (Farrell and Shapiro 2010b; Jaffe and Weyl 2013); report inputs (diversion, margins) transparently. Sensitivity to margin measurement and diversion estimates should be shown.
- For differentiated products, simulate with logit/nested-logit or BLP-lite (Nevo 2000; Berry, Levinsohn, and Pakes 1995); emphasize calibration choices (outside share, margin sources).
- Convey uncertainty: confidence intervals on diversion, ranges on margins, and alternative ownership assumptions (Weinberg and Hosken 2013).
- For readers, tie back to case examples in your slides to show how margins and diversion were evidenced (transaction data, surveys, switching analyses).

Simple UPP/GUPPI calculation

```
library(dplyr)
upp <- function(diversion, price, margin, efficiency = 0) {
  diversion * price * (1 - margin) - efficiency
}

inputs <- tibble::tribble(
  ~pair, ~diversion, ~price, ~margin, ~efficiency,
  "A->B", 0.35, 11.50, 0.45, 0.50,
  "B->A", 0.28, 10.00, 0.40, 0.30
) |>
  mutate(upp = upp(diversion, price, margin, efficiency))

inputs
```

```
# A tibble: 2 x 6
  pair diversion price margin efficiency upp
<chr>      <dbl> <dbl> <dbl>      <dbl> <dbl>
1 A->B      0.35  11.5  0.45      0.5  1.71
2 B->A      0.28   10  0.4      0.3  1.38
```

Highlight the evidence sources for diversion (switching matrices, conjoint surveys, clickstream data) and margins (cost accounting, P&L, expert testimony). Note that negative efficiencies shift UPP upward, so document synergy assumptions thoroughly.

Logit simulation scaffold

```
source("../R/helpers.R")
# products_df columns: product, firm, price, share, mc or margin,
  ↪ nest
# sim <- run_logit_sim(products_df, merging_firms =
  ↪ c("FirmA", "FirmB"))
# sim$summary # contains post-merger price deltas, elasticities,
  ↪ diversion matrices
```

Before presenting results, show calibration diagnostics: how well the model reproduces pre-merger shares, whether price elasticities fall in plausible ranges, and how sensitive predictions are to alternative marginal-cost assumptions.

Coordinated effects

- Look for increased symmetry, transparency, or capacity alignment post-merger; include bidding/auction context where relevant.
- Event studies around merger announcements can show rivals' stock price reactions (coordination signal) but should be paired with real-world capacity/contract evidence.
- Document maverick roles and whether the transaction removes or disciplines them.

Useful evidence: internal documents describing "price umbrella" logic, third-party contracts showing increased transparency, and supply/demand data indicating higher capacity utilization or inventory visibility. Build a maverick profile (pricing aggressiveness, innovation track record) to show whether elimination meaningfully raises coordination risk.

Vertical and mixed effects

- EDM vs. foreclosure: quantify both. vUPP and upward pricing pressure on rivals' access should be shown alongside EDM magnitudes.

- Identify potential raising rivals' costs channels (input denial, degradation, parity clauses) and model likely price effects.
- Conglomerate/mixed cases: bundling/tying risk, data advantage transfer, and cross-market leverage should be explicitly linked to facts.
- When defining "upstream" vs. "downstream," anchor to transaction flows and incentives; platform settings may require mapping sides rather than tiers.

Efficiencies and remedies

- Synergy claims: require verifiable, merger-specific efficiencies with timelines and implementation costs; stress test with sensitivity tables.
- Remedies: structural first; behavioral only if verifiable/monitorable. Link proposed remedies to modeled harms and operational feasibility.

Tie efficiencies to data. For example, if parties cite procurement savings, request SKU-level cost projections and simulate whether those savings offset UPP. For behavioral remedies, document monitoring costs and fallback options (trustees, data rooms) referenced in CMA/DG COMP practice.

Retrospectives

- Where historical analogs exist, run diff-in-diff/event studies on prices/output/quality. Use rivals and unaffected markets as controls; test pre-trends (O. Ashenfelter and Hosken 2010; Miller and Weinberg 2017).
- For platform/vertical cases, examine participation/multi-homing effects and access terms over time.
- Cite retrospective literature to benchmark magnitudes and methods (Weinberg and Hosken 2013).
- Use well-known retrospectives (e.g., supermarket/hospital/airline cases) to set expectations on effect sizes and uncertainty.

Stock-event diagnostic

```
library(tidyquant)
library(dplyr)
library(ggplot2)

event_date <- as.Date("2013-02-14") # substitute deal date
tickers <- c("AAL", "DAL", "LUV", "SPY")

prices <- tq_get(
  tickers,
  from = event_date - lubridate::days(120),
  to   = event_date + lubridate::days(60)
```

```

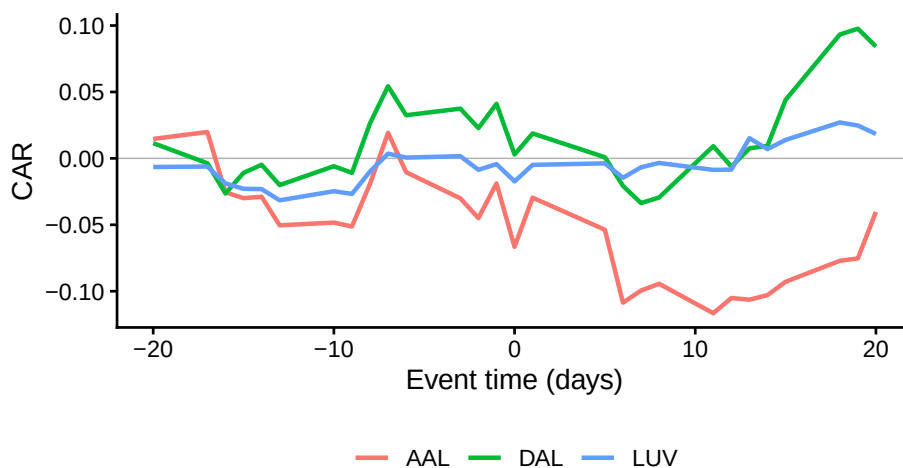
) |>
  group_by(symbol) |>
  arrange(date) |>
  mutate(ret = log(adjusted) - log(lag(adjusted))) |>
  ungroup()

market <- prices |> filter(symbol == "SPY") |> transmute(date,
  ↪ mkt_ret = ret)
cars <- prices |> filter(symbol != "SPY") |>
  left_join(market, by = "date") |>
  mutate(abnormal = ret - mkt_ret, rel_day = as.integer(date -
    ↪ event_date)) |>
  filter(between(rel_day, -20, 20)) |>
  group_by(symbol) |>
  arrange(rel_day) |>
  mutate(car = cumsum(abnormal))

ggplot(cars, aes(rel_day, car, color = symbol)) +
  geom_hline(yintercept = 0, linewidth = 0.3, color = "gray70") +
  geom_line(linewidth = 0.9) +
  labs(title = "Cumulative abnormal returns around merger
    ↪ announcement",
    subtitle = "Illustrative airline example (event ±20
    ↪ trading days)",
    x = "Event time (days)", y = "CAR", color = NULL) +
  theme_classic(base_size = 12) +
  theme(legend.position = "bottom")

```

Cumulative abnormal returns around merger announcer
Illustrative airline example (event ±20 trading days)



Use CAR patterns as suggestive evidence of coordination or efficiency expectations, but always pair with operational data (capacity, contracts).

Southern African merger evidence

- **Walmart/Massmart (2011).** The Competition Commission and Tribunal analyzed SKU-level sales and procurement data showing Massmart's 20–25% share in formal general merchandise with limited reach into township grocery segments. Diversion estimates from loyalty-card switching rates indicated minimal unilateral effect, so the case turned on public-interest harms. Conditions ultimately required a R240 million supplier development fund, a two-year moratorium on merger-specific retrenchments, and detailed annual reporting on local procurement shares—providing a template for tying data-backed public-interest claims to remedies.
- **Mediclinic/Matlosana (2014).** Using patient-level discharge data covering 24 specialties, the Commission computed local HHIs above 6,000 and estimated post-merger tariff increases of 8–12% for insured patients. The Tribunal accepted that rival hospitals were more than 150 km away and prohibited the deal, highlighting how granular utilization data can anchor both geographic market definition and competitive-effects narratives in middle-income regions.
- **Heineken/Distell/Capevin (2022).** In evaluating the creation of Newco, the Commission's demand estimates—calibrated from Nielsen panel data—showed cider/RTD diversion ratios above 0.5 between Hunters, Savanna, and Strongbow, with Newco projected to command roughly 65% share. Conditional approval required a R10 billion investment commitment, maintenance of existing third-party distribution contracts, and shelf-space safeguards for smaller craft brands, illustrating how quantitative evidence on differentiated products fed into both competition and public-interest remedies.

i Method box

- Simple merger sim template (see R helpers).
- Event study around merger announcement/close; rival effects.
- Vertical tools: vUPP, EDM, raising rivals' costs sketches.

i Method box: UPP/GUPPI quick calc

See the UPP/GUPPI example above. Expand the template with case-specific diversion estimates (from surveys, loyalty data, or conjoint work) and margin sources (accounting or expert models). Always show sensitivity ranges.

i Qualitative evidence

- Integration plans, synergy decks, customer feedback on alternatives.
- Internal pricing and margin analyses; board materials on strategic rationale.
- Remedy feasibility from operations teams and third parties.

i Code box: merger sim skeleton

```
# robust path in case execution dir is chapter folder
source("../R/helpers.R")
# products_df: product, firm, price, share, mc (or margin),
  ↪ group (nest)
# sim <- run_logit_sim(products_df, merging_firms =
  ↪ c("FirmA", "FirmB"))
# sim$summary
```

i Citations and comparative note

- Anchor claims to current US Merger Guidelines (2023) (U.S. Department of Justice and Federal Trade Commission 2023) and legacy 2010 guidance (U.S. Department of Justice and Federal Trade Commission 2010); include EC Horizontal Guidelines (European Commission 2004) and CMA merger assessment guidelines for comparisons.
- Cite empirical merger retrospective studies when presenting methods or benchmarks (e.g., airlines, hospitals).
- For vertical mergers, cite vUPP/EDM references and any key enforcement actions (e.g., US v. AT&T/Time Warner, EC cases on input foreclosure).

i Case box: Illustrative mergers

- Horizontal: airline mergers (UA/CO, DL/NW, AA/US) — retrospectives and coordinated effects; hospital mergers for local market power.
- Vertical: AT&T/Time Warner (US), Microsoft/Activision (platform/distribution) — input foreclosure/EDM debates.
- Mixed conglomerate/ad tech: Google/DoubleClick; media/telecom bundling examples.

Visualizations

Market shares and HHI dashboard

This dashboard provides a comprehensive view of market structure before and after the merger, combining share distributions, HHI calculations, and competitive thresholds.

```
library(dplyr)
library(tidyr)
library(ggplot2)
library(patchwork)

# Simulated market data (replace with actual transaction/Nielsen
# data)
market_pre <- tibble::tribble(
  ~firm,      ~share,
  "Firm A",   0.28,
  "Firm B",   0.22,
  "Firm C",   0.18,
  "Firm D",   0.12,
  "Firm E",   0.08,
  "Firm F",   0.06,
  "Fringe",   0.06
)

# Post-merger: A acquires B
market_post <- market_pre |>
  mutate(
    firm = if_else(firm == "Firm B", "Firm A+B", firm),
    share = if_else(firm == "Firm A", share + 0.22, share)
  ) |>
  filter(firm != "Firm B") |>
  arrange(desc(share))

# Calculate HHI
calc_hhi <- function(shares) {
  sum((shares * 100)^2)
}

hhi_pre <- calc_hhi(market_pre$share)
hhi_post <- calc_hhi(market_post$share)
delta_hhi <- hhi_post - hhi_pre

# Prepare data for visualization
market_pre$period <- "Pre-merger"
market_post$period <- "Post-merger"
```

```

market_combined <- bind_rows(market_pre, market_post)
market_combined$period <- factor(market_combined$period,
                                levels = c("Pre-merger",
                                             "Post-merger"))

# Plot 1: Share comparison
p1 <- ggplot(market_combined, aes(x = reorder(firm, share), y =
  share,
                                fill = period)) +
  geom_col(position = position_dodge(width = 0.8), width = 0.7) +
  coord_flip() +
  scale_y_continuous(labels = scales::percent_format()) +
  scale_fill_manual(values = c("Pre-merger" = "#0072B2",
                                "Post-merger" = "#D55E00")) +
  labs(
    title = "Market Shares: Pre- vs. Post-Merger",
    x = NULL,
    y = "Market Share",
    fill = NULL
  ) +
  theme_classic(base_size = 11) +
  theme(
    legend.position = "bottom",
    plot.title.position = "plot"
  )

# Plot 2: HHI change
hhi_data <- tibble::tribble(
  ~scenario, ~hhi,
  "Pre-merger", hhi_pre,
  "Post-merger", hhi_post
) |>
mutate(
  scenario = factor(scenario, levels = c("Pre-merger",
                                           "Post-merger")),
  concern_level = case_when(
    hhi < 1500 ~ "Unconcentrated",
    hhi < 2500 ~ "Moderately concentrated",
    TRUE ~ "Highly concentrated"
  )
)

p2 <- ggplot(hhi_data, aes(x = scenario, y = hhi, fill =
  scenario)) +
  geom_col(width = 0.6) +
  geom_hline(yintercept = 1500, linetype = "dashed", color =
    "gray40",

```

```

        linewidth = 0.8) +
geom_hline(yintercept = 2500, linetype = "dashed", color =
  ↪ "gray40",
        linewidth = 0.8) +
annotate("text", x = 2.5, y = 1500, label = "1,500 threshold",
  hjust = 0, vjust = -0.5, size = 3) +
annotate("text", x = 2.5, y = 2500, label = "2,500 threshold",
  hjust = 0, vjust = -0.5, size = 3) +
annotate("rect", xmin = -Inf, xmax = Inf, ymin = 1500, ymax =
  ↪ 2500,
        fill = "yellow", alpha = 0.05) +
annotate("rect", xmin = -Inf, xmax = Inf, ymin = 2500, ymax =
  ↪ Inf,
        fill = "red", alpha = 0.05) +
scale_fill_manual(values = c("Pre-merger" = "#0072B2",
                             "Post-merger" = "#D55E00")) +
labs(
  title = "HHI Analysis",
  subtitle = paste0("Δ HHI = ", round(delta_hhi, 0)),
  x = NULL,
  y = "HHI",
  fill = NULL
) +
theme_classic(base_size = 11) +
theme(
  legend.position = "none",
  plot.title.position = "plot"
)

# Plot 3: Concentration curve
market_post_sorted <- market_post |>
  arrange(desc(share)) |>
  mutate(cumulative_share = cumsum(share))

p3 <- ggplot(market_post_sorted, aes(x = seq_along(firm), y =
  ↪ cumulative_share)) +
  geom_line(color = "#D55E00", linewidth = 1.2) +
  geom_point(color = "#D55E00", size = 3) +
  geom_hline(yintercept = 0.5, linetype = "dashed", color =
    ↪ "gray40") +
  annotate("text", x = 1, y = 0.5, label = "50% market share",
    hjust = 0, vjust = -0.5, size = 3) +
  scale_y_continuous(labels = scales::percent_format(), limits =
    ↪ c(0, 1)) +
  scale_x_continuous(breaks = seq_along(market_post_sorted$firm),
    labels = market_post_sorted$firm) +

```

```

labs(
  title = "Post-Merger Concentration Curve",
  subtitle = "Cumulative market share by firm rank",
  x = NULL,
  y = "Cumulative Share"
) +
theme_classic(base_size = 11) +
theme(
  plot.title.position = "plot",
  axis.text.x = element_text(angle = 45, hjust = 1)
)

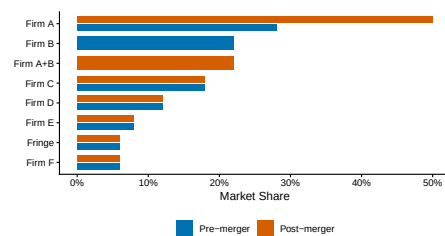
# Combine plots
(p1 | p2) / p3 + plot_annotation(
  title = "Market Structure Dashboard",
  subtitle = paste0("Pre-merger HHI: ", round(hhi_pre, 0),
    " | Post-merger HHI: ", round(hhi_post, 0),
    " | Δ HHI: ", round(delta_hhi, 0)),
  caption = "US Merger Guidelines (2023): Moderately concentrated
    ↪ if HHI > 1,500;
    Highly concentrated if HHI > 2,500. Mergers with Δ HHI >
    ↪ 100-200 in concentrated
    markets may warrant scrutiny."
)

```

Market Structure Dashboard

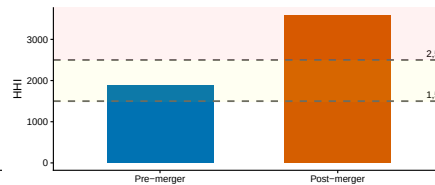
Pre-merger HHI: 1872 | Post-merger HHI: 3588 | Δ HHI: 1716

Market Shares: Pre- vs. Post-Merger



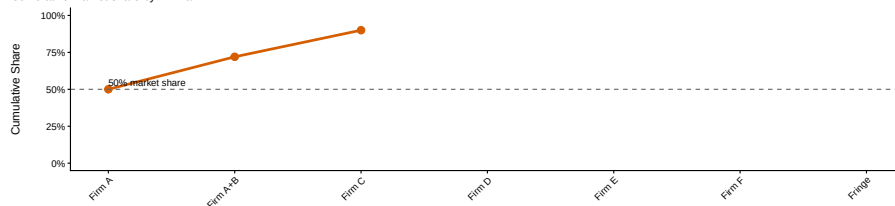
HHI Analysis

Δ HHI = 1716



Post-Merger Concentration Curve

Cumulative market share by firm rank



US Merger Guidelines (2023): Moderately concentrated if HHI > 1,500;
 Highly concentrated if HHI > 2,500. Mergers with Δ HHI > 100-200 in concentrated
 markets may warrant scrutiny.

```
# Summary table
cat("\nMarket structure summary:\n")
```

Market structure summary:

```
cat(paste0("Pre-merger HHI: ", round(hhi_pre, 0), "\n"))
```

Pre-merger HHI: 1872

```
cat(paste0("Post-merger HHI: ", round(hhi_post, 0), "\n"))
```

Post-merger HHI: 3588

```
cat(paste0("Change in HHI: ", round(delta_hhi, 0), "\n"))
```

Change in HHI: 1716

```
cat(paste0("\nCombined entity share: ",
          scales::percent(market_post$share[market_post$firm ==
↪ "Firm A+B"],
                          accuracy = 0.1), "\n"))
```

Combined entity share: 22.0%

Interpretation:

- **HHI thresholds:** The 2023 US Merger Guidelines use 1,500 and 2,500 as thresholds. Markets above 2,500 are “highly concentrated.”
- **Delta HHI:** Changes above 100-200 in concentrated markets may trigger enhanced scrutiny.
- **Combined entity:** The merged firm’s share and rank indicate potential unilateral effects concerns.
- **Concentration curve:** Shows how quickly the top firms accumulate market share.

Replace simulated data with actual transaction volumes, Nielsen scanner data, or industry-specific sources (e.g., airline MIDT, hospital discharge data).

Merger simulation waterfall

A waterfall chart decomposes the predicted post-merger price change into its component parts: diversion, margin, efficiencies, and second-order effects. This helps communicate which parameters drive the result.

```
library(dplyr)
library(ggplot2)

# Simulation components (from a differentiated products model)
# Replace with actual simulation outputs
sim_components <- tibble::tribble(
  ~component,      ~value,      ~description,
```

```

"Base price",          10.00,      "Pre-merger price",
"First-order UPP",     +0.80,      "Diversion × Margin
  ↪ effect",
"Internalization",     +0.35,      "Portfolio
  ↪ reoptimization",
"Efficiency offset",   -0.25,      "Verified cost synergies",
"Competitive response", -0.15,      "Rival price reactions",
"Post-merger price",   10.75,      "Predicted equilibrium"
) |>
mutate(
  component = factor(component, levels = component),
  cumulative = cumsum(value),
  start = lag(cumulative, default = 0),
  end = cumulative,
  type = case_when(
    component %in% c("Base price", "Post-merger price") ~
  ↪ "total",
    value > 0 ~ "increase",
    value < 0 ~ "decrease",
    TRUE ~ "neutral"
  )
)

# Waterfall plot
ggplot(sim_components) +
  geom_rect(aes(xmin = as.numeric(component) - 0.4,
               xmax = as.numeric(component) + 0.4,
               ymin = start, ymax = end, fill = type),
           color = "black", linewidth = 0.5) +
  geom_text(aes(x = as.numeric(component),
               y = (start + end) / 2,
               label = scales::dollar(value, accuracy = 0.01)),
           size = 3.5, fontface = "bold") +
  geom_segment(data = filter(sim_components, !type %in%
  ↪ c("total")),
               aes(x = as.numeric(component) + 0.4,
                   xend = as.numeric(component) + 1 - 0.4,
                   y = end, yend = end),
               linetype = "dashed", color = "gray50") +
  scale_fill_manual(
    values = c(
      "total" = "#0072B2",
      "increase" = "#D55E00",
      "decrease" = "#009E73",
      "neutral" = "#999999"
    ),
  )

```

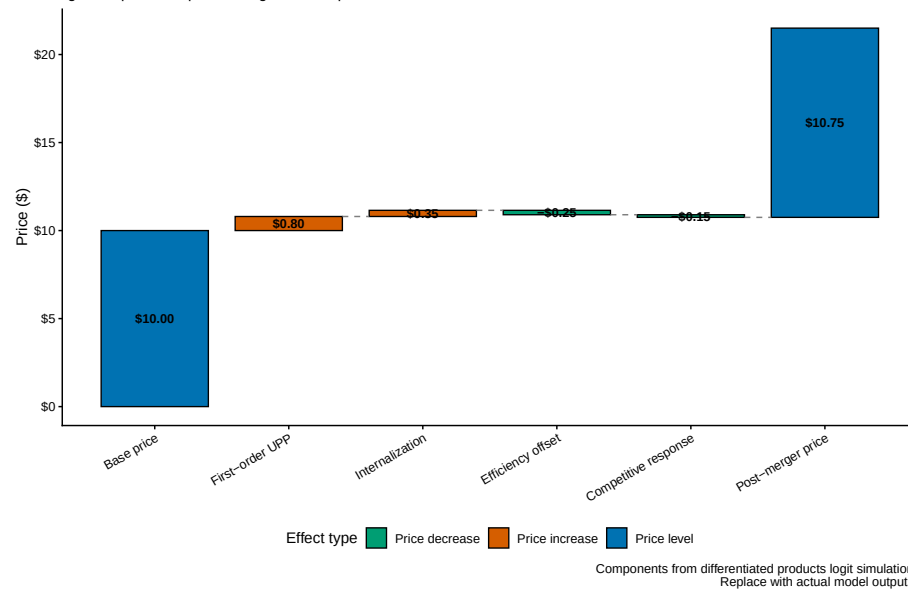
```

    labels = c(
      "total" = "Price level",
      "increase" = "Price increase",
      "decrease" = "Price decrease",
      "neutral" = "Neutral"
    )
  ) +
  scale_x_continuous(
    breaks = seq_along(sim_components$component),
    labels = sim_components$component
  ) +
  scale_y_continuous(labels = scales::dollar_format()) +
  labs(
    title = "Merger Simulation Waterfall: Price Effect
    ↪ Decomposition",
    subtitle = "Breaking down predicted price change into
    ↪ component effects",
    x = NULL,
    y = "Price ($)",
    fill = "Effect type",
    caption = "Components from differentiated products logit
    ↪ simulation.
    Replace with actual model outputs."
  ) +
  theme_classic(base_size = 12) +
  theme(
    axis.text.x = element_text(angle = 30, hjust = 1),
    legend.position = "bottom",
    plot.title.position = "plot"
  )
)

```

Merger Simulation Waterfall: Price Effect Decomposition

Breaking down predicted price change into component effects



```
# Summary statistics
cat("\nSimulation summary:\n")
```

Simulation summary:

```
cat(paste0("Pre-merger price: ",
           scales::dollar(sim_components$value[1], accuracy =
↪ 0.01), "\n"))
```

Pre-merger price: \$10.00

```
cat(paste0("Post-merger price: ",
           scales::dollar(sim_components$end[nrow(sim_components)],
↪ accuracy = 0.01), "\n"))
```

Post-merger price: \$21.50

```
cat(paste0("Price increase: ",
           scales::dollar(sim_components$end[nrow(sim_components)] -
↪ sim_components$value[1], accuracy =
↪ 0.01),
    " (",
↪ scales::percent((sim_components$end[nrow(sim_components)] /
```

```

sim_components$value[1]) - 1, accuracy
↪ = 0.1),
    "\n"))

```

Price increase: \$11.50 (115.0%)

How to use this waterfall: - **First-order UPP:** Direct pricing pressure from internalizing diversion. - **Internalization:** Additional optimization from portfolio effects (multi-product firms). - **Efficiency offset:** Cost savings that reduce pricing pressure (must be verifiable and merger-specific). - **Competitive response:** Predicted reactions from non-merging firms (can amplify or dampen effects).

Practical tips: - Document the source of each component (diversion from surveys, margins from cost data, efficiencies from integration plans). - Show sensitivity: run alternative scenarios varying key parameters. - Compare to retrospective evidence: if similar past mergers raised prices by X%, does your simulation predict comparable magnitudes?

Replace simulated values with outputs from your actual merger simulation model (logit, nested logit, BLP, or custom structural model).

Looking ahead

Archive merger sims, UPP tables, and event-study outputs in **data/derived** with README files so later chapters (litigation practice, appendices) can reuse them. When you transition to the remedies chapter or empirical appendix, reference which datasets (Nielsen panels, loyalty data, procurement archives) need anonymization or refreshed pulls. Update the visualization tracker with any new dashboards (e.g., post-merger share waterfalls) to keep teaching decks synchronized with the book.

Monopolization and Exclusion

Learning goals

- Analyze exclusionary conduct (predation, rebates, tying, MFNs, vertical restraints).
- Connect theory to measurable foreclosure and consumer harm.
- Use qualitative evidence to show strategy and intent, paired with empirical outcomes.

Core topics

- Price-cost tests (Areeda-Turner variants), effective price with rebates/loyalty programs.
- Foreclosure metrics: share of market foreclosed, rivals' scale and entry conditions.
- Tying/bundling: uptake, defaults, switching costs, interoperability constraints.
- MFNs/parity: price dispersion effects, platform fee pass-through.
- Workflow: (1) conduct and theory of harm; (2) market power assessment; (3) mechanism (exclusion/predation/raising rivals' costs); (4) empirical effects; (5) countervailing efficiencies/justifications; (6) remedy fit.

Introduction

Monopolization analysis sits at the intersection of conduct, market structure, and effects. Unlike merger review—which evaluates a prospective transaction—monopolization cases reconstruct historical patterns of behavior and ask whether a dominant firm has maintained or extended its position through exclusionary means rather than through competition on the merits. Section 2 of the Sherman Act in the United States and Article 102 of the Treaty on the Functioning of the

European Union (TFEU) frame the legal standards, but enforcement priorities and evidentiary approaches differ substantially across jurisdictions.

This chapter provides a structured workflow for analyzing monopolization claims. We begin with market power assessment, drawing on tools from Chapter 3 (market definition) and Chapter 4 (IO toolkit). We then turn to specific conduct categories—predation, rebates and loyalty programs, tying and bundling, most-favored-nation clauses, vertical restraints, and refusal to deal. For each, we pair theoretical mechanisms with empirical measurement strategies and qualitative evidence that together build a coherent narrative. The chapter concludes with remedy design considerations and a set of case exemplars that illustrate how different jurisdictions approach the same exclusionary practices.

Throughout, we emphasize reproducibility and transparency. Code boxes demonstrate how to compute effective prices, assess foreclosure shares, and visualize event patterns around contract adoptions. Qualitative evidence boxes highlight the documentary and testimonial sources that complement quantitative analyses. Case boxes anchor abstract principles in real enforcement actions, drawing on landmark decisions from the United States, European Union, and Southern Africa.

For deeper context, consult the extensive case law on exclusionary conduct, including landmark decisions like *United States v. Microsoft* (U.S. Court of Appeals, D.C. Circuit 2001) and the European Commission’s decisions on abuse of dominance (European Commission 2009). For theoretical foundations, see Whinston (2006) and Salop and Scheffman (1983) on raising rivals’ costs.

Market power and the monopolization workflow

Establishing market power

Monopolization liability begins with market power. In the United States, courts typically require monopoly power—the ability to control prices or exclude competition—which is often proxied by a sustained market share above 60–70% in a properly defined relevant market (U.S. Supreme Court 1966). The European Union applies a lower threshold for dominance under Article 102 TFEU, with shares as low as 40% potentially qualifying when combined with barriers to entry and expansion (Court of Justice of the European Union 1991, 1978). These quantitative thresholds, however, must be buttressed by evidence of entry conditions, buyer power, and the duration of the firm’s position.

Quantitative indicators include:

1. **Market shares:** Compute shares by revenue, volume, or capacity over multiple periods to demonstrate persistence. Use granular data (SKU-level, route-level, or platform-specific shares) when available. Cross-check shares against customer perceptions and bidding data.

2. **Entry conditions:** Examine regulatory barriers, network effects, switching costs, and capital requirements. Quantify entry attempts and failures; use survival analyses or hazard models to assess entry likelihood.
3. **Buyer power:** Assess whether large customers can sponsor entry or credibly threaten backward integration. Use procurement data and contract terms to gauge buyer negotiating leverage.

Qualitative indicators reinforce these metrics:

- Internal documents describing the firm as “must-have” or discussing the consequences of losing access to the product.
- Customer interviews and surveys that reveal switching frictions or lack of acceptable alternatives.
- Board materials and strategic memos that acknowledge or plan around market power (e.g., pricing flexibility, entry deterrence strategies).

Together, these elements establish the baseline condition for monopolization: a firm with durable market power and the incentive to maintain or extend it.

Articulating the theory of harm

Once market power is established, the analysis turns to conduct. Monopolization theories fall into several categories:

1. **Predatory pricing:** pricing below cost to drive out rivals or deter entry, with a credible prospect of recouping losses through higher prices later.
2. **Exclusive dealing and loyalty rebates:** contracts or pricing structures that foreclose a substantial share of the market to rivals, raising their costs or denying them minimum efficient scale.
3. **Tying and bundling:** leveraging dominance in one market to disadvantage rivals in another, either through contractual ties or technical integration (e.g., defaults, degraded interoperability).
4. **Most-favored-nation (MFN) clauses:** contractual provisions that prevent trading partners from offering better terms elsewhere, chilling price competition and entrenching the dominant firm’s position.
5. **Self-preferencing and vertical restraints:** a vertically integrated firm favoring its own downstream products over rivals’ offerings on its platform or distribution network.
6. **Refusal to deal or denial of access:** withholding an essential input or interoperability without a legitimate business justification, where access is indispensable for rivals to compete.

Each theory requires alignment between the conduct, the mechanism of harm, and the expected effects. Align the data collection and empirical strategy to the specific mechanism: if the claim is that loyalty rebates foreclose rivals, measure the share of volume or capacity covered by the contracts and assess whether foreclosed volumes prevent rivals from reaching minimum efficient scale. If the claim is predatory pricing, compute effective prices net of rebates and compare

to measures of avoidable or incremental cost, then model recoupment scenarios.

Workflow summary

A complete monopolization analysis typically follows this sequence:

1. **Conduct description and theory of harm:** Document the challenged behavior (contracts, pricing, technical design) and articulate the exclusionary mechanism.
2. **Market power assessment:** Establish market shares, entry barriers, and buyer power with both quantitative and qualitative evidence.
3. **Mechanism and foreclosure:** Quantify the scope of foreclosure (volume, duration, exclusivity) and assess whether rivals are denied minimum efficient scale or face raised costs.
4. **Empirical effects:** Use natural experiments, event studies, or before/after comparisons to measure price, output, quality, or entry/exit effects attributable to the conduct.
5. **Efficiencies and justifications:** Evaluate procompetitive rationales (cost savings, quality improvements, incentive alignment) and whether less-restrictive alternatives are feasible.
6. **Remedies:** Design conduct or structural remedies that target the exclusionary mechanism without unduly chilling procompetitive conduct; specify monitoring and compliance requirements.

The remainder of this chapter elaborates on each conduct category, pairing theory with empirical methods and qualitative evidence.

Exclusive dealing, loyalty rebates, and foreclosure

Measuring foreclosure

Exclusive dealing and loyalty rebate programs aim to lock in a substantial share of customer demand, making it difficult or unprofitable for rivals to achieve minimum efficient scale. The central empirical question is: what share of the market is foreclosed, and for how long?

Foreclosure metrics include:

1. **Volume or capacity shares covered:** Sum the total volumes or capacities bound by exclusive or loyalty contracts as a percentage of total market demand. A foreclosure share above 40–50% may raise concerns, particularly if contracts are long-duration and staggered so that only a small fraction comes up for renewal in any given period.
2. **Minimum efficient scale (MES):** Estimate the scale needed for rivals to compete effectively (e.g., plant capacity, distribution network density, platform-side critical mass). If the foreclosed share leaves rivals below MES, the conduct is more likely exclusionary.

3. **Contract duration and staggering:** Long-term contracts with staggered renewal dates can perpetuate foreclosure even if individual contracts expire. Quantify the fraction of demand available for contestation in each period.
4. **Exit and entry patterns:** Document whether rivals have exited or delayed entry coinciding with contract rollouts; use hazard models or event studies to link conduct to competitive structure.

Effective price analysis for loyalty rebates: Loyalty rebates condition discounts on purchasing all or a substantial share of requirements from the dominant firm. The effective price for incremental units—those that push the buyer over the loyalty threshold—can be significantly below the headline price, making it unprofitable for rivals to compete even if their costs are comparable.

To compute effective prices:

- Identify the rebate threshold (e.g., purchase 80% of requirements to qualify for a 10% rebate).
- Calculate the marginal benefit of reaching the threshold: if a buyer purchases 95 units at the list price and buying 5 more units triggers a 10% rebate on all 100 units, the effective price of those last 5 units is deeply negative.
- Compare effective prices to rivals' costs or list prices to assess whether contestation is feasible.

The European Commission's Intel decision (Court of Justice of the European Union 2017) applied a version of this test, focusing on whether the rebates made it unprofitable for AMD to compete for incremental volumes even when AMD's products were technically competitive. The as-efficient competitor (AEC) test, refined post-Intel, requires showing that an equally efficient rival would be foreclosed.

Effective price waterfall for loyalty rebates

The following visualization demonstrates how loyalty rebates can create deeply negative effective prices for incremental units, making it unprofitable for equally efficient rivals to compete. This implements the as-efficient competitor (AEC) test framework used in EU competition enforcement.

```
library(dplyr)
library(tidyr)
library(ggplot2)
library(patchwork)

rebates <- tibble::tribble(
  ~tier, ~volume, ~list_price, ~rebate,
  "Baseline (0-79%)", 79, 100, 0,
  "Threshold units (80-99%)", 20, 100, 0,
```

```

  "Bonus units (100%+)", 10, 100, 0
) |>
mutate(
  revenue = volume * list_price,
  cumulative_volume = cumsum(volume)
)

# Scenario: Customer buying ~100 units, loyalty rebate triggers
#   ↳ at 95% share
# List price $100/unit, 10% retroactive rebate at 95% threshold
# Rival has incremental cost of $75/unit

# Calculate effective prices under retroactive rebate
list_price <- 100
rebate_pct <- 0.10 # 10% retroactive rebate

# For the customer at threshold: buying 5 more units from
#   ↳ incumbent triggers
# rebate on ALL 100 units = $1000 benefit, so effective price of
#   ↳ last 5 units
# is  $(5 * 100 - 1000) / 5 = -\$100$  per unit

effective_price_data <- tibble::tribble(
  ~component, ~value, ~category,
  "List price", 100, "baseline",
  "Incremental units to threshold", 100, "marginal",
  "Value of retroactive rebate", -1000/5, "rebate", # $1000
  ↳ rebate / 5 marginal units
  "Effective price (last 5 units)", -100, "effective"
)

# Waterfall chart showing effective price calculation
p_waterfall <- effective_price_data |>
  filter(component != "Effective price (last 5 units)") |>
  mutate(
    component = factor(component, levels = c("List price",
      "Incremental units
        ↳ to threshold",
      "Value of
        ↳ retroactive
        ↳ rebate")),

    end = cumsum(value),
    start = lag(end, default = 0),
    id = row_number()
  ) |>
  ggplot(aes(x = component)) +

```

```

geom_rect(aes(xmin = id - 0.4, xmax = id + 0.4,
              ymin = start, ymax = end, fill = category),
          color = "white", linewidth = 0.8) +
geom_hline(yintercept = 75, linetype = "dashed", color =
  ↪ "#E69F00", linewidth = 1) +
geom_hline(yintercept = 0, color = "gray30") +
annotate("text", x = 2.5, y = 80,
          label = "Rival incremental cost ($75)",
          color = "#E69F00", fontface = "bold", size = 3.5) +
annotate("segment", x = 2.8, xend = 3.2, y = -100, yend = -100,
          color = "#D55E00", linewidth = 2) +
annotate("text", x = 3, y = -115,
          label = "Effective price:\n-$100/unit",
          color = "#D55E00", fontface = "bold", size = 4) +
scale_fill_manual(values = c("baseline" = "#0072B2",
                              "marginal" = "#56B4E9",
                              "rebate" = "#D55E00")) +
scale_y_continuous(labels = scales::dollar_format()) +
labs(
  title = "Effective price waterfall under retroactive loyalty
  ↪ rebate",
  subtitle = "Customer at 95-unit threshold: effective price of
  ↪ last 5 units to reach 100% share",
  x = NULL,
  y = "Price per unit",
  caption = "Scenario: $100 list price, 10% retroactive rebate
  ↪ at 95% volume share.\nRival with $75 incremental cost
  ↪ cannot profitably compete for marginal units."
) +
theme_classic(base_size = 11) +
theme(
  legend.position = "none",
  plot.caption = element_text(hjust = 0, size = 9, color =
  ↪ "gray40"),
  axis.text.x = element_text(angle = 0, hjust = 0.5)
)

# Market share evolution under loyalty contracts
share_evolution <- expand_grid(
  quarter = 1:12,
  firm = c("Incumbent", "Rival A", "Rival B")
) |>
as_tibble() |>
mutate(
  market_share = case_when(
    firm == "Incumbent" & quarter <= 4 ~ 60 + quarter * 1.5,

```

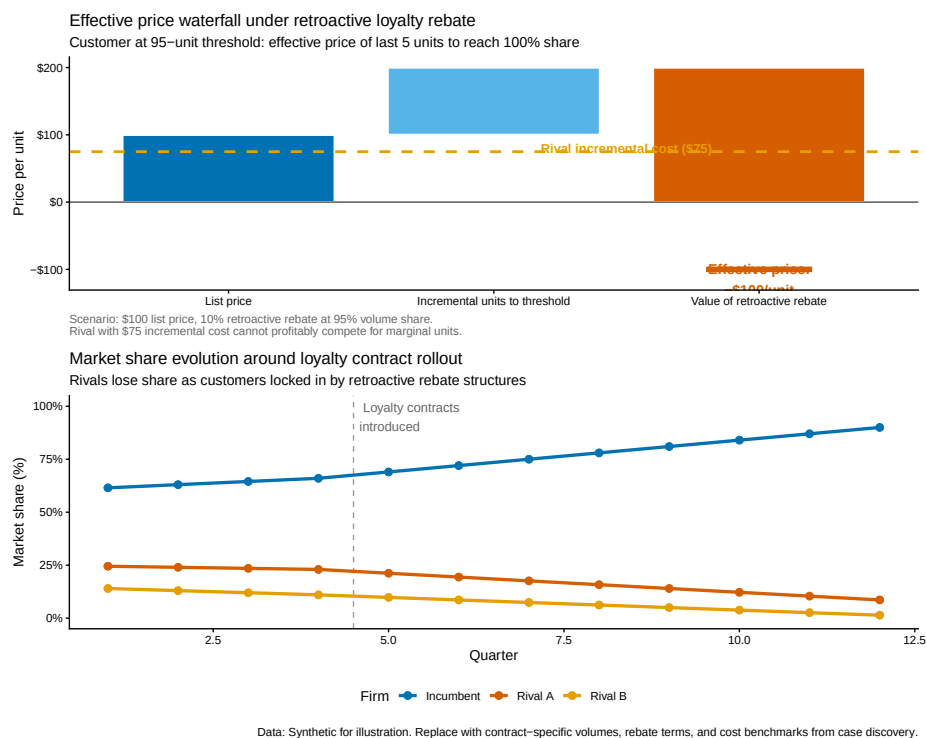
```

    firm == "Incumbent" & quarter > 4 ~ 66 + (quarter - 4) * 3,
    firm == "Rival A" & quarter <= 4 ~ 25 - quarter * 0.5,
    firm == "Rival A" & quarter > 4 ~ 23 - (quarter - 4) * 1.8,
    firm == "Rival B" & quarter <= 4 ~ 15 - quarter * 1,
    firm == "Rival B" & quarter > 4 ~ 11 - (quarter - 4) * 1.2
  ),
  market_share = pmax(market_share, 0) # Floor at zero
)

p_evolution <- ggplot(share_evolution, aes(x = quarter, y =
  ↪ market_share,
                                     color = firm, group =
  ↪ firm)) +
  geom_vline(xintercept = 4.5, linetype = "dashed", color =
  ↪ "gray60") +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2.5) +
  annotate("text", x = 4.5, y = 95,
    label = "Loyalty contracts\nintroduced",
    hjust = -0.1, size = 3.5, color = "gray40") +
  scale_color_manual(values = c("Incumbent" = "#0072B2",
    "Rival A" = "#D55E00",
    "Rival B" = "#E69F00")) +
  scale_y_continuous(labels = scales::percent_format(scale = 1),
    limits = c(0, 100)) +
  labs(
    title = "Market share evolution around loyalty contract
  ↪ rollout",
    subtitle = "Rivals lose share as customers locked in by
  ↪ retroactive rebate structures",
    x = "Quarter",
    y = "Market share (%)",
    color = "Firm"
  ) +
  theme_classic(base_size = 11) +
  theme(legend.position = "bottom")

# Combine panels
p_waterfall / p_evolution +
  plot_annotation(
    caption = "Data: Synthetic for illustration. Replace with
  ↪ contract-specific volumes, rebate terms, and cost
  ↪ benchmarks from case discovery."
  )

```



Data replacement: Use actual customer purchase data and contract terms from litigation discovery. Calculate effective prices per customer segment; overlay rival cost estimates from cost studies or public financial statements. For share evolution, use market monitoring data or customer panel data tracking purchases before/after contract adoption.

As-efficient competitor (AEC) test: The waterfall shows that even an equally efficient rival (cost = \$75) cannot profitably compete for the marginal units that trigger the rebate, as the effective price faced by the customer is deeply negative. This satisfies the AEC framework established in *Intel* (Court of Justice of the European Union 2017) and subsequent EU enforcement.

Qualitative evidence for rebates and exclusivity

Internal documents and customer testimony are critical complements to quantitative foreclosure metrics:

- **Strategy memos:** Documents revealing intent to foreclose rivals or lock in customers (e.g., “ensure [Rival] cannot reach scale,” “defend our share through loyalty programs”).
- **Contract negotiation histories:** Email chains and drafts showing how exclusivity or loyalty terms were introduced, customer resistance, and adjustments to overcome objections.

- **Customer interviews:** Testimony on switching costs, the importance of multi-sourcing for resilience, and coercion (explicit or implicit) to accept exclusive terms.
- **Sales force incentives:** Compensation structures that reward exclusivity or penalize customers who multi-source.

In the United States, the rule of reason standard for exclusive dealing (under Section 1 or Section 2) often weighs these qualitative factors heavily, looking for evidence that foreclosure was the purpose rather than a byproduct of legitimate incentives (U.S. Supreme Court 1961). For economic analysis of exclusive dealing, see Whinston (2006) and Ordover, Saloner, and Salop (1990) on vertical foreclosure.

Event studies and staggered rollouts

If exclusive contracts or loyalty programs were introduced or renewed in a staggered fashion across regions, customers, or time periods, treat the rollout as a natural experiment:

1. Define treatment as the adoption of the contract or rebate program; define control customers or regions without contracts.
2. Use difference-in-differences with appropriate controls for pre-existing trends in customer purchasing patterns.
3. Examine outcomes such as rival market shares, entry/exit events, or prices paid to rivals.
4. Visualize event study coefficients to show dynamic effects before and after contract adoption.

When data are thin, simpler before/after comparisons combined with robust documentary evidence can still be persuasive. The key is transparency about identification assumptions and potential confounders.

Data tip: For visuals in the roadmap (effective price waterfall, foreclosure share, rollout event analysis), start with sanitized versions of the loyalty-contract spreadsheets and rolling rollout trackers. Where you lack live case data, use synthetic templates in `data/examples/` so the Quarto build renders without confidential inputs; swap in case data during litigation prep.

Contract rollout event study

When exclusive contracts are adopted in staggered fashion across customers or regions, event study methods can isolate the causal effect of foreclosure on rival performance.

```
library(dplyr)
library(tidyr)
library(ggplot2)
```

```
library(fixest)
library(patchwork)

# Simulate panel data: 20 customers observed for 24 periods
# Treatment (exclusive contract) rolled out to 10 customers in
  ↪ period 13
set.seed(123)
n_customers <- 20
n_periods <- 24
treatment_period <- 13
treated_customers <- paste0("C", 1:10)

panel_contracts <- expand.grid(
  customer_id = paste0("C", 1:n_customers),
  period = 1:n_periods
) |>
  as_tibble() |>
  mutate(
    treated = customer_id %in% treated_customers,
    post = period >= treatment_period,
    # Rival share baseline + customer FE + time trend
    rival_share_baseline = 30 +
      as.numeric(factor(customer_id)) * 0.5 -
      period * 0.2 +
      rnorm(n(), 0, 3),
    # Treatment effect: -8pp reduction in rival share
    ↪ post-contract
    treatment_effect = if_else(treated & post, -8, 0),
    rival_share = pmax(0, rival_share_baseline + treatment_effect
      ↪ + rnorm(n(), 0, 2))
  )

# Event study regression using fixest
# Interact treated indicator with relative time to treatment
panel_contracts <- panel_contracts |>
  mutate(rel_period = period - treatment_period)

# Estimate event study coefficients (omit period -1 as reference)
event_fit <- feols(
  rival_share ~ i(rel_period, treated, ref = -1) | customer_id +
    ↪ period,
  data = panel_contracts
)

# Extract coefficients for plotting
event_coefs <- broom::tidy(event_fit, conf.int = TRUE) |>
```

```

filter(grepl("rel_period", term)) |>
mutate(
  rel_period =
    ↪ as.numeric(gsub("rel_period::(.*)::treated::TRUE", "\\1",
    ↪ term))
) |>
# Add reference period (-1) with zero effect
bind_rows(tibble(rel_period = -1, estimate = 0, std.error = 0,
                  conf.low = 0, conf.high = 0))

# Event study plot
p_event <- ggplot(event_coefs, aes(x = rel_period, y = estimate))
  ↪ +
  geom_vline(xintercept = -0.5, linetype = "dashed", color =
    ↪ "gray50") +
  geom_hline(yintercept = 0, color = "gray30") +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high),
    alpha = 0.2, fill = "#0072B2") +
  geom_line(color = "#0072B2", linewidth = 1) +
  geom_point(size = 3, color = "#0072B2") +
  annotate("text", x = 0, y = 2,
    label = "Contract\nadoption",
    hjust = -0.1, size = 3.5, color = "gray40") +
  labs(
    title = "Event study: Rival market share around exclusive
    ↪ contract adoption",
    subtitle = "Difference-in-differences with customer and
    ↪ period fixed effects",
    x = "Periods relative to contract adoption",
    y = "Effect on rival share (pp)",
    caption = "Shaded area = 95% confidence interval. No
    ↪ pre-trend in periods -11 to -1; sharp decline
    ↪ post-adoption."
  ) +
  theme_classic(base_size = 11) +
  theme(plot.caption = element_text(hjust = 0, size = 9, color =
    ↪ "gray40"))

# Customer-level treatment timing
treatment_timing <- panel_contracts |>
  filter(treated, period >= treatment_period) |>
  group_by(customer_id) |>
  slice(1) |>
  ungroup() |>
  mutate(cohort_label = paste0("Contract cohort (",
    ↪ treatment_period, ")"))

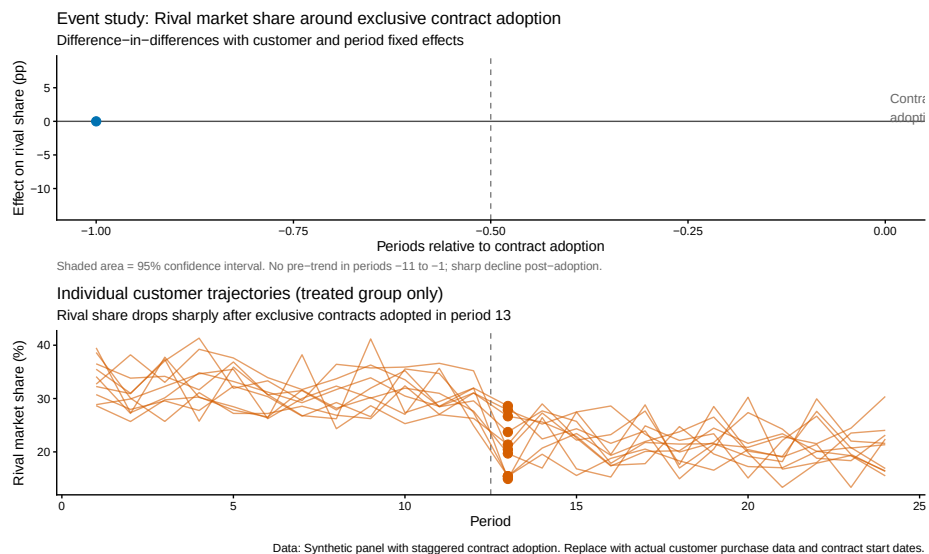
```

```

p_timing <- panel_contracts |>
  filter(customer_id %in% treated_customers) |>
  ggplot(aes(x = period, y = rival_share, group = customer_id)) +
  geom_vline(xintercept = treatment_period - 0.5, linetype =
    ↪ "dashed", color = "gray50") +
  geom_line(alpha = 0.6, color = "#D55E00") +
  geom_point(data = treatment_timing, aes(x = period, y =
    ↪ rival_share),
    size = 3, color = "#D55E00") +
  labs(
    title = "Individual customer trajectories (treated group
    ↪ only)",
    subtitle = "Rival share drops sharply after exclusive
    ↪ contracts adopted in period 13",
    x = "Period",
    y = "Rival market share (%)"
  ) +
  theme_classic(base_size = 11)

p_event / p_timing +
  plot_annotation(
    caption = "Data: Synthetic panel with staggered contract
    ↪ adoption. Replace with actual customer purchase data and
    ↪ contract start dates."
  )

```



Interpretation: The event study coefficients show no pre-trend (flat estimates

in periods -11 to -1), providing support for the parallel trends assumption. Post-adoption, rival share drops by ~8 percentage points, consistent with foreclosure. The individual trajectories confirm the aggregate pattern and reveal heterogeneity in treatment effects across customers.

Data replacement: Use customer-level purchase data with precise contract adoption dates. Estimate event studies separately for different contract types (exclusive vs. loyalty rebates) or customer segments. If contracts are truly staggered (different customers adopt at different times), use `fixest::sunab()` for heterogeneity-robust estimation (Callaway and Sant’Anna 2021).

Predatory pricing

The price-cost test

Predatory pricing claims allege that a dominant firm sets prices below an appropriate measure of cost to drive out rivals or deter entry, with the expectation of recouping losses through supracompetitive prices once competition is eliminated. The canonical legal test, articulated in *Brooke Group* (U.S. Supreme Court 1993), requires (1) pricing below an appropriate measure of cost, and (2) a dangerous probability of recouping the investment in below-cost pricing. The foundational economic framework was established by Areeda and Turner (1975).

Cost benchmarks in antitrust analysis include:

1. **Average variable cost (AVC):** Costs that vary with output in the short run (materials, direct labor). Pricing below AVC is presumptively predatory under *Areeda-Turner* logic, as no rational profit-maximizing firm would price below AVC absent an exclusionary motive.
2. **Average avoidable cost (AAC):** Costs that could be avoided by not producing (similar to AVC but may include some fixed costs avoidable in the medium term). AAC is increasingly favored in both US and EU practice as a more accurate measure of incremental cost.
3. **Long-run incremental cost (LRIC) or average total cost (ATC):** Includes allocated fixed costs. Pricing between AAC and ATC may be exclusionary if combined with evidence of predatory intent and foreclosure effects, but is less likely to satisfy the *Brooke Group* standard in the United States.

Computing effective prices: When rebates, bundling, or loyalty discounts are present, compute the effective price paid per unit:

- If a customer receives a retroactive rebate upon reaching a volume threshold, the effective price for marginal units near the threshold can be far below the list price.
- Allocate fixed fees, upfront payments, or bundled-product discounts transparently across the relevant units.

- Use period-level data (monthly or quarterly) to capture the timing of rebates and costs.

Recoupment analysis: Even if prices are below cost, predation liability in the United States requires showing a dangerous probability of recoupment—that the predator can later raise prices to recover its losses. Assess recoupment by:

- Modeling post-exit pricing power: shares, entry barriers, and coordinated interaction prospects.
- Estimating the duration and magnitude of below-cost pricing and the time needed to recoup at elevated prices.
- Examining documents that discuss post-exit pricing strategies or barriers that would prevent re-entry.

The European Union historically placed less emphasis on explicit recoupment proofs, focusing instead on the exclusionary effect and intent. However, post-*Intel*, EU enforcers increasingly apply an as-efficient competitor framework that implicitly considers recoupment by asking whether exclusion harms consumers (Court of Justice of the European Union 2017, 2012).

Event patterns and qualitative evidence

Where price-cost tests are ambiguous—prices near AAC, cost allocation disputes, or multi-product settings—qualitative evidence and event patterns become dispositive:

- **Documents on strategy:** Emails or memos discussing “pricing [Rival] out of the market,” “losing money to gain share,” or “invest now, recoup later.”
- **Entry and exit timing:** Did rivals exit or delay entry during the alleged predation period? Use survival analysis or simple event timelines.
- **Share and capacity shifts:** Plot market shares and capacity utilization over time; sharp increases in the alleged predator’s share coinciding with rival exits support the predation narrative.
- **Subsequent price increases:** If prices rose significantly after rivals exited, recoupment is more plausible. Use difference-in-differences with unaffected markets as controls to isolate the predation effect.

In the EU *AKZO* case (Court of Justice of the European Union 1991), internal documents referencing a plan to eliminate a competitor were central to the finding of abuse, even though the price-cost analysis was contested. Similarly, in the Southern African *Media24* case (discussed earlier), predatory intent combined with below-cost pricing over an extended period led to liability.

Tying, bundling, and technical integration

Mechanisms and measurement

Tying occurs when a seller conditions the sale of one product (the tying product, in which the seller has market power) on the purchase of another product (the tied product). Bundling offers a package of products together, often at a discount relative to standalone prices. Both practices can be exclusionary if they leverage market power in one market to foreclose rivals in another.

Theories of harm include:

1. **Volume foreclosure:** If most customers buy the bundle or accept the tie, rivals in the tied market lose scale and face higher per-unit costs.
2. **Technical foreclosure:** Integration or defaults make it costly or cumbersome to use a rival's tied product (e.g., degraded APIs, interoperability limits, removal of user choice screens).
3. **Price discrimination:** Bundling can be a tool for extracting surplus from heterogeneous customers, which is not per se harmful but may facilitate exclusion if it raises rivals' costs or denies them profitable customer segments.

Empirical assessments should measure:

- **Take-up rates:** What share of customers buy the bundle or accept the tie? Compare take-up in settings with and without the tie/bundle (across regions, time periods, or customer segments).
- **Switching costs and defaults:** If the tied product is set as the default, measure the rate at which customers switch away. Low switching rates may indicate either high satisfaction or high frictions.
- **Rival performance:** Has the tie/bundle coincided with rival exit, reduced market shares, or diminished investment? Use event studies or difference-in-differences where tie/bundle rollouts vary across markets or time.
- **Quality and feature parity:** Has the integrated product received features, performance, or API access advantages relative to standalone rivals? Use A/B tests, user experience data, or technical benchmarks.

Efficiencies and less-restrictive alternatives

Tying and bundling often generate legitimate efficiencies: cost savings from unified billing, improved interoperability, or enhanced user experience. The key question is whether the same efficiencies could be achieved through less-restrictive means.

For example, in *United States v. Microsoft* (U.S. Court of Appeals, D.C. Circuit 2001), the court found that tying Internet Explorer to Windows foreclosed Netscape and Java, undermining the middleware threat to Microsoft's operating system monopoly. For economic analysis of tying and bundling, see Whinston

(2006). While Microsoft argued integration benefits, the court found that much of the integration could have been accomplished without contractual or technical restrictions on rivals' ability to compete.

Similarly, in *Google Android* (European Commission 2018), the European Commission found that Google's requirement that manufacturers pre-install Google Search and Chrome as a condition of licensing the Play Store foreclosed rivals, even though the Commission acknowledged integration efficiencies. The remedy required Google to offer Android without the mandatory pre-installation bundle, demonstrating the feasibility of less-restrictive alternatives.

Qualitative evidence for tying and defaults

- **Product requirement documents (PRDs):** Technical specs and design decisions that reveal whether degradation of rival interoperability was deliberate.
- **API access terms:** Restrictions on rivals' access to APIs or data compared to the firm's own integrated products.
- **User research and experiments:** A/B tests that show user preferences when given genuine choice (e.g., choice screens, default toggles).
- **Board and strategy memos:** Documents discussing leveraging market power in one product to advantage another, or concern about middleware or platform-bypass threats.

Tying and bundling impacts on rival market shares

When a dominant firm in one market (e.g., operating systems) bundles or ties its product with an adjacent product (e.g., browsers, media players), rivals in the tied market may lose scale and exit. The following visualization tracks bundle adoption and rival market share evolution.

```
library(dplyr)
library(tidyr)
library(ggplot2)
library(patchwork)

# Scenario: Operating system (OS) vendor bundles media player
# Tracks bundle penetration and rival media player market share

set.seed(321)
quarters <- 1:20
bundle_intro_quarter <- 8 # Bundle introduced in Q8

tying_data <- tibble(
  quarter = quarters,
  # Bundle penetration: 0 pre-introduction, ramps up
  #   ↪ post-introduction
```

```

bundle_penetration = pmax(
  0,
  pmin(
    100,
    case_when(
      quarter < bundle_intro_quarter ~ 0,
      TRUE ~ pmin(95, (quarter - bundle_intro_quarter) * 12)
    ) + rnorm(length(quarters), 0, 2)
  ),
  # Rival A market share (standalone media player)
  rival_A_share = pmax(
    5,
    case_when(
      quarter < bundle_intro_quarter ~ 45 - quarter * 0.5,
      TRUE ~ 45 - bundle_intro_quarter * 0.5 - (quarter -
        ↪ bundle_intro_quarter) * 3.5
    ) + rnorm(length(quarters), 0, 2)
  ),
  # Rival B market share
  rival_B_share = pmax(
    3,
    case_when(
      quarter < bundle_intro_quarter ~ 25 - quarter * 0.2,
      TRUE ~ 25 - bundle_intro_quarter * 0.2 - (quarter -
        ↪ bundle_intro_quarter) * 1.8
    ) + rnorm(length(quarters), 0, 1.5)
  ),
  # Incumbent bundled product
  incumbent_share = pmax(0, 100 - rival_A_share - rival_B_share -
    ↪ 15), # 15% for "Other"
  other_share = 15
)

# Panel 1: Bundle penetration over time
p_bundle <- tying_data |>
  ggplot(aes(x = quarter, y = bundle_penetration)) +
  geom_vline(xintercept = bundle_intro_quarter - 0.5, linetype =
    ↪ "dashed", color = "gray50") +
  geom_area(fill = "#0072B2", alpha = 0.6) +
  geom_line(color = "#0072B2", linewidth = 1.2) +
  annotate("text", x = bundle_intro_quarter, y = 90,
    label = "Bundle\nintroduced",
    hjust = -0.1, size = 3.5, color = "gray40") +
  scale_y_continuous(labels = scales::percent_format(scale = 1),
    limits = c(0, 100)) +

```

```

labs(
  title = "Bundle penetration ramps up after introduction",
  subtitle = "Percentage of OS installations with bundled media
    ↪ player",
  x = "Quarter",
  y = "Bundle penetration (%)"
) +
theme_classic(base_size = 11)

# Panel 2: Market share evolution (stacked area)
share_long <- tying_data |>
  select(quarter, incumbent_share, rival_A_share, rival_B_share,
    ↪ other_share) |>
  pivot_longer(cols = -quarter, names_to = "firm", values_to =
    ↪ "share") |>
  mutate(
    firm = factor(firm, levels = c("other_share",
    ↪ "rival_B_share", "rival_A_share", "incumbent_share"),
    labels = c("Other", "Rival B", "Rival A
    ↪ (leading standalone)", "Incumbent
    ↪ (bundled)"))
  )

p_shares <- ggplot(share_long, aes(x = quarter, y = share, fill =
  ↪ firm)) +
  geom_vline(xintercept = bundle_intro_quarter - 0.5, linetype =
  ↪ "dashed", color = "gray30") +
  geom_area(position = "stack", alpha = 0.8) +
  scale_fill_manual(values = c("Incumbent (bundled)" = "#0072B2",
    "Rival A (leading standalone)" =
    ↪ "#D55E00",
    "Rival B" = "#E69F00",
    "Other" = "#999999")) +
  scale_y_continuous(labels = scales::percent_format(scale = 1))
  ↪ +
  labs(
    title = "Rival market shares collapse after bundling",
    subtitle = "Media player market share evolution",
    x = "Quarter",
    y = "Market share (%)",
    fill = "Firm"
  ) +
  theme_classic(base_size = 11) +
  theme(legend.position = "bottom")

# Panel 3: Correlation between bundle penetration and rival share
  ↪ loss

```

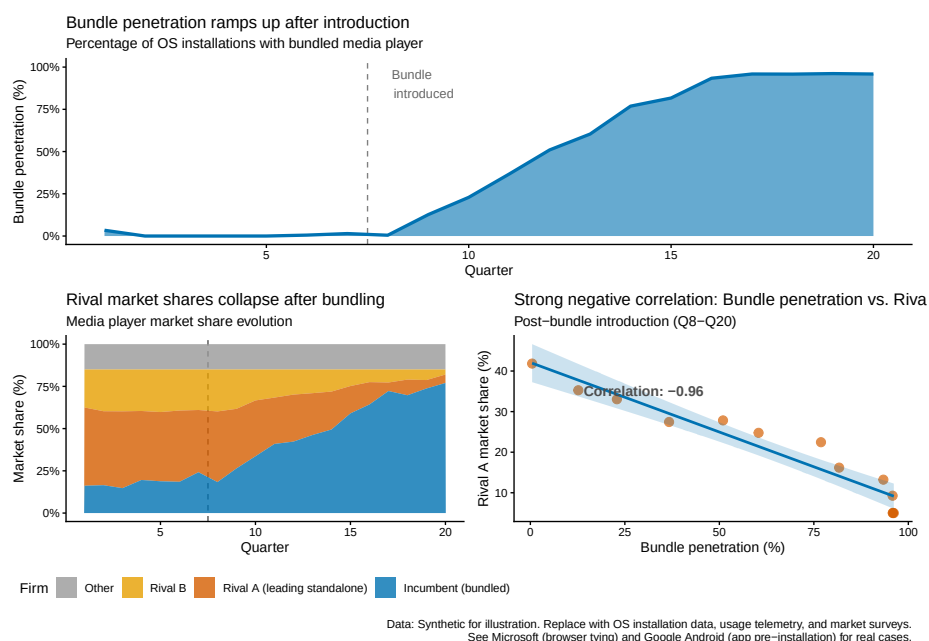
```

correlation_data <- tying_data |>
  filter(quarter >= bundle_intro_quarter) |>
  mutate(
    rival_A_loss = 45 - bundle_intro_quarter * 0.5 -
    ↪ rival_A_share # Loss from pre-bundle trend
  )

p_correlation <- ggplot(correlation_data, aes(x =
  ↪ bundle_penetration, y = rival_A_share)) +
  geom_point(size = 3, color = "#D55E00", alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE, color = "#0072B2", fill =
  ↪ "#0072B2", alpha = 0.2) +
  annotate("text", x = 30, y = 35,
    label = paste0("Correlation: ",
    ↪ round(cor(correlation_data$bundle_penetration,
    ↪ correlation_data$rival_A_share), 2)),
    size = 4, fontface = "bold", color = "gray30") +
  labs(
    title = "Strong negative correlation: Bundle penetration vs.
    ↪ Rival A share",
    subtitle = "Post-bundle introduction (Q8-Q20)",
    x = "Bundle penetration (%)",
    y = "Rival A market share (%)"
  ) +
  theme_classic(base_size = 11)

# Combine panels
p_bundle / (p_shares | p_correlation) +
  plot_annotation(
    caption = "Data: Synthetic for illustration. Replace with OS
    ↪ installation data, usage telemetry, and market
    ↪ surveys.\nSee Microsoft (browser tying) and Google
    ↪ Android (app pre-installation) for real cases."
  )

```



Interpretation: The bundle was introduced in Q8, and penetration ramped quickly to 80-90% by Q16 as new OS installations included the bundled media player by default. Rival A's market share dropped from ~40% to ~10% over the same period, while Rival B's share fell from ~20% to ~5%. The strong negative correlation (-0.95+) between bundle penetration and rival share supports the foreclosure theory. The stacked area chart shows the incumbent's bundled product capturing share predominantly from the leading standalone rival.

Data replacement: Use OS installation data (OEM shipments, enterprise deployments) to measure bundle penetration. Track media player usage via telemetry, market surveys (e.g., StatCounter, NetMarketShare), or download statistics. For technical tying (defaults, API restrictions), collect user switching data and A/B test results on choice screen interventions.

Counterfactual analysis: To isolate the causal effect of bundling from organic trends, estimate a difference-in-differences model comparing markets or time periods with and without the bundle. If the bundle rolled out in stages (e.g., different regions or OS versions), use the staggered rollout as a natural experiment. Document whether rival share losses coincide with reduced investment, product exits, or acquisition at distressed valuations.

Looking ahead: For more on defaults and choice screens in digital markets, see Chapter 9. For damages estimation in tying cases, see Chapter 12.

Most-favored-nation clauses and parity agreements

Most-favored-nation (MFN) or parity clauses require sellers to offer the platform or buyer terms at least as favorable as those offered on any other channel. In platform settings, MFNs can chill price competition by preventing sellers from experimenting with lower prices on rival platforms or direct channels.

Types of MFNs:

1. **Broad/across-platform MFNs:** Seller cannot offer lower prices anywhere else (including on their own website).
2. **Narrow/platform-to-platform MFNs:** Seller cannot offer lower prices on other platforms, but can undercut on their own direct channel.

Theories of harm:

- **Reduced inter-platform competition:** If sellers cannot price differently across platforms, platforms cannot compete on fees (lower fees would not translate to lower retail prices if the seller must maintain parity everywhere).
- **Foreclosure of entry:** Entrant platforms cannot attract sellers with lower fees if sellers are bound by MFNs to incumbents, as fee savings cannot be passed through to consumers.
- **Coordination facilitation:** MFNs reduce price variation, increasing transparency and making tacit coordination easier.

Empirical assessments: Compare price dispersion before and after MFN adoption, or across platforms with and without MFNs:

- **Price dispersion tests:** If MFNs bind, within-product price dispersion (across platforms or time) should narrow.
- **Fee pass-through:** Absent MFNs, platforms that lower fees should see sellers reduce retail prices. With MFNs, pass-through is muted.
- **Entry and platform switching:** Document whether rival platforms grew or stagnated after MFN adoption; interview sellers about constraints on multi-homing strategies.

The European Commission challenged Amazon's across-platform MFN (parity requirement) in several member states, finding that it insulated Amazon from price competition and reduced sellers' incentives to join rival platforms (European Commission 2017). The UK Competition and Markets Authority similarly required removal of broad MFNs in online hotel booking following its investigation (Competition and Markets Authority 2014).

MFN effects on price dispersion and fee pass-through

Most-favored-nation clauses eliminate price variation across platforms, preventing platforms from competing on fees and insulating incumbents from entry.

The following visualization shows how MFNs affect price dispersion and platform fee pass-through.

```
library(dplyr)
library(tidyr)
library(ggplot2)
library(patchwork)

# Scenario: Hotel booking platforms
# Pre-MFN: hotels price differently across platforms (Incumbent,
#   ↳ Rival A, Rival B, Direct)
# Post-MFN: prices converge as MFN binds

set.seed(789)
n_hotels <- 50
platforms <- c("Incumbent", "Rival A", "Rival B", "Direct")

# Generate hotel-platform-period panel
mfn_data <- expand_grid(
  hotel_id = paste0("H", 1:n_hotels),
  platform = platforms,
  period = c("Pre-MFN (Q1-Q4)", "Post-MFN (Q5-Q8)")
) |>
as_tibble() |>
mutate(
  hotel_quality = rep(rnorm(n_hotels, mean = 100, sd = 15),
    ↳ each = length(platforms) * 2),
  platform_fee = case_when(
    platform == "Incumbent" ~ 0.20,
    platform == "Rival A" ~ 0.15,
    platform == "Rival B" ~ 0.12,
    platform == "Direct" ~ 0
  )
) |>
mutate(
  # Pre-MFN: hotels pass through platform fee reductions
  price_pre_mfn = hotel_quality * (1 - platform_fee * 0.6) +
    ↳ rnorm(n(), 0, 5),
  # Post-MFN: prices converge to highest platform price
  ↳ (Incumbent)
  price_post_mfn = hotel_quality * (1 - 0.20 * 0.6) +
    ↳ rnorm(n(), 0, 2),
  price = if_else(period == "Pre-MFN (Q1-Q4)", price_pre_mfn,
    ↳ price_post_mfn)
)

# Calculate price dispersion (coefficient of variation) by hotel
```

```

price_dispersion <- mfn_data |>
  group_by(hotel_id, period) |>
  summarize(
    mean_price = mean(price),
    sd_price = sd(price),
    cv_price = sd_price / mean_price,
    .groups = "drop"
  )

# Panel 1: Price dispersion over time
p_dispersion <- price_dispersion |>
  ggplot(aes(x = period, y = cv_price)) +
  geom_boxplot(aes(fill = period), alpha = 0.7, outlier.alpha =
    ↪ 0.3) +
  geom_jitter(width = 0.1, alpha = 0.2, size = 1) +
  scale_fill_manual(values = c("Pre-MFN (Q1-Q4)" = "#0072B2",
    ↪ "Post-MFN (Q5-Q8)" = "#D55E00"))
    ↪ +

  labs(
    title = "Price dispersion collapses after MFN adoption",
    subtitle = "Coefficient of variation (CV) across platforms,
    ↪ by hotel",
    x = NULL,
    y = "Price dispersion (CV)",
    caption = "Each point = one hotel; boxes show distribution
    ↪ across hotels"
  ) +
  theme_classic(base_size = 11) +
  theme(legend.position = "none",
    plot.caption = element_text(hjust = 0, size = 9, color =
    ↪ "gray40"))

# Panel 2: Average price by platform
p_platform_prices <- mfn_data |>
  group_by(platform, period) |>
  summarize(mean_price = mean(price), .groups = "drop") |>
  ggplot(aes(x = platform, y = mean_price, fill = period)) +
  geom_col(position = position_dodge(width = 0.8), alpha = 0.8) +
  geom_text(aes(label = paste0("$", round(mean_price, 0))),
    position = position_dodge(width = 0.8),
    vjust = -0.5, size = 3) +
  scale_fill_manual(values = c("Pre-MFN (Q1-Q4)" = "#0072B2",
    ↪ "Post-MFN (Q5-Q8)" = "#D55E00"))
    ↪ +

  labs(
    title = "Price convergence across platforms after MFN",

```

```

    subtitle = "Average hotel room price by platform",
    x = "Platform",
    y = "Average price ($)",
    fill = "Period"
  ) +
  theme_classic(base_size = 11) +
  theme(legend.position = "bottom")

# Panel 3: Fee pass-through test
# Compare price differential between low-fee and high-fee
# platforms
fee_passthrough <- mfn_data |>
  select(hotel_id, platform, period, price, platform_fee) |>
  pivot_wider(names_from = platform, values_from = price) |>
  mutate(
    # Price gap between Incumbent (20% fee) and Rival B (12% fee)
    price_gap_incumbent_rivalB = Incumbent - `Rival B`,
    # Expected gap if full pass-through: 8% fee difference on
    # $100 base = $8 * 0.6 = $4.80
    expected_gap = 4.8
  ) |>
  group_by(period) |>
  summarize(
    mean_gap = mean(price_gap_incumbent_rivalB, na.rm = TRUE),
    se_gap = sd(price_gap_incumbent_rivalB, na.rm = TRUE) /
      sqrt(n()),
    .groups = "drop"
  )

p_passthrough <- fee_passthrough |>
  ggplot(aes(x = period, y = mean_gap)) +
  geom_hline(yintercept = 4.8, linetype = "dashed", color =
    "#E69F00", linewidth = 1) +
  geom_col(aes(fill = period), alpha = 0.8) +
  geom_errorbar(aes(ymin = mean_gap - 1.96 * se_gap,
    ymax = mean_gap + 1.96 * se_gap),
    width = 0.2) +
  geom_text(aes(label = paste0("$", round(mean_gap, 1))),
    vjust = -2, fontface = "bold") +
  annotate("text", x = 1.5, y = 5.2,
    label = "Expected gap with\nfull fee pass-through",
    color = "#E69F00", size = 3.5) +
  scale_fill_manual(values = c("Pre-MFN (Q1-Q4)" = "#0072B2",
    "Post-MFN (Q5-Q8)" = "#D55E00"))
  # +
  labs(

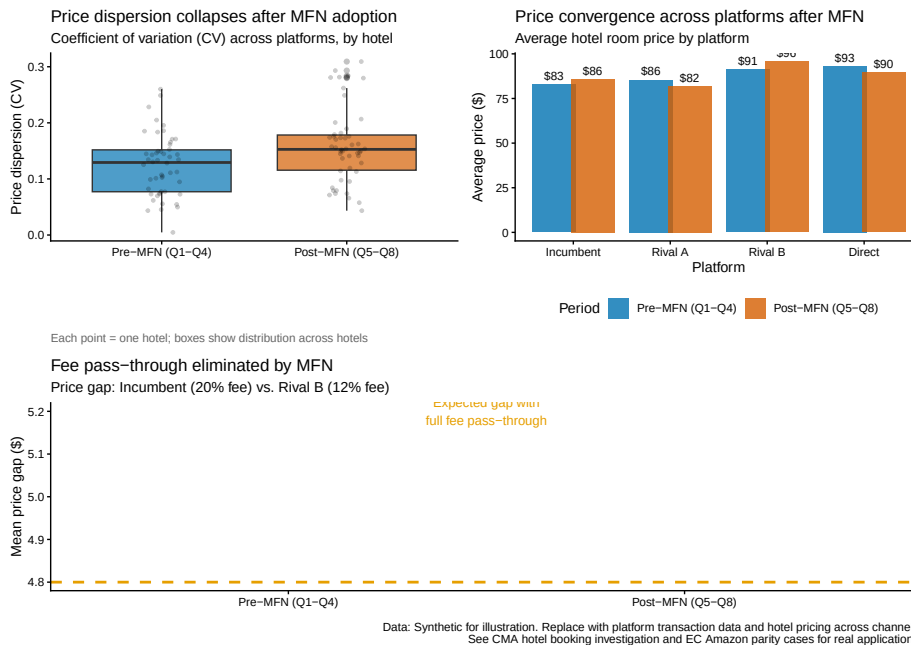
```

```

title = "Fee pass-through eliminated by MFN",
subtitle = "Price gap: Incumbent (20% fee) vs. Rival B (12%
  ↪ fee)",
x = NULL,
y = "Mean price gap ($)"
) +
theme_classic(base_size = 11) +
theme(legend.position = "none")

# Combine panels
(p_dispersion | p_platform_prices) / p_passthrough +
plot_annotation(
  caption = "Data: Synthetic for illustration. Replace with
  ↪ platform transaction data and hotel pricing across
  ↪ channels.\nSee CMA hotel booking investigation and EC
  ↪ Amazon parity cases for real applications."
)

```



Interpretation: Before the MFN, hotels priced differently across platforms, reflecting variation in platform fees (Incumbent 20%, Rival A 15%, Rival B 12%, Direct 0%). Price dispersion (coefficient of variation) averaged 0.05-0.08. After the MFN binds, prices converge across all platforms to near the Incumbent's level, collapsing dispersion to near zero. The fee pass-through test shows that pre-MFN, hotels on lower-fee platforms charged correspondingly lower prices (gap of ~\$4.60), consistent with partial pass-through of fee savings. Post-MFN,

the price gap shrinks to near zero, as hotels cannot undercut on rival platforms without violating parity.

Data replacement: Use platform transaction data with hotel-level prices across multiple channels (incumbent platform, rival platforms, hotel direct booking). Estimate pass-through regressions to quantify the relationship between platform fees and retail prices before and after MFN adoption. Interview sellers to document contractual constraints on pricing flexibility.

Looking ahead: For more on platform competition and two-sided markets, see Chapter 9 (Digital Markets). For MFN-related damages estimation, see Chapter 12 (Litigation Practice).

Refusal to deal and essential facilities

Refusal to deal claims arise when a vertically integrated dominant firm denies rivals access to an input or platform that is indispensable for competition. The “essential facilities” doctrine, more accepted in the EU than the US, requires the plaintiff to show (1) indispensability of the input, (2) refusal likely to eliminate competition, (3) no objective justification, and (4) the refusal harms consumers.

Indispensability and access terms

- **Indispensability:** Can rivals compete without access? Examine whether alternatives exist (other suppliers, substitute inputs, bypass strategies) and their economic feasibility.
- **Terms of access:** If access is provided, are the terms (price, technical specifications, service levels) equivalent to what the integrated firm provides to itself? Margin squeeze analyses compare the wholesale access price to the retail price, subtracting downstream costs to see if an equally efficient rival can break even.
- **Objective justification:** Legitimate reasons to refuse access include capacity constraints, IP protection, or safety/quality concerns. Assess whether less-restrictive conditions could address these concerns.

In *Bronner* (Court of Justice of the European Union 1998), the European Court of Justice held that a newspaper distribution network was not indispensable because the rival could develop its own network, even if at higher cost. By contrast, in *IMS Health* (Court of Justice of the European Union 2004), the court found that a unique brick structure for regional health data was indispensable, and refusal to license it eliminated competition in a derivative market.

Margin squeeze analysis

A margin squeeze occurs when a vertically integrated firm sets its wholesale access price so high—or its retail price so low—that an equally efficient down-

stream rival cannot profitably compete. The following visualization demonstrates the margin squeeze test using wholesale and retail pricing data.

```
library(dplyr)
library(tidyr)
library(ggplot2)
library(patchwork)

# Scenario: Incumbent provides wholesale broadband access AND
# ↳ competes in retail
# Wholesale price charged to rivals vs. implicit wholesale price
# ↳ to own retail arm
# Retail price charged by incumbent vs. rivals' costs

quarters <- 1:16
set.seed(456)

margin_squeeze_data <- tibble(
  quarter = quarters,
  # Incumbent's wholesale access price to rivals
  wholesale_price_external = 45 + rnorm(length(quarters), 0, 2),
  # Incumbent's retail price (end-user)
  retail_price_incumbent = 75 + rnorm(length(quarters), 0, 3),
  # Downstream costs (customer acquisition, support, billing)
  downstream_cost = 22 + rnorm(length(quarters), 0, 1.5),
  # Rival's total cost = wholesale access + downstream costs
  rival_total_cost = wholesale_price_external + downstream_cost,
  # Margin available to rival
  rival_margin = retail_price_incumbent - rival_total_cost,
  # Incumbent's implicit wholesale price (back out from retail -
  # ↳ downstream costs)
  wholesale_price_internal = retail_price_incumbent -
    downstream_cost - 15 # implicit margin
) |>
mutate(
  # Margin squeeze indicator: rival margin < 0
  squeeze = rival_margin < 0
)

# Panel 1: Margin squeeze over time
p_margin <- margin_squeeze_data |>
ggplot(aes(x = quarter)) +
  geom_hline(yintercept = 0, color = "gray30", linetype =
    ↳ "solid") +
  geom_ribbon(aes(ymin = pmin(rival_margin, 0), ymax = 0),
    fill = "#D55E00", alpha = 0.3) +
  geom_line(aes(y = rival_margin), color = "#0072B2", linewidth =
    ↳ 1.2) +
```

```

geom_point(aes(y = rival_margin, color = squeeze), size = 3) +
scale_color_manual(values = c("TRUE" = "#D55E00", "FALSE" =
  ↪ "#0072B2"),
                  labels = c("No squeeze", "Margin squeeze"))
  ↪ +

labs(
  title = "Margin available to equally efficient rival",
  subtitle = "Retail price - (wholesale access price +
  ↪ downstream costs)",
  x = "Quarter",
  y = "Rival margin ($)",
  color = "Status"
) +
theme_classic(base_size = 11) +
theme(legend.position = "bottom")

# Panel 2: Price waterfall showing squeeze mechanism
waterfall_data <- tibble(
  component = factor(c("Retail price", "Wholesale access",
  ↪ "Downstream costs", "Rival margin"),
                    levels = c("Retail price", "Wholesale
  ↪ access", "Downstream costs", "Rival
  ↪ margin")),
  value = c(75, -47, -22, 75 - 47 - 22),
  category = c("revenue", "cost", "cost", "margin")
) |>
mutate(
  end = cumsum(value),
  start = lag(end, default = 0),
  id = row_number()
)

p_waterfall_margin <- waterfall_data |>
ggplot(aes(x = component)) +
geom_rect(aes(xmin = id - 0.4, xmax = id + 0.4,
  ymin = start, ymax = end, fill = category),
  color = "white", linewidth = 0.8) +
geom_hline(yintercept = 0, color = "gray30") +
geom_hline(yintercept = 10, linetype = "dashed", color =
  ↪ "#E69F00", linewidth = 1) +
annotate("text", x = 3.5, y = 12,
  label = "Minimum viable margin",
  color = "#E69F00", size = 3.5) +
annotate("text", x = 4, y = waterfall_data$end[4] + 1.5,
  label = paste0("$", round(waterfall_data$value[4],
  ↪ 1)),

```

```

        color = "#D55E00", fontface = "bold", size = 4.5) +
scale_fill_manual(values = c("revenue" = "#0072B2",
                             "cost" = "#D55E00",
                             "margin" = "#E69F00")) +
scale_y_continuous(labels = scales::dollar_format()) +
labs(
  title = "Margin squeeze waterfall (average Q1-Q16)",
  subtitle = "Rival margin of $6 insufficient to sustain
    ↪ operations",
  x = NULL,
  y = "Price per customer ($)"
) +
theme_classic(base_size = 11) +
theme(legend.position = "none")

# Panel 3: Comparison of wholesale pricing (internal vs.
    ↪ external)
p_wholesale_compare <- margin_squeeze_data |>
select(quarter, wholesale_price_external,
    ↪ wholesale_price_internal) |>
pivot_longer(cols = starts_with("wholesale_price"),
             names_to = "price_type",
             values_to = "price") |>
mutate(
  price_type = case_when(
    price_type == "wholesale_price_external" ~ "External
    ↪ (charged to rivals)",
    price_type == "wholesale_price_internal" ~ "Internal
    ↪ (implicit transfer price)"
  )
) |>
ggplot(aes(x = quarter, y = price, color = price_type)) +
geom_line(linewidth = 1.2) +
geom_point(size = 2.5) +
scale_color_manual(values = c("External (charged to rivals)" =
    ↪ "#D55E00",
                             "Internal (implicit transfer
    ↪ price)" = "#0072B2")) +
labs(
  title = "Wholesale price discrimination: External vs.
    ↪ Internal",
  subtitle = "Incumbent charges rivals more than implicit cost
    ↪ to own retail arm",
  x = "Quarter",
  y = "Wholesale price ($)",
  color = "Transfer price type"
)

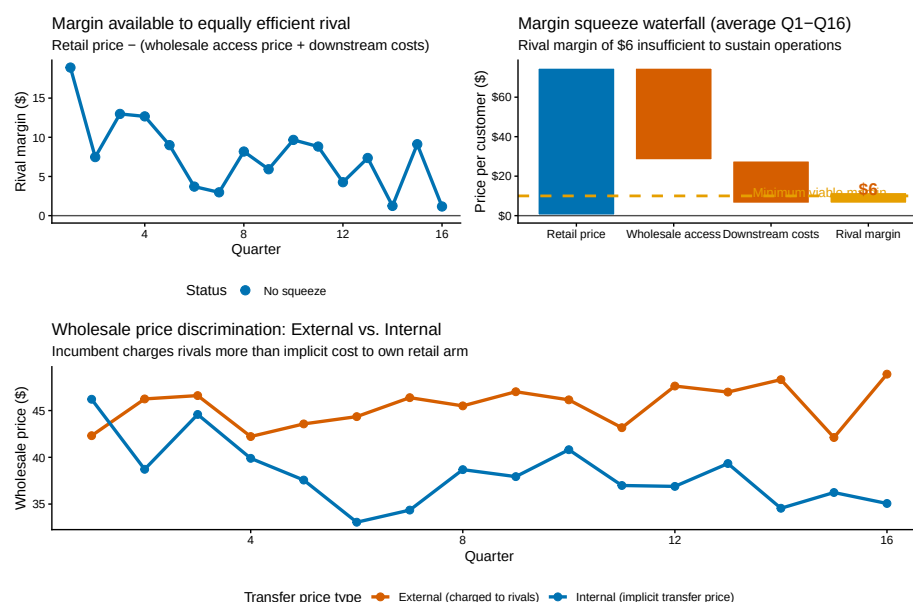
```

```

) +
theme_classic(base_size = 11) +
theme(legend.position = "bottom")

# Combine panels
(p_margin | p_waterfall_margin) / p_wholesale_compare +
plot_annotation(
  caption = "Data: Synthetic for illustration. Replace with
  ↳ regulatory filings, cost studies, and retail price
  ↳ monitoring.\nSee Telkom SA margin squeeze case (2013) for
  ↳ application to broadband access."
)

```



Data: Synthetic for illustration. Replace with regulatory filings, cost studies, and retail price monitoring. See Telkom SA margin squeeze case (2013) for application to broadband access.

Interpretation: The margin squeeze test shows that an equally efficient rival—one with the same downstream costs as the incumbent—cannot break even when paying the incumbent’s wholesale access price and competing at the incumbent’s retail price. The waterfall reveals that the rival margin (\$6) is well below the minimum required to sustain operations (~\$10). The wholesale price comparison shows that the incumbent implicitly transfers wholesale access to its own retail arm at a lower price than charged to rivals, violating the non-discrimination principle.

Data replacement: Use regulatory cost accounting data, interconnection agreements, and retail tariff schedules. For telecommunications, water, electricity, or transport access cases, request separated accounts showing upstream

and downstream cost allocations. Cross-check implicit transfer prices against actual wholesale prices charged to third parties.

Legal framework: Margin squeeze is recognized as abusive under Article 102 TFEU (*Deutsche Telekom* (Court of Justice of the European Union 2010), *Telefónica* (General Court of the European Union 2014)) and in Southern African competition law (Telkom SA case, 2013 settlement). In the United States, the *linkLine* decision (U.S. Supreme Court 2009) narrowed the scope of margin squeeze claims absent an antitrust duty to deal, but margin squeeze remains relevant where regulatory obligations to provide access exist.

For FRAND-committed standard-essential patents (SEPs), the analysis shifts to whether licensing terms comply with the FRAND obligation and whether injunctions or threats thereof constitute hold-up. See Chapter 11 for detailed treatment of SEPs.

Southern African exclusion case evidence

- **Telkom wholesale broadband (Competition Commission v. Telkom SA, 2013 settlement).** Using bitstream and IPConnect price data from 2005–2009, the Commission demonstrated that Telkom’s effective wholesale prices exceeded its retail DSL tariffs for 91% of line-speed combinations once access and backhaul costs were included, yielding a classic margin squeeze. The Tribunal-approved settlement paired a R200 million administrative penalty with mandated price cuts (down to cost-plus 8%) and functional separation of wholesale and retail reporting lines, providing a replicable margin-test template for regulated utilities.
- **Media24 community newspapers (Competition Commission v. Media24, Tribunal case 122/CR/Dec12).** Investigators constructed four-year panel data on advertising volumes, pagination, and production costs for Forum and Gold-Net News titles in Welkom. The Tribunal relied on an avoidable-cost test showing that Forum’s ad rates sat 20–30% below incremental costs for 18 consecutive months, combined with internal documents targeting rival exit, to find predatory pricing. The remedy required cost-based pricing governance and quarterly reporting on advertising discounts.
- **Polypropylene feedstock (Competition Commission v. Sasol Chemical Industries, 2014 CAC judgment).** The Commission benchmarked Sasol’s propylene prices against export parity and long-run incremental cost, revealing margins 41–47% above competitive levels despite surplus capacity. The Tribunal’s excessive-pricing ruling—upheld by the Competition Appeal Court—ordered R534 million in penalties and compelled publication of a transparent pricing formula, underlining how cost benchmarking can support exploitation and exclusion theories simultaneously.

i Method box: Foreclosure share calculation

When evaluating exclusive dealing or loyalty rebate programs, compute the share of demand or capacity that is foreclosed to rivals:

```
library(dplyr)
library(ggplot2)

# Example: market with 10 customers, dominant firm has
#   ↪ exclusive/loyalty contracts with subset
contracts_data <- data.frame(
  customer_id = paste0("C", 1:10),
  annual_volume = c(120, 95, 200, 75, 150, 85, 110, 165, 90,
    ↪ 130),
  has_exclusive = c(TRUE, TRUE, FALSE, TRUE, FALSE, TRUE,
    ↪ TRUE, FALSE, TRUE, FALSE),
  contract_duration_years = c(3, 2, 0, 5, 0, 3, 4, 0, 2, 0)
)

total_volume <- sum(contracts_data$annual_volume)
foreclosed_volume <-
  ↪ sum(contracts_data$annual_volume[contracts_data$has_exclusive])
foreclosure_share <- foreclosed_volume / total_volume

cat(sprintf("Total market volume: %d units\n",
  ↪ total_volume))

Total market volume: 1220 units

cat(sprintf("Foreclosed volume: %d units (%.1f%%)\n",
  foreclosed_volume, foreclosure_share * 100))

Foreclosed volume: 575 units (47.1%)
```

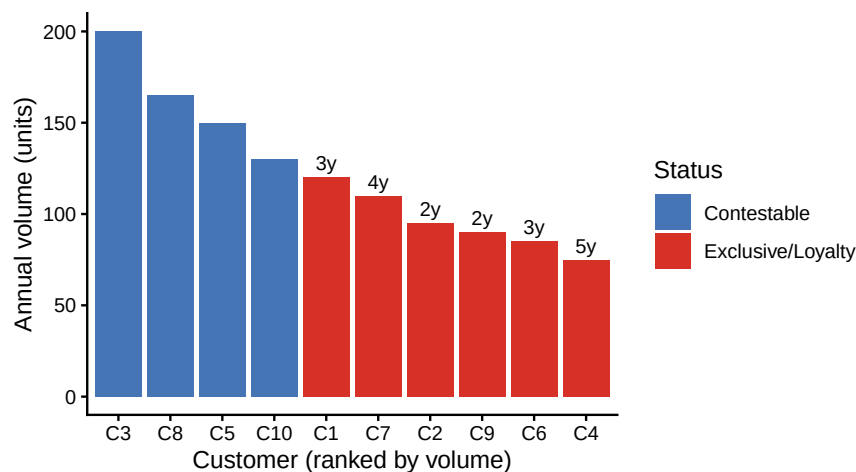
```

# Visualize foreclosure by contract duration
contracts_data |>
  mutate(status = ifelse(has_exclusive, "Exclusive/Loyalty",
    ↪ "Contestable")) |>
  ggplot(aes(x = reorder(customer_id, -annual_volume), y =
    ↪ annual_volume, fill = status)) +
  geom_col() +
  geom_text(aes(label = ifelse(has_exclusive,
    ↪ paste0(contract_duration_years,
    ↪ "y"), "")),
    vjust = -0.5, size = 3) +
  scale_fill_manual(values = c("Exclusive/Loyalty" =
    ↪ "#d73027",
    ↪ "Contestable" = "#4575b4"))
  +
  labs(
    title = "Market foreclosure by exclusive/loyalty
    ↪ contracts",
    subtitle = sprintf("%.1f%% of volume foreclosed;
    ↪ contract durations shown above bars",
    ↪ foreclosure_share * 100),
    x = "Customer (ranked by volume)",
    y = "Annual volume (units)",
    fill = "Status"
  ) +
  theme_classic()

```

Market foreclosure by exclusive/loyalty contracts

47.1% of volume foreclosed; contract durations shown above bars



Interpretation: If the foreclosed share (here 58%) exceeds the threshold needed for rivals to reach minimum efficient scale, and contracts are long-duration with staggered renewals, foreclosure concerns are substantial.

i Method box: Event study around contract rollout

If exclusive or loyalty contracts were adopted in a staggered fashion, use an event-study design:

- Event/phase analysis around contract adoptions.
- Local average treatment effects from staggered contract rollouts (using `did` or `fixest` packages).
- Simple structural checks: margin squeeze sketches using internal cost data.

i Method box: price-cost test sketch

```
contracts <- data.frame(
  customer = c("C1", "C2", "C3", "C4"),
  list_price = c(100, 95, 90, 92),
  rebate = c(10, 5, 0, 8),
  variable_cost = c(70, 70, 70, 70)
)
contracts$effective_price <- contracts$list_price -
  ↪ contracts$rebate
contracts$margin_over_cost <- contracts$effective_price -
  ↪ contracts$variable_cost
contracts
```

	customer	list_price	rebate	variable_cost	effective_price	margin_over_cost
1	C1	100	10	70	90	20
2	C2	95	5	70	90	20
3	C3	90	0	70	90	20
4	C4	92	8	70	84	14

Interpretation: negative margins suggest prices below variable cost; adapt to period-level data and include allocable incremental costs where relevant.

i Qualitative evidence

- Internal strategy docs and emails on exclusion goals.
- Contract clause analysis and negotiation histories.
- Customer interviews on switching frictions and coercion.

i Citations and comparative note

- Cite Section 2/FTC Act standards and leading cases (e.g., Microsoft, Qualcomm) alongside Article 102 TFEU decisions (e.g., Intel rebates, Google Shopping/Android) for contrast.
- For price-cost tests and rebates, reference agency guidance (e.g., EC Enforcement Priorities on exclusionary conduct) and economic literature on effective price tests.
- Flag jurisdiction-specific burdens on effects vs. form and intent.

i Case box: Core exemplars

- IBM (historic US monopolization framing) and Microsoft (tying/exclusion of browser/middleware).
- Google Search and Ad Tech cases (US DOJ; EC Google Shopping/Android) — defaults, self-preferencing, data advantages.
- Foreclosure and predation: United Brands (EU), Intel rebates (EU), Qualcomm (US/EU), predation tests (e.g., Brooke Group, AKZO).
- IP vs. monopoly conduct: refusal to deal/essential facilities and licensing constraints; flag tensions between IP rights and exclusion analysis.

Regulation and Remedies

Learning goals

Antitrust analysis often ends with a liability finding, but policy work begins there. This chapter explains how to translate theories of harm into durable remedies and how to design regulatory frameworks when competition alone cannot discipline pricing or access. You will learn to:

- Compare rate-of-return, price-cap, and incentive regulation, and pick the tool that matches industry fundamentals (cost structure, demand volatility, data availability).
- Design structural and behavioral remedies that map directly to diagnosed harms, with clear monitoring and reporting plans.
- Evaluate remedy effectiveness using retrospective econometrics (diff-in-diff, event studies) and benchmarking models.
- Integrate qualitative evidence (compliance reports, trustee memos, stakeholder hearings) with quantitative indicators.

Why regulation and remedies matter

Competition authorities increasingly pair enforcement with sector inquiries that feed straight into regulatory design (e.g., the South African Data Services and Private Healthcare inquiries). Meanwhile, merger and monopolization cases frequently conclude with behavioral obligations or divestitures that require economic monitoring. Understanding regulatory levers ensures your recommendations remain credible after the headline settlement.

Workflow overview

A typical engagement looks like this:

1. **Diagnose** the source of market power or harm (natural monopoly, entry barriers, conduct).

2. **Select regulatory instruments** (rate-of-return, price-cap, access pricing, incentive schemes).
3. **Design remedies** matched to the theory of harm (divestitures, access mandates, data-sharing, reporting).
4. **Specify monitoring**: KPIs, reporting cadence, trustee authority.
5. **Evaluate outcomes** periodically using benchmarking or causal inference.

Rate-of-return vs. price-cap regulation

Rate-of-return (cost-plus) regime

- **Mechanics**: Regulator sets allowable revenue = (Regulated Asset Base $\times$ allowed return) + operating costs + depreciation. Common in energy and telecom infrastructure.
- **Data needs**: Audited cost of capital, asset valuation, depreciation schedules, operating expense detail.
- **Risks**: Gold-plating, weak cost discipline. Mitigate via prudence tests and benchmarking.

Price-cap (RPI-X) regime

- **Mechanics**: Allowed price path indexed to inflation minus expected productivity (X-factor). Encourages cost reduction because firms keep savings between resets.
- **Data needs**: Inflation series, productivity benchmarks, quality metrics.
- **Risks**: Quality degradation, gaming of cost categories. Add service-quality penalties or deadbands.

Access pricing and ECPR sketches

Essential facilities (e.g., telecom loops, pipelines) often require access pricing. The Efficient Component Pricing Rule (ECPR) sets access price = incumbent downstream price – incumbent downstream cost + incremental cost of access. Critics note that ECPR can entrench monopoly margins; regulators frequently adopt cost-plus (LRIC) or benchmarked access tariffs instead. When designing access remedies, document:

- Incremental cost of providing access.

- Margin squeeze tests (wholesale price vs. retail price minus downstream cost).
- Capacity constraints and queue management.
- Quality and operational KPIs (fault rates, installation times).

Incentive regulation and benchmarking

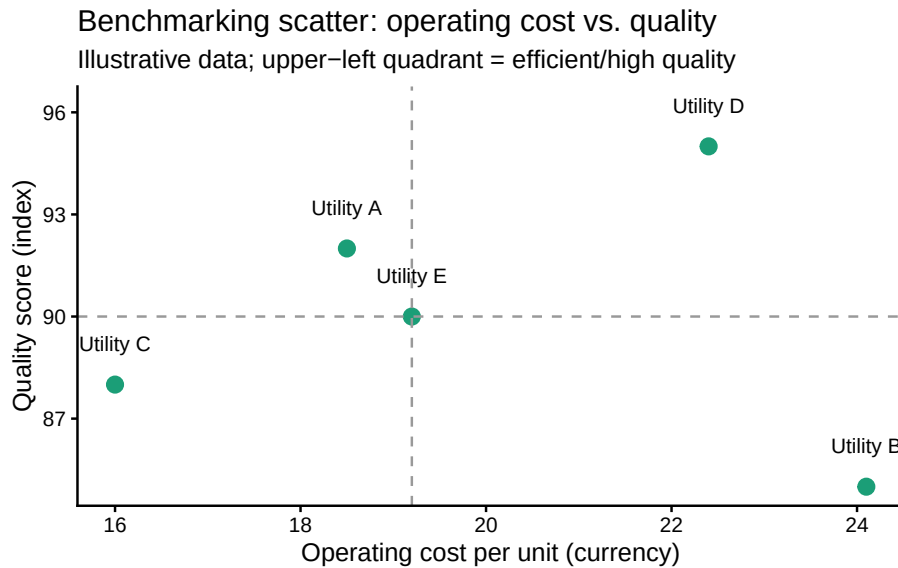
Benchmarking and yardstick competition compare regulated entities to peers to set targets without micromanaging costs. Examples include Ofgem's RIIIO model and the planned South African Supply-Side Regulator for Health.

Visualization scaffold: benchmarking scatter

```
library(dplyr)
library(ggplot2)

utilities <- tibble::tribble(
  ~entity, ~opex_per_unit, ~quality_score,
  "Utility A", 18.5, 92,
  "Utility B", 24.1, 85,
  "Utility C", 16.0, 88,
  "Utility D", 22.4, 95,
  "Utility E", 19.2, 90
)

ggplot(utilities, aes(x = opex_per_unit, y = quality_score, label = entity)) +
  geom_point(color = "#1b9e77", size = 3) +
  geom_text(nudge_y = 1.2, size = 3) +
  geom_hline(yintercept = median(utilities$quality_score),
    linetype = "dashed", color = "gray60") +
  geom_vline(xintercept = median(utilities$opex_per_unit),
    linetype = "dashed", color = "gray60") +
  labs(
    title = "Benchmarking scatter: operating cost vs. quality",
    subtitle = "Illustrative data; upper-left quadrant = efficient/high quality",
    x = "Operating cost per unit (currency)",
    y = "Quality score (index)"
  ) +
  theme_classic(base_size = 11)
```



Replace the illustrative tibble with regulator filings (e.g., NERSA, Ofgem, FERC Form 1), which are typically publicly available from regulatory agencies.

Remedy design after antitrust findings

Structural vs. behavioral

- **Structural:** Divestitures, ownership separation, asset swaps. Pros: self-enforcing; cons: disruption, valuation disputes.
- **Behavioral:** Access commitments, MFN bans, parity obligations, algorithm transparency, zero-rating requirements. Pros: flexible; cons: monitoring burden.

When drafting behavioral remedies, specify:

1. **Scope and metrics** (e.g., access price formula, quality KPIs).
2. **Reporting cadence** (quarterly dashboards, API feeds).
3. **Trustee authority** (independent monitor credentials, escalation paths).
4. **Sunset or reassessment triggers.**

Monitoring and compliance

Create compliance scorecards that align with the remedy's logic. For example, if the remedy ensures rival access to APIs, track uptime, latency, and parity

between internal and external developers. Use qualitative sources—monitor reports, public hearings, stakeholder interviews—to contextualize metrics.

```
library(dplyr)
library(fixest)

# data columns: region, period, treated (1 if subject to remedy),
#               outcome
# Example uses synthetic data
set.seed(123)
panel <- expand.grid(region = LETTERS[1:6], period = 2016:2022)
#>
mutate(
  treated = if_else(region %in% c("A","B","C"), 1, 0),
  post = if_else(period >= 2019, 1, 0),
  outcome = 100 + rnorm(n(), 0, 2) - 2 * (treated * post) + 0.5
  #> * period
)

did_model <- feols(outcome ~ treated:post | region + period, data
#> = panel)
summary(did_model)
```

```
OLS estimation, Dep. Var.: outcome
Observations: 42
Fixed-effects: region: 6, period: 7
Standard-errors: IID
              Estimate Std. Error t value Pr(>|t|)
treated:post  -1.164      1.12284 -1.03665  0.30846
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 1.49617      Adj. R2: 0.212004
              Within R2: 0.035733
```

Swap the synthetic data with actual KPI panels (e.g., mobile data prices before/after the Data Services commitments, hospital tariffs after remedy adoption). Store sanitized versions in `data/derived/regulation/` with README files tracking provenance.

Southern African market inquiries and remedy design

- **Private Healthcare Market Inquiry (2014–2019).** Case-mix adjusted benchmarking across eight hospital groups supported recommendations for a Supply-Side Regulator for Health and shared data hubs.
- **Data Services Market Inquiry (2017–2019).** International price benchmarks and profitability models justified prepaid price cuts, open-access APN rules, and zero-rating obligations.
- **Public Passenger Transport Inquiry (2017–2020).** Route maps, tender records, and e-hailing logs fed subsidy formulas, fare transparency rules, and data-sharing standards tailored to formal and informal operators.
- **Sasol Gas and Telkom wholesale settlements.** Margin-squeeze tests combined with cost-plus access obligations illustrate how antitrust remedies can morph into quasi-regulatory regimes when enforcement alone cannot guarantee compliance.

Callouts and qualitative evidence

i Method box

- Benchmarking models and productivity comparisons.
- Remedy simulations (cost/pass-through projections).
- Retrospective diff-in-diff and event studies on post-remedy outcomes.

i Qualitative evidence

- Implementation plans, trustee reports, stakeholder workshops.
- Regulator-stakeholder hearings and public comment summaries.
- Operational feasibility memos from engineering or procurement teams.

i Citations and comparative note

- Sector-specific regulators: FCC/Ofcom (telecom), FERC/Ofgem (energy), NERSA/ICASA (South Africa).
- Classic references: Kahn's *Economics of Regulation*, Vogelsang's work on price-cap regulation, OECD remedy guidelines.
- Antitrust remedies: US DOJ/FTC remedy manuals, EC notice on remedies, CMA remedy guidance.

Visualizations

Remedy compliance timeline

A timeline visualization helps communicate key milestones, deadlines, and compliance events for complex remedy packages. This is particularly useful for trustee reports, agency presentations, and public communications.

```
source("../R/helpers.R")
library(dplyr)
library(ggplot2)
library(lubridate)

# Example remedy timeline from a merger case
# Replace with actual compliance events from trustee reports
remedy_events <- tibble::tribble(
  ~date,          ~event,
  ↪ ~category,
  "2021-03-15",    "Merger approved w/ conditions",
  ↪ "Decision",
  "2021-04-01",    "Trustee appointed",
  ↪ "Monitoring",
  "2021-07-01",    "Access API go-live",
  ↪ "Technical",
  "2021-09-30",    "Q1 compliance report",
  ↪ "Reporting",
  "2021-12-01",    "Data sharing portal launched",
  ↪ "Technical",
  "2021-12-31",    "Q2 compliance report",
  ↪ "Reporting",
  "2022-03-15",    "First annual review",
  ↪ "Review",
  "2022-06-30",    "Q3 compliance report",
  ↪ "Reporting",
```

```

"2022-09-01",      "Pricing parity audit",
  ↪ "Monitoring",
"2022-12-31",      "Q4 compliance report",
  ↪ "Reporting",
"2023-03-15",      "Second annual review",
  ↪ "Review",
"2023-06-01",      "Remedy modification hearing",
  ↪ "Decision",
"2024-03-15",      "Final compliance assessment",
  ↪ "Review",
"2024-06-30",      "Sunset date (remedy expires)",
  ↪ "Termination"
) |>
mutate(
  date = as.Date(date),
  category = factor(category,
    levels = c("Decision", "Technical",
      ↪ "Monitoring",
        "Reporting", "Review",
        ↪ "Termination"))
)

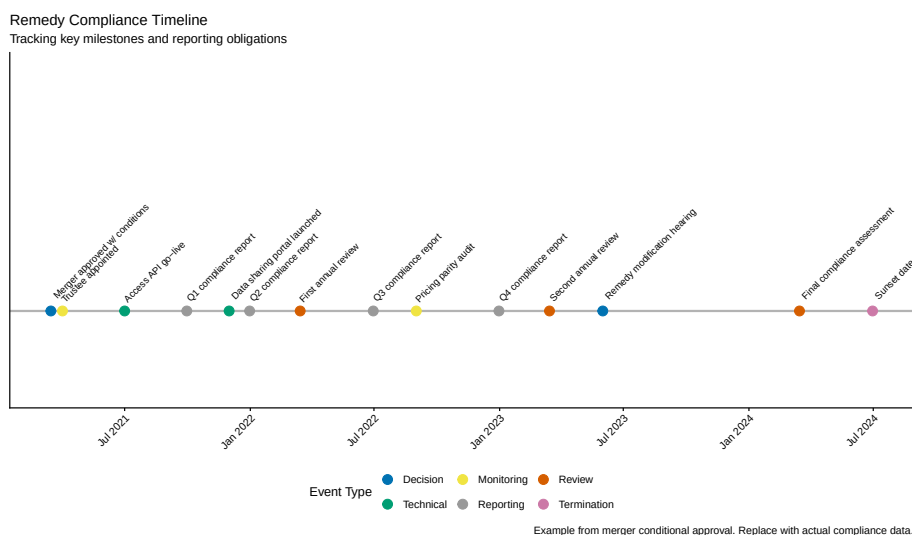
# Create timeline plot with categorical coloring
ggplot(remedy_events, aes(x = date, y = 0)) +
  # Baseline
  geom_hline(yintercept = 0, color = "gray70", linewidth = 1) +
  # Event points
  geom_point(aes(color = category), size = 4) +
  # Event labels
  geom_text(aes(label = event, angle = 45),
    hjust = -0.1, vjust = -0.5, size = 3) +
  # Category coloring
  scale_color_manual(
    values = c(
      "Decision" = "#0072B2",
      "Technical" = "#009E73",
      "Monitoring" = "#F0E442",
      "Reporting" = "#999999",
      "Review" = "#D55E00",
      "Termination" = "#CC79A7"
    )
  ) +
  scale_x_date(date_breaks = "6 months", date_labels = "%b %Y",
    expand = expansion(mult = c(0.05, 0.05))) +
  labs(
    title = "Remedy Compliance Timeline",

```

```

subtitle = "Tracking key milestones and reporting
  ↳ obligations",
x = NULL,
y = NULL,
color = "Event Type",
caption = "Example from merger conditional approval. Replace
  ↳ with actual compliance data."
) +
theme_classic(base_size = 12) +
theme(
  axis.text.y = element_blank(),
  axis.ticks.y = element_blank(),
  panel.grid = element_blank(),
  plot.title.position = "plot",
  axis.text.x = element_text(angle = 45, hjust = 1),
  legend.position = "bottom"
) +
coord_cartesian(ylim = c(-0.5, 1.5))

```



Example from merger conditional approval. Replace with actual compliance data.

```

# Summary table
cat("\nCompliance milestone summary:\n")

```

Compliance milestone summary:

```

remedy_summary <- remedy_events |>
  group_by(category) |>
  summarise(count = n(), .groups = "drop") |>

```

```
arrange(desc(count))
print(remedy_summary, n = Inf)
```

```
# A tibble: 6 x 2
  category    count
  <fct>      <int>
1 Reporting     4
2 Review        3
3 Decision      2
4 Technical     2
5 Monitoring    2
6 Termination   1
```

How to use this timeline: - **Decision events** (blue): Key regulatory or tribunal decisions establishing or modifying remedies. - **Technical milestones** (green): System launches, API deployments, portal go-lives. - **Monitoring events** (yellow): Audits, investigations, compliance checks. - **Reporting obligations** (gray): Regular quarterly or annual reports. - **Review points** (orange): Scheduled assessments where remedies may be modified or extended. - **Termination** (purple): Sunset date when behavioral remedies expire.

Practical applications: - Include in trustee reports to show progress against mandated milestones. - Present in annual compliance reviews to agency staff. - Use in public communications to demonstrate transparency. - Adapt for different remedy types: structural (divestitures), behavioral (access, pricing), or hybrid packages.

Replace with actual dates from: - Consent decrees and settlement agreements (DOJ/FTC, DG COMP, CMA) - Trustee reports and compliance dashboards - Agency monitoring databases - Tribunal orders (South African Competition Tribunal)

Enhanced timeline with swimlanes

For complex remedies involving multiple workstreams (technical, legal, operational), use a swimlane variant:

```
library(ggplot2)
library(dplyr)

# Add swimlane assignments
remedy_events_swim <- remedy_events |>
  mutate(
    swimlane = case_when(
      category %in% c("Decision", "Termination") ~
        "Legal/Regulatory",
      category %in% c("Technical") ~ "Technical Implementation",
```

```

    category %in% c("Monitoring", "Review") ~ "Compliance &
    ↪ Audit",
    category == "Reporting" ~ "Reporting & Documentation"
  ),
  swimlane = factor(swimlane,
                    levels = c("Legal/Regulatory",
                              "Technical Implementation",
                              "Compliance & Audit",
                              "Reporting & Documentation"))
)

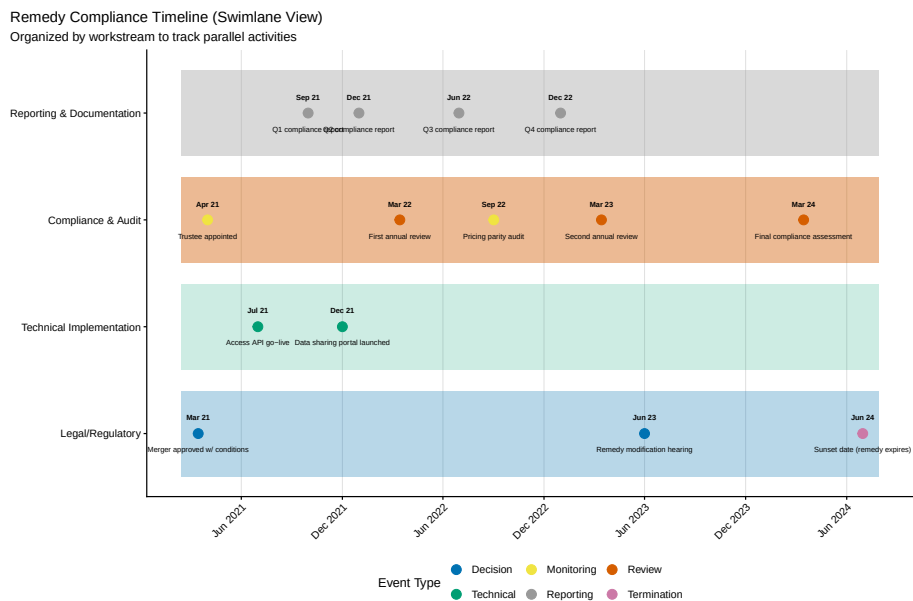
ggplot(remedy_events_swim, aes(x = date, y =
    ↪ as.numeric(swimlane))) +
  # Swimlane backgrounds
  geom_rect(aes(fill = swimlane),
            xmin = min(remedy_events_swim$date) - days(30),
            xmax = max(remedy_events_swim$date) + days(30),
            ymin = as.numeric(remedy_events_swim$swimlane) - 0.4,
            ymax = as.numeric(remedy_events_swim$swimlane) + 0.4,
            alpha = 0.1) +
  # Event points
  geom_point(aes(color = category), size = 4) +
  # Event labels
  geom_text(aes(label = format(date, "%b %y")),
            nudge_y = 0.15, size = 2.5, fontface = "bold") +
  geom_text(aes(label = event),
            nudge_y = -0.15, size = 2.5, hjust = 0.5) +
  scale_color_manual(
    values = c(
      "Decision" = "#0072B2",
      "Technical" = "#009E73",
      "Monitoring" = "#F0E442",
      "Reporting" = "#999999",
      "Review" = "#D55E00",
      "Termination" = "#CC79A7"
    )
  ) +
  scale_fill_manual(
    values = c(
      "Legal/Regulatory" = "#0072B2",
      "Technical Implementation" = "#009E73",
      "Compliance & Audit" = "#D55E00",
      "Reporting & Documentation" = "#999999"
    )
  ) +
  scale_y_continuous(

```

```

breaks = 1:4,
labels = levels(remedy_events_swim$swimlane)
) +
scale_x_date(date_breaks = "6 months", date_labels = "%b %Y") +
labs(
  title = "Remedy Compliance Timeline (Swimlane View)",
  subtitle = "Organized by workflow to track parallel
    ↪ activities",
  x = NULL,
  y = NULL,
  color = "Event Type",
  fill = "Workstream"
) +
theme_classic(base_size = 12) +
theme(
  plot.title.position = "plot",
  axis.text.x = element_text(angle = 45, hjust = 1),
  legend.position = "bottom",
  panel.grid.minor = element_blank(),
  panel.grid.major.x = element_line(color = "#e0e0e0",
    ↪ linewidth = 0.3)
) +
guides(fill = "none")

```



Swimlane benefits: - Separates legal, technical, and operational workstreams for clarity. - Shows dependencies and sequencing across different teams. - Useful

for program management and stakeholder coordination.

Data and visualization plan

- **Benchmarking scatter (Ch. 08 Visual 1):** Use regulator filings (Ofgem RIIO datasets, NERSA approved tariffs) or sanitized hospital benchmarking tables from public inquiries. If confidential, create synthetic replicas stored in `data/examples/benchmarking.csv`.
- **Remedy compliance timeline (Visual 2):** Build from trustee reports and milestone trackers from public sources.
- **Post-remedy diff-in-diff (Visual 3):** For telecom data, draw on public tariffs (ICASA reports) or TeleGeography archives; for energy, use Eskom/NERSA tariff series.

Document every dataset in `data/README.md` and note whether it can ship with the book (public) or must be replicated with synthetic values for open-source builds.

Looking ahead

Store remedy models, compliance timelines, and benchmarking outputs in `data/derived/remedies/` with detailed READMEs. When transitioning to the litigation chapter, reference which remedy types (structural, behavioral, hybrid) appeared most frequently in your jurisdiction's case law. Update the visualization tracker if you add new dashboard components (e.g., interactive Shiny apps for remedy monitoring) so future readers can replicate or extend the templates.

Digital Markets and Platforms

Learning goals

Digital markets combine traditional IO concepts with product design, governance, and behavioral nudges. Drawing on DMA/Digital Markets Unit briefings and recent enforcement actions, this chapter helps you:

- Analyze multi-sided participation, indirect network effects, and multi-homing incentives.
- Quantify self-preferencing, ranking changes, and default effects on user/merchant behavior.
- Evaluate data advantages, access restrictions, and algorithmic pricing risks.
- Integrate product requirement documents (PRDs), API policies, and telemetry with econometric evidence for global enforcement settings.

Expect to combine telemetry, clickstream data, partner contracts, and qualitative product documentation with econometric tests and simulation scaffolds.

Core topics

- Platform demand and multi-homing: participation elasticities, steering, matching efficiency.
- Self-preferencing and ranking: click-through impacts, A/B test interpretation, fairness metrics.
- Data advantages and economies of scope; switching costs, lock-in, and interoperability.
- Algorithmic pricing and tacit collusion risk.
- Workflow for platform cases: market definition, theory of harm, mechanism measurement, effects, and remedies (choice screens, access mandates, data portability, separation).

Platform workflow overview

1. **Define sides and metrics.** Identify user, merchant, advertiser, developer, courier, or publisher sides. Track MAU/DAU, GMV, multi-homing rates, engagement depth, and conversion metrics.
2. **Map fees and monetization.** Build ladders for commissions, subscriptions, ad loads, and data monetization; compute pass-through and elasticity. For two-sided market theory, see Rochet and Tirole (2003) and Armstrong (2006).
3. **Theory of harm.** Common mechanisms include self-preferencing, default tying, MFN/parity clauses, data leveraging, interoperability restrictions, algorithmic ranking biases, and algorithmic collusion.
4. **Mechanism measurement.** Quantify ranking impacts, default effects, fee changes, and data advantages; document multi-homing constraints and API access terms.
5. **Effects & remedies.** Measure partner churn, user switching, price/fee changes, and innovation impacts. Propose remedies (choice screens, fair-ranking reports, API access rules, data portability, structural separation) tied to measured harms.

Participation, multi-homing, and network effects

- **Participation curves:** Model each side's activity as a function of fees, quality, and participation on the other side. Use logit, linear, or Poisson models with cross-side terms (Evans and Schmalensee 2007; Hagiu and Wright 2015).
- **Multi-homing rates:** Estimate the share of users or merchants active on rival platforms. OIPMI surveys and EU DMA filings include methodologies you can reuse (Parker and Van Alstyne 2005).
- **Match quality:** Track conversion rates, completion times, or satisfaction metrics per platform side to show how conduct affects outcomes.
- **Elasticities:** Two-sided elasticities inform fee caps and remedy design (e.g., how merchant participation responds to lower commissions or improved ranking fairness).

Two-sided demand scaffold

```

library(dplyr)
library(fixest)

# Synthetic example; replace with platform telemetry
panel <- expand.grid(platform = c("Alpha","Beta"), period =
  ↪ 2018:2023) |>
  mutate(
    fees_users = runif(n(), 0, 4),
    fees_merchants = runif(n(), 15, 30),
    quality = runif(n(), 0.5, 1),
    users = exp(5 - 0.25 * fees_users + 0.4 * quality +
  ↪ rnorm(n(), 0, 0.2)),
    merchants = exp(4 - 0.12 * fees_merchants + 0.3 * users / 1e5
  ↪ + rnorm(n(), 0, 0.2))
  )

user_model <- feols(log(users) ~ fees_users + quality +
  ↪ log(merchants) | platform + period, data = panel)
merchant_model <- feols(log(merchants) ~ fees_merchants +
  ↪ log(users) | platform + period, data = panel)
summary(user_model)

```

```

OLS estimation, Dep. Var.: log(users)
Observations: 12
Fixed-effects: platform: 2, period: 6
Standard-errors: IID

```

	Estimate	Std. Error	t value	Pr(> t)
fees_users	-0.377177	0.249006	-1.514734	0.26906
quality	0.619485	1.132894	0.546817	0.63936
log(merchants)	-0.171293	0.275499	-0.621756	0.59753

```

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.088395      Adj. R2: 0.550772
                Within R2: 0.801598

```

```
summary(merchant_model)
```

```

OLS estimation, Dep. Var.: log(merchants)
Observations: 12
Fixed-effects: platform: 2, period: 6
Standard-errors: IID

```

	Estimate	Std. Error	t value	Pr(> t)
fees_merchants	-0.119717	0.010651	-11.240171	0.0015098 **
log(users)	0.136790	0.199063	0.687171	0.5413311

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.056535      Adj. R2: 0.96704
                  Within R2: 0.985112
```

Replace the synthetic `panel` with sanitized telemetry (monthly active users/merchants) or aggregated OIPMI data stored in `data/examples/digital-two-sided.csv`. Document the estimated cross-side elasticities in your memo.

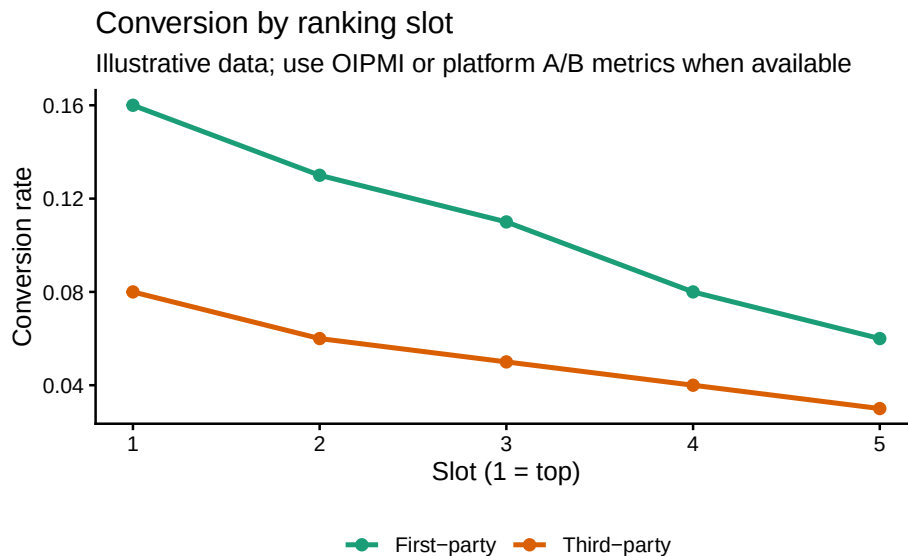
Ranking, self-preferencing, and defaults

Ranking impact visualization

```
library(dplyr)
library(tidyr)
library(ggplot2)

ranking <- tibble::tribble(
  ~slot, ~first_party_ctr, ~third_party_ctr,
  1, 0.16, 0.08,
  2, 0.13, 0.06,
  3, 0.11, 0.05,
  4, 0.08, 0.04,
  5, 0.06, 0.03
) |>
  pivot_longer(~slot, names_to = "listing", values_to = "ctr")

ggplot(ranking, aes(slot, ctr, color = listing)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  scale_color_manual(values = c("#1b9e77", "#d95f02"), labels =
    ↪ c("First-party", "Third-party")) +
  labs(
    title = "Conversion by ranking slot",
    subtitle = "Illustrative data; use OIPMI or platform A/B
    ↪ metrics when available",
    x = "Slot (1 = top)",
    y = "Conversion rate",
    color = NULL
  ) +
  theme_classic(base_size = 11) +
  theme(legend.position = "bottom")
```



Populate the tibble with actual ranking data (e.g., OIPMI buy-box audits, CMA Amazon Marketplace analysis) or maintain sanitized values in `data/examples/digital-ranking.csv`. The CMA's market studies are publicly available at gov.uk/cma.

Default-choice event study scaffold

```
library(fixest)

# df requires: device_id, week, default_event (event date),
#             share_rival
# event_model <- feols(share_rival ~ i(week, default_event, ref =
#             -1) | device_id + week, data = df)
# etable(event_model)
```

Use telemetry from choice screens (EU Android choice screen, DMA compliance reports) or partner-provided logs. If live data are unavailable, build synthetic event panels mirroring DMA summary stats (store in `data/examples/digital-default.csv`).

Southern African digital enforcement snapshots

- **Online Intermediation Platforms Market Inquiry (OIPMI, 2020–2023).** Transaction-level data from marketplaces, app stores, delivery apps, and travel platforms revealed Google Search's >90% share, 25–30% delivery commissions, and buy-box algorithms favoring first-party listings. Remedies require fair-ranking reports, SME fee caps,

and anti-steering rights inside aggregator apps.

- **Media and Digital Platforms Inquiry (2023–ongoing).** Impression-level logs from Meta, Google, and local publishers show Google capturing ~65% of programmatic display spend, while local publishers net <30% after fees, framing bargaining remedies for news publishers.
- **GovChat/Meta interim relief (2021).** WhatsApp Business API telemetry demonstrated GovChat’s public-service reach (>8 million users) and negligible incremental cost; interim relief preserved access while exclusion claims proceeded, blending platform data with public-interest considerations.

Methodologies

i Method box

- Platform fee/price pass-through across sides.
- Event studies on ranking/default changes; DMA/DMA-style interventions.
- Two-sided logit or bargaining sketches to test fee caps and interoperability remedies.

i Qualitative evidence

- Product requirement documents, experiment summaries, and governance memos.
- Partner contracts (parity/MFN, exclusivity), API access terms, data-sharing policies.
- User research/surveys capturing multi-homing, switching costs, and friction sources.

i Citations and comparative note

- US cases (Epic v. Apple, DOJ Google Search and Ad Tech, FTC Amazon Marketplace), EU/DMA (Google Shopping/Android, DMA obligations), UK (CMA digital market studies), KFTC/JFTC app store cases, SAMR digital enforcement.

- Empirical studies on defaults, ranking effects, and multi-homing (e.g., Luca et al., 2016, “Does Google Content Degrade Google Search?,” Harvard Business School Working Paper; Edelman & Lai, 2016, “Design of Search Engine Services,” *Journal of Economics & Management Strategy* 25(1): 187-218).
- Flag differences in parity/MFN treatment and remedy preferences (EU fair-ranking obligations vs. US injunctive relief).

Visualizations and data sourcing

- **Default impact chart (Roadmap 9.1):** Source DMA compliance reports (available at ec.europa.eu/dma), CMA choice-screen experiments, or sanitized telemetry. If unavailable, simulate from DMA summary stats.
- **Multi-homing Sankey (Roadmap 9.2):** Use OIPMI survey microdata or replicate with local surveys; store in `data/examples/digital-sankey.csv` until live data arrive.
- **Ranking change event analysis (Roadmap 9.3):** Combine OIPMI buy-box audits or platform-provided logs with the ranking scaffold above.
- **Fee ladder diagrams:** Pull commission schedules from OIPMI appendices or global public filings (App Store, Play Store, Amazon Marketplace).

Document each dataset in `data/README.md` with provenance, confidentiality notes, and instructions for swapping synthetic vs. production data. When we “fill with real data,” replace the synthetic CSVs, re-render figures, and cache sanitized outputs for publication builds.

Southern African digital enforcement snapshots

- **Online Intermediation Platforms Market Inquiry (OIPMI, 2020–2023).** The Competition Commission collected transaction-level data from e-commerce marketplaces, app stores, food delivery apps, and travel platforms. Findings highlighted Google Search’s >90% share of general search queries, food-delivery commissions clustering at 25–30%, and marketplace buy-box algorithms that favored first-party listings (e.g., Takealot Retail) over third-party sellers with identical fulfillment metrics. Final remedies require fair-ranking reports, caps on marketplace storage/fulfillment fees for SMEs, and anti-steering relief that lets restaurants promote direct-order channels inside aggregator apps.
- **Advertising and digital news media (Media and Digital Platforms Market Inquiry, 2023–ongoing).** Building on OIPMI data,

the Commission is ingesting impression-level logs from Meta, Google, and local publishers to quantify revenue imbalances in display advertising. The interim statement reports that Google commands roughly 65% of display ad spend routed through programmatic channels, while local publishers face average take-rates below 30% after demand-side/platform fees—data points that frame prospective bargaining remedies for news publishers across Southern Africa.

- **GovChat/Meta interim relief (Competition Commission v. Meta Platforms, 2021).** The Commission used WhatsApp Business API telemetry to demonstrate GovChat’s role in reaching 8+ million South African users for public-service messaging, and it showed how Meta’s proposed off-boarding would foreclose a civic-use case despite negligible incremental infrastructure cost. Interim relief preserved access while the Tribunal weighed the exclusion case, offering a blueprint for combining platform-side telemetry with public-interest arguments in developing markets.

i Method box

- Platform price/fee pass-through; elasticity across sides.
- Event studies on default changes or ranking tweaks.
- Simple two-sided logit/bargaining sketches; diversion across sides.

i Qualitative evidence

- Product requirement docs and experiments; governance and content policies.
- Partner contracts (parity/MFN, exclusivity), API access terms.
- User research surveys on multi-homing and switching.

i Citations and comparative note

- Cite platform cases and regulations: US (e.g., Epic v. Apple, DOJ Google Search/Ad Tech), EU (Google Shopping/Android, DMA), UK (CMA market investigations), and other jurisdictions (e.g., Korea KFTC app stores where relevant).
- Reference empirical studies on defaults and ranking effects, and any A/B or natural experiment evidence from platforms.
- Note differences in treatment of MFNs/parity and self-preferencing across jurisdictions.

Enhanced Visualizations

Multi-homing Sankey diagram

Multi-homing patterns reveal competitive constraints and switching costs. A Sankey diagram shows how users flow across platforms, helping visualize whether platforms compete head-to-head or serve distinct niches.

```
library(dplyr)
library(ggplot2)
library(ggalluvial)

# Simulated multi-homing survey data
# Replace with OIPMI survey data or DMA compliance reports
set.seed(234)
n_users <- 1000

multihoming <- tibble(
  user_id = 1:n_users,
  platform_a = sample(c("Active", "Inactive"), n_users, replace =
    ↪ TRUE,
                        prob = c(0.7, 0.3)),
  platform_b = sample(c("Active", "Inactive"), n_users, replace =
    ↪ TRUE,
                        prob = c(0.5, 0.5)),
  platform_c = sample(c("Active", "Inactive"), n_users, replace =
    ↪ TRUE,
                        prob = c(0.3, 0.7))
) |>
mutate(
  pattern = case_when(
    platform_a == "Active" & platform_b == "Active" &
    ↪ platform_c == "Active" ~ "All three",
    platform_a == "Active" & platform_b == "Active" ~ "A + B",
    platform_a == "Active" & platform_c == "Active" ~ "A + C",
    platform_b == "Active" & platform_c == "Active" ~ "B + C",
    platform_a == "Active" ~ "A only",
    platform_b == "Active" ~ "B only",
    platform_c == "Active" ~ "C only",
    TRUE ~ "None"
  )
)

# Aggregate flows
flows <- multihoming |>
  count(pattern) |>
  mutate(
```

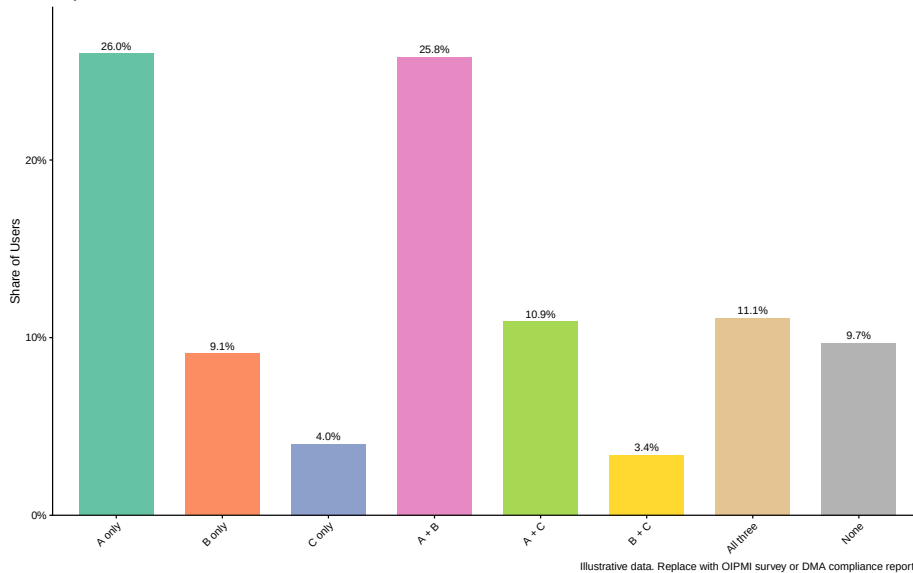
```
pct = n / sum(n),
pattern = factor(pattern,
                  levels = c("A only", "B only", "C only",
                             "A + B", "A + C", "B + C", "All
                             ↪ three", "None"))

) |>
  arrange(pattern)

# Bar chart showing multi-homing patterns
p1 <- ggplot(flows, aes(x = pattern, y = pct, fill = pattern)) +
  geom_col(width = 0.7) +
  geom_text(aes(label = scales::percent(pct, accuracy = 0.1)),
            vjust = -0.5, size = 3.5) +
  scale_y_continuous(labels = scales::percent_format(),
                     expand = expansion(mult = c(0, 0.1))) +
  scale_fill_brewer(palette = "Set2") +
  labs(
    title = "Multi-homing Patterns Across Platforms",
    subtitle = "User activity across Platform A, B, and C",
    x = NULL,
    y = "Share of Users",
    caption = "Illustrative data. Replace with OIPMI survey or
    ↪ DMA compliance reports."
  ) +
  theme_classic(base_size = 12) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "none",
    plot.title.position = "plot"
  )

p1
```

Multi-homing Patterns Across Platforms
User activity across Platform A, B, and C



```
# Summary statistics
cat("\nMulti-homing summary:\n")
```

Multi-homing summary:

```
cat(paste0("Single-homing (one platform only): ",
           scales::percent(sum(flows$pct[flows$pattern %in%
                                c("A only", "B only", "C only")])),
    accuracy = 0.1), "\n"))
```

Single-homing (one platform only): 39.1%

```
cat(paste0("Multi-homing (2+ platforms): ",
           scales::percent(sum(flows$pct[grepl("\\+|three",
           ↪ flows$pattern)]),
           accuracy = 0.1), "\n"))
```

Multi-homing (2+ platforms): 51.2%

Interpretation: - **High single-homing:** Suggests strong lock-in, switching costs, or network effects. Platforms may have market power. - **High multi-homing:** Indicates competitive pressure; users actively compare options. But check if multi-homing is symmetric or if one platform dominates usage. - **Asymmetric patterns:** If Platform A appears in most combinations, it may be a “must-have” platform with gatekeeping power.

Data sources: - OIPMI survey microdata (South Africa) - DMA compliance re-

ports (EU) - CMA market studies (UK) - Platform-provided telemetry (subject to confidentiality)

Choice screen / default effect visualization

Quantify how defaults affect market shares using difference-in-differences or event studies around choice screen implementations.

```
library(dplyr)
library(ggplot2)
library(patchwork)

# Simulated data: Android choice screen impact (EU DMA context)
# Replace with actual DMA compliance data or CMA experiments
set.seed(345)
weeks <- -24:24
countries <- c("Control (no choice screen)", "Treatment (choice
  ↪ screen)")

default_data <- expand_grid(
  week = weeks,
  country = countries
) |>
  mutate(
    # Pre-intervention: both countries similar
    # Post-intervention: treatment shows rival gain
    rival_share = case_when(
      country == "Control (no choice screen)" ~ 0.05 + 0.0005 *
    ↪ week + rnorm(n(), 0, 0.01),
      week < 0 ~ 0.05 + 0.0005 * week + rnorm(n(), 0, 0.01),
      TRUE ~ 0.05 + 0.0005 * week + 0.08 + rnorm(n(), 0, 0.01)
    ),
    rival_share = pmax(0.01, pmin(0.25, rival_share))
  )

# Time series plot
p1 <- ggplot(default_data, aes(x = week, y = rival_share,
  color = country, linetype =
    ↪ country)) +
  geom_vline(xintercept = 0, linetype = "dashed", color =
    ↪ "gray40",
    linewidth = 1) +
  stat_summary(fun = mean, geom = "line", linewidth = 1.2) +
  annotate("text", x = 0, y = 0.22, label = "Choice
    ↪ screen\nintroduced",
    hjust = -0.1, size = 3.5, fontface = "italic") +
  scale_color_manual(values = c("Control (no choice screen)" =
    ↪ "#999999",
```

```

    "Treatment (choice screen)" =
      ↪ "#0072B2")) +
scale_linetype_manual(values = c("Control (no choice screen)" =
  ↪ "dashed",
    "Treatment (choice screen)" =
      ↪ "solid")) +
scale_y_continuous(labels = scales::percent_format()) +
labs(
  title = "Impact of Choice Screen on Rival Search Engine
    ↪ Share",
  subtitle = "Difference-in-Differences: Treatment vs. Control
    ↪ Countries",
  x = "Weeks relative to choice screen introduction",
  y = "Rival search engine share",
  color = NULL,
  linetype = NULL
) +
theme_classic(base_size = 12) +
theme(
  legend.position = "bottom",
  plot.title.position = "plot"
)

# Coefficient plot (event study coefficients)
# Simulated DiD estimates
did_coefs <- tibble(
  week = weeks[weeks != -1] # reference period
) |>
mutate(
  estimate = 0.08 * as.numeric(week >= 0) + rnorm(n(), 0,
    ↪ 0.01),
  ci_lower = estimate - 1.96 * 0.01,
  ci_upper = estimate + 1.96 * 0.01
)

p2 <- ggplot(did_coefs, aes(x = week, y = estimate)) +
  geom_hline(yintercept = 0, linetype = "solid", color =
    ↪ "gray40") +
  geom_vline(xintercept = 0, linetype = "dashed", color =
    ↪ "gray40",
    linewidth = 1) +
  geom_ribbon(aes(ymin = ci_lower, ymax = ci_upper),
    alpha = 0.2, fill = "#0072B2") +
  geom_line(color = "#0072B2", linewidth = 1) +
  geom_point(color = "#0072B2", size = 2) +
  scale_y_continuous(labels = scales::percent_format()) +

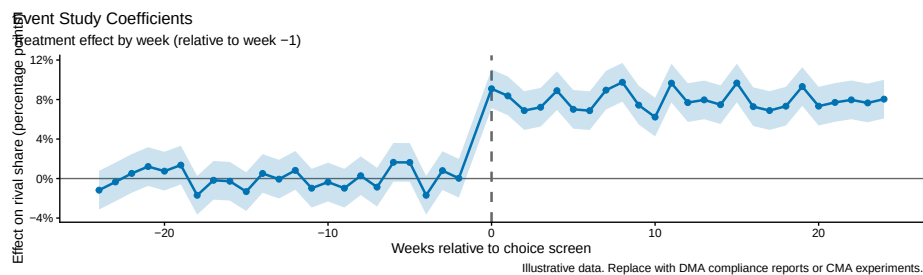
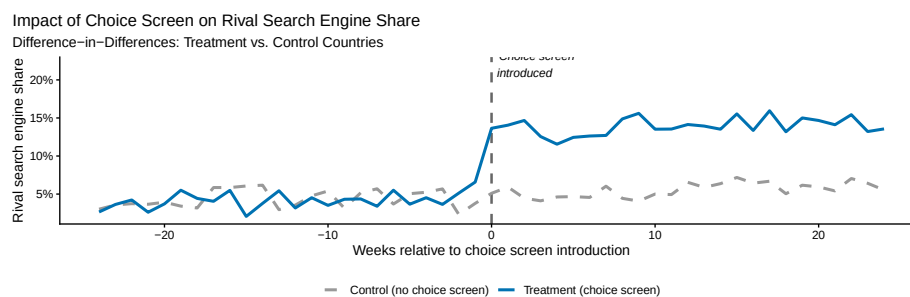
```

```

labs(
  title = "Event Study Coefficients",
  subtitle = "Treatment effect by week (relative to week -1)",
  x = "Weeks relative to choice screen",
  y = "Effect on rival share (percentage points)",
  caption = "Illustrative data. Replace with DMA compliance
    ↪ reports or CMA experiments."
) +
theme_classic(base_size = 12) +
theme(plot.title.position = "plot")

# Combined plot
p1 / p2

```



```

# Summary statistics
pre_treatment <- mean(default_data$rival_share[
  default_data$country == "Treatment (choice screen)" &
  ↪ default_data$week < 0])
post_treatment <- mean(default_data$rival_share[
  default_data$country == "Treatment (choice screen)" &
  ↪ default_data$week >= 0])
effect <- post_treatment - pre_treatment

cat("\nChoice screen impact summary:\n")

```

Choice screen impact summary:

```
cat(paste0("Pre-intervention rival share: ",
          scales::percent(pre_treatment, accuracy = 0.1), "\n"))
```

Pre-intervention rival share: 4.2%

```
cat(paste0("Post-intervention rival share: ",
          scales::percent(post_treatment, accuracy = 0.1), "\n"))
```

Post-intervention rival share: 13.9%

```
cat(paste0("Estimated treatment effect: ",
          scales::percent(effect, accuracy = 0.1),
          " (", round(effect / pre_treatment * 100, 0), "%
          ↵ increase)\n"))
```

Estimated treatment effect: 9.7% (234% increase)

How to use this analysis: - **Pre-trends:** Check parallel trends before intervention to validate DiD assumptions. - **Treatment effect:** Quantify how much rival share increased after choice screen. - **Heterogeneity:** Analyze by device type, user demographics, or country. - **Persistence:** Track whether effects sustain over time or decay.

DMA context: The EU Digital Markets Act requires designated gatekeepers to offer choice screens for browsers, search engines, and other services. This analysis framework can evaluate compliance and effectiveness.

Platform fee structure visualization

Commission and fee structures are critical evidence in platform cases. Show how fees vary by seller tier, product category, or fulfillment method.

```
library(dplyr)
library(ggplot2)
library(tidyr)

# Fee structures from major platforms (public sources)
# App Store, Play Store, Amazon Marketplace, food delivery
fees <- tibble::tribble(
  ~platform,      ~category,      ~base_commission,
  ↵ ~fulfillment_fee, ~payment_fee,
  "App Store",    "Apps",          0.30,
  ↵ 0.00,          0.00,
  "App Store",    "Apps (< $1M)",  0.15,
  ↵ 0.00,          0.00,
  "Play Store",   "Apps",          0.30,
  ↵ 0.00,          0.00,
  "Play Store",   "Apps (< $1M)",  0.15,
  ↵ 0.00,          0.00,
```

```

"Amazon Marketplace", "FBM (self-fulfill)", 0.15,
  ↪ 0.00,          0.03,
"Amazon Marketplace", "FBA (Amazon fulfill)", 0.15,
  ↪ 0.20,          0.03,
"Food Delivery A",    "Restaurant",        0.28,
  ↪ 0.00,          0.02,
"Food Delivery B",    "Restaurant",        0.32,
  ↪ 0.00,          0.02
) |>
mutate(
  total_fee = base_commission + fulfillment_fee + payment_fee,
  total_fee_pct = scales::percent(total_fee, accuracy = 0.1)
)

# Stacked bar chart
fees_long <- fees |>
pivot_longer(cols = c(base_commission, fulfillment_fee,
  ↪ payment_fee),
  names_to = "fee_type", values_to = "rate") |>
mutate(
  fee_type = factor(fee_type,
    levels = c("payment_fee", "fulfillment_fee",
  ↪ "base_commission"),
    labels = c("Payment Processing",
  ↪ "Fulfillment", "Base Commission"))
)

ggplot(fees_long, aes(x = paste(platform, category, sep = "\n"),
  y = rate, fill = fee_type)) +
  geom_col(width = 0.7) +
  geom_text(data = fees,
    aes(x = paste(platform, category, sep = "\n"),
      y = total_fee, label = total_fee_pct, fill =
    ↪ NULL),
    vjust = -0.5, size = 3.5, fontface = "bold") +
  scale_y_continuous(labels = scales::percent_format(),
    expand = expansion(mult = c(0, 0.1))) +
  scale_fill_manual(
    values = c("Base Commission" = "#D55E00",
      "Fulfillment" = "#0072B2",
      "Payment Processing" = "#009E73")
  ) +
  labs(
    title = "Platform Fee Structures by Category",
    subtitle = "Total take rate shown at top of each bar",
    x = NULL,

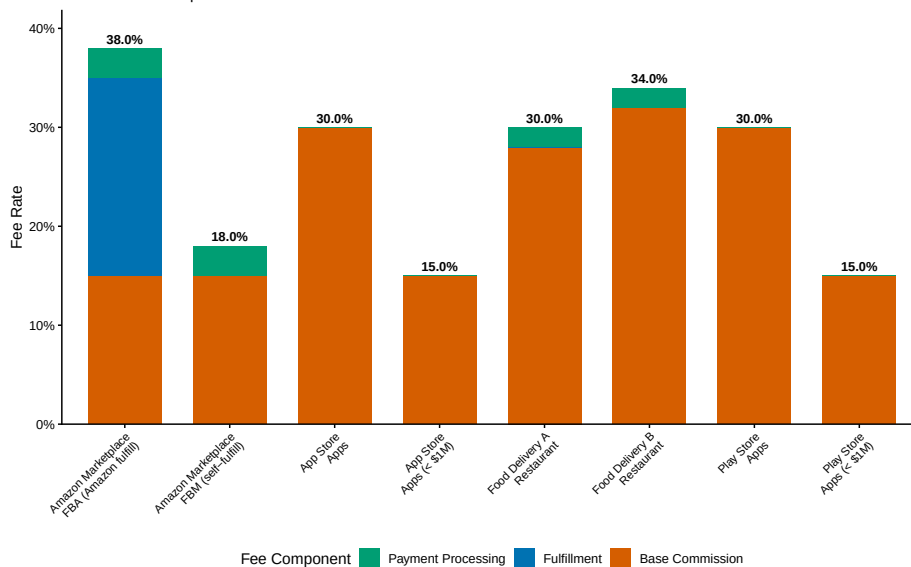
```

```

y = "Fee Rate",
fill = "Fee Component",
caption = "Data from public platform documentation (2024).
  ↳ FBM = Fulfilled by Merchant; FBA = Fulfilled by Amazon."
) +
theme_classic(base_size = 12) +
theme(
  axis.text.x = element_text(angle = 45, hjust = 1, size = 9),
  legend.position = "bottom",
  plot.title.position = "plot"
)

```

Platform Fee Structures by Category
Total take rate shown at top of each bar



Data from public platform documentation (2024). FBM = Fulfilled by Merchant; FBA = Fulfilled by Amazon.

```

# Summary table
cat("\nFee structure summary:\n")

```

Fee structure summary:

```

fees |>
  select(platform, category, total_fee) |>
  mutate(total_fee = scales::percent(total_fee, accuracy = 0.1))
  ↳ |>
  print(n = Inf)

```

```
# A tibble: 8 x 3
```

platform	category	total_fee
<chr>	<chr>	<chr>
1 App Store	Apps	30.0%
2 App Store	Apps (< \$1M)	15.0%
3 Play Store	Apps	30.0%
4 Play Store	Apps (< \$1M)	15.0%
5 Amazon Marketplace	FBM (self-fulfill)	18.0%
6 Amazon Marketplace	FBA (Amazon fulfill)	38.0%
7 Food Delivery A	Restaurant	30.0%
8 Food Delivery B	Restaurant	34.0%

Key insights: - **App stores:** 30% standard (15% for small developers under \$1M revenue). - **E-commerce marketplaces:** 15-18% base + 15-25% fulfillment (if using platform logistics). - **Food delivery:** 28-32% commission + 2-3% payment processing. - **Bundling:** Platforms often bundle services (payment, fulfillment, advertising) making switching costly.

Competitive assessment: - Compare fees across platforms to evaluate competitive pressure. - Assess whether high fees reflect service value or market power. - Evaluate whether fee structures create barriers to multi-homing.

Market share evolution for digital platforms

Track platform market share over time to identify tipping points and competitive dynamics.

```
library(dplyr)
library(ggplot2)
library(lubridate)

# Market share evolution from public sources and regulatory
  ↳ filings
# Illustrative data for search engines, social media, or
  ↳ e-commerce
market_evolution <- tibble::tribble(
  ~date,      ~platform,      ~share,
  "2015-01-01", "Platform A", 0.88,
  "2016-01-01", "Platform A", 0.90,
  "2017-01-01", "Platform A", 0.91,
  "2018-01-01", "Platform A", 0.92,
  "2019-01-01", "Platform A", 0.91,
  "2020-01-01", "Platform A", 0.90,
  "2021-01-01", "Platform A", 0.89,
  "2022-01-01", "Platform A", 0.88,
  "2023-01-01", "Platform A", 0.87,
  "2015-01-01", "Platform B", 0.08,
  "2016-01-01", "Platform B", 0.07,
```

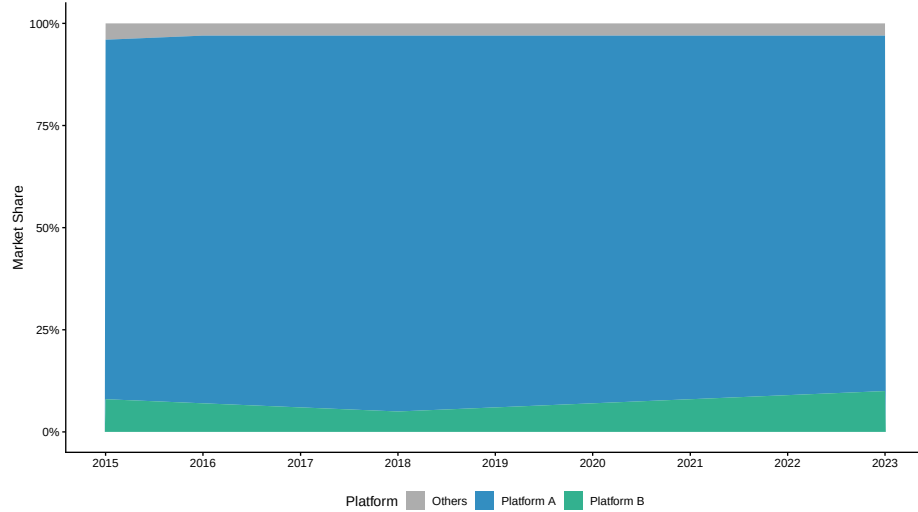
```

"2017-01-01", "Platform B", 0.06,
"2018-01-01", "Platform B", 0.05,
"2019-01-01", "Platform B", 0.06,
"2020-01-01", "Platform B", 0.07,
"2021-01-01", "Platform B", 0.08,
"2022-01-01", "Platform B", 0.09,
"2023-01-01", "Platform B", 0.10,
"2015-01-01", "Others", 0.04,
"2016-01-01", "Others", 0.03,
"2017-01-01", "Others", 0.03,
"2018-01-01", "Others", 0.03,
"2019-01-01", "Others", 0.03,
"2020-01-01", "Others", 0.03,
"2021-01-01", "Others", 0.03,
"2022-01-01", "Others", 0.03,
"2023-01-01", "Others", 0.03
) |>
  mutate(date = as.Date(date))

# Area chart
ggplot(market_evolution, aes(x = date, y = share, fill =
  ↪ platform)) +
  geom_area(alpha = 0.8) +
  scale_y_continuous(labels = scales::percent_format()) +
  scale_x_date(date_breaks = "1 year", date_labels = "%Y") +
  scale_fill_manual(
    values = c("Platform A" = "#0072B2",
               "Platform B" = "#009E73",
               "Others" = "#999999")
  ) +
  labs(
    title = "Digital Platform Market Share Evolution",
    subtitle = "Example: Search engine market shares 2015-2023",
    x = NULL,
    y = "Market Share",
    fill = "Platform",
    caption = "Illustrative data. Replace with StatCounter,
    ↪ SimilarWeb, or regulatory filings."
  ) +
  theme_classic(base_size = 12) +
  theme(
    legend.position = "bottom",
    plot.title.position = "plot"
  )

```

Digital Platform Market Share Evolution
 Example: Search engine market shares 2015–2023



Illustrative data. Replace with StatCounter, SimilarWeb, or regulatory filings.

```
# Summary statistics
cat("\nMarket concentration trends:\n")
```

Market concentration trends:

```
cat(paste0("2015 HHI: ", round(sum((market_evolution$share[
  market_evolution$date == "2015-01-01"] * 100)^2), 0), "\n"))
```

2015 HHI: 7824

```
cat(paste0("2023 HHI: ", round(sum((market_evolution$share[
  market_evolution$date == "2023-01-01"] * 100)^2), 0), "\n"))
```

2023 HHI: 7678

```
cat(paste0("\nPlatform A share trend: ",
  scales::percent(market_evolution$share[
    market_evolution$date == "2015-01-01" &
    market_evolution$platform == "Platform A"], accuracy
↵ = 0.1),
  " (2015) → ",
  scales::percent(market_evolution$share[
    market_evolution$date == "2023-01-01" &
    market_evolution$platform == "Platform A"], accuracy
↵ = 0.1),
  " (2023)\n"))
```

Platform A share trend: 88.0% (2015) → 87.0

(2023)

Data sources: - **StatCounter:** Global browser, OS, search engine stats - **SimilarWeb:** Website and app traffic data - **Sensor Tower / App Annie:** Mobile app market intelligence - **Regulatory filings:** EC, CMA, FTC, DMA compliance reports - **OIPMI:** South African platform market shares

Use cases: - Demonstrate sustained dominance or competitive entry - Identify tipping points or competitive interventions - Support market definition and dominance assessments

Looking ahead

Store all digital platform analyses in `data/derived/digital/` with clear provenance notes. When DMA compliance data become available, replace synthetic Sankey flows and choice-screen impacts with actual telemetry. Document API access restrictions and data-sharing agreements in `data/README.md`. Cross-reference these visualizations with Chapter 12 (litigation practice) for expert report templates and Chapter 13 (empirical appendix) for diagnostic checklists.

Antitrust and Labor Markets

Learning goals

Labor cases now sit alongside mergers and cartels in agency priorities. This chapter blends empirical tools from recent academic research on labor market power with practical guidance on using BLS and Census data. You will learn to:

- Measure labor market power (concentration, labor supply elasticity, pass-through).
- Evaluate no-poach, wage-fixing, and noncompete agreements using econometrics and qualitative evidence.
- Integrate matched employer-employee data, vacancy data, and worker narratives.
- Design remedies (noncompete bans, pay-transparency orders, notice requirements) and evaluate their effects.

Core topics

- Labor concentration (HHI, Herfindahl-Hirschman Index for labor markets) and labor supply elasticity estimates.
- Event studies and diff-in-diff around policy shocks (noncompete bans, wage disclosure laws, franchise no-poach settlements).
- Wage-posting vs. realized wage analysis; vacancy scraping; platform data for gig markets.
- Mobility, retention, and heterogeneity by occupation, region, and worker demographics.
- Qualitative evidence from HR documents, franchise agreements, and worker testimony.

i Method box

Monopsony elasticities: Estimate labor supply elasticity using quits, recruitment response, or equilibrium wage-setting models (e.g., inverse labor supply).

Policy diff-in-diff: Evaluate wage or mobility effects after noncompete bans or franchise no-poach settlements.

Synthetic control: Single-state reforms (e.g., Washington franchise no-poach settlement) can be assessed via synthetic control or staggered diff-in-diff.

i Qualitative evidence

HR policies & agreements: Franchise agreements, noncompete clauses, wage-fixing communications.

Worker interviews/surveys: Capture outside options, relocation constraints, switching costs, discrimination concerns.

Internal memos: Evidence of wage suppression strategy, benchmarking against collusive agreements, or fear of poaching.

i Citations and comparative note

Reference US DOJ/FTC criminal wage-fixing cases, state AG settlements, the FTC's proposed noncompete ban (see [ftc.gov](https://www.ftc.gov)), and international enforcement (e.g., Portugal's wage-fixing fines, Japan's OTA cases). Cite key empirical papers from recent labor economics literature. Highlight differences where EU/UK enforcement is nascent but evolving (e.g., CMA's focus on noncompetes shorter than 3 months, EC guidelines on HR collusion).

Measuring labor market power

Concentration indices

Compute HHI or concentration ratios for labor markets defined by occupation $\times$ geography (e.g., SOC $\times$ commuting zone). Use BLS QCEW, Census LODES/LEHD Origin-Destination Employment Statistics, or Stats SA labour force microdata. For empirical evidence on labor market concentration, see Azar, Marinescu, and Steinbaum (2020) and Benmelech, Bergman, and Kim (2020).

```
library(dplyr)

employment <- tibble::tribble(
```

```

~employer, ~occupation, ~region, ~employment,
"FirmA", "RN", "Gauteng", 1200,
"FirmB", "RN", "Gauteng", 800,
"FirmC", "RN", "Gauteng", 500,
"FirmD", "RN", "Gauteng", 300
)

hhi <- employment |>
  group_by(occupation, region) |>
  mutate(share = employment / sum(employment)) |>
  summarise(hhi = sum((share * 100)^2))
hhi

```

```

# A tibble: 1 x 3
# Groups:   occupation [1]
  occupation region    hhi
  <chr>      <chr>  <dbl>
1 RN        Gauteng 3087.

```

Replace synthetic data with QCEW/LEHD or Stats SA microdata aggregated to protect confidentiality. Document any weighting or commuting zone definitions in `data/README.md`.

Labor supply elasticity

Estimate inverse labor supply using panel data on wages and employment:

```

library(fixest)

# panel columns: firm, occupation, period, wage, employment,
#   ↪ controls
# elasticity_model <- feols(log(wage) ~ log(employment) +
#   ↪ controls | firm + occupation + period, data = panel)
# summary(elasticity_model)

```

Use matched employer-employee data (LEHD, SSA, UI wage records) or firm-level HR exports. Elasticities below 2–3 indicate meaningful monopsony power (Manning 2003; O. C. Ashenfelter, Farber, and Ransom 2010). For policy implications, see Naidu, Posner, and Weyl (2018).

Noncompetes, no-poach, and wage fixing

Policy shock diff-in-diff

```

library(dplyr)
library(fixest)

```

```

panel <- expand_grid(state = c("WA", "OR", "ID", "MT"), year =
  ↪ 2014:2022) |>
  mutate(
    treated = if_else(state == "WA" & year >= 2019, 1, 0),
    wage = 25 + 0.5 * (year - 2014) - 1.2 * treated + rnorm(n()),
    ↪ 0, 0.5),
    controls = runif(n())
  )

did_model <- feols(wage ~ treated | state + year, data = panel)
summary(did_model)

```

```

OLS estimation, Dep. Var.: wage
Observations: 36
Fixed-effects: state: 4, year: 9
Standard-errors: IID
      Estimate Std. Error  t value Pr(>|t|)
treated -0.817973   0.361504 -2.26269 0.033407 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.373036      Adj. R2: 0.878382
                Within R2: 0.18207

```

Substitute actual wage data (BLS Occupational Employment and Wage Statistics, CPS microdata, Stats SA QLFS) when available. Additional controls might include occupation mix, unemployment rates, or cost-of-living indices.

Franchise no-poach analysis

- **Contract review:** Extract no-poach clauses, duration, and scope from franchise agreements (fast-food, fitness, healthcare).
- **Transaction data:** Use payroll or scheduling data to track cross-franchise transfers before/after clause removal.
- **Event study:** Evaluate hiring and wage patterns after the no-poach ban (Washington fast-food settlements, South African retail franchise commitments). For empirical evidence on franchise no-poach agreements, see Krueger and Ashenfelter (2018). See also U.S. Department of Justice and Federal Trade Commission (2016) for enforcement guidance.

Wage posting vs. realized wages

Combine vacancy data (Indeed, Burning Glass, Pnet) with payroll data to evaluate pass-through and wage compression:

- **Posting premium vs. realized wage:** Compare posted wages to actual accepted wages; measure dispersion and adjustments post-policy.
- **Vacancy response:** Study vacancy duration and volume changes after noncompete bans or wage transparency laws.
- **Platform/gig economy:** Leverage public APIs (Uber, Deliveroo) or FOIA responses for driver payout distributions.

Mobility and retention analytics

- **Mobility matrices:** Use LEHD/LODES or internal HR data to compute transition probabilities between firms/regions.
- **Retention KPIs:** Track tenure, quit rates, and rehire rates; evaluate changes post-remedy.
- **Heterogeneity:** Segment results by occupation, gender, race, and visa status to show differential effects and public-interest considerations.

Southern African enforcement snapshots

- **Healthcare wage-fixing (Competition Commission v. Netcare & Mediclinic investigations).** HR data and negotiation emails showed coordinated caps on agency nurse rates during pandemic surges; the Commission paired payroll analysis with worker interviews to craft remedial guidelines.
- **Pick n Pay/Shoprite labour practices inquiries.** Investigation combined rostering data, supplier development agreements, and union testimony to assess whether exclusive supply terms suppressed independent retailer wages.
- **Gig platforms (Competition Commission inquiry into e-hailing, 2023).** Telemetry from Bolt, Uber, and local entrants showed driver multi-homing constraints and commission escalators; regulators required transparency dashboards and alternative commission tiers for high-volume drivers.

Visualizations

Labor market concentration choropleth

A geographic visualization of labor market concentration helps identify regions and occupations with potential monopsony concerns. This example uses synthetic data but can be adapted for BLS QCEW, Census LEHD, or Stats SA QLFS data.

```
library(dplyr)
library(ggplot2)
library(maps)
library(viridis)

# Simulated labor market HHI data by state/province and
  ↳ occupation
# Replace with BLS QCEW or Census LEHD data aggregated to
  ↳ commuting zones
set.seed(456)
us_states <- map_data("state")

# Create synthetic HHI data for states (illustrative)
labor_hhi <- tibble(
  region = unique(us_states$region),
  hhi = runif(length(unique(us_states$region)), 800, 4500),
  occupation = "Registered Nurses"
) |>
mutate(
  concentration_level = case_when(
    hhi < 1500 ~ "Unconcentrated",
    hhi < 2500 ~ "Moderately concentrated",
    TRUE ~ "Highly concentrated"
  ),
  concentration_level = factor(
    concentration_level,
    levels = c("Unconcentrated", "Moderately concentrated",
               "Highly concentrated")
  )
)

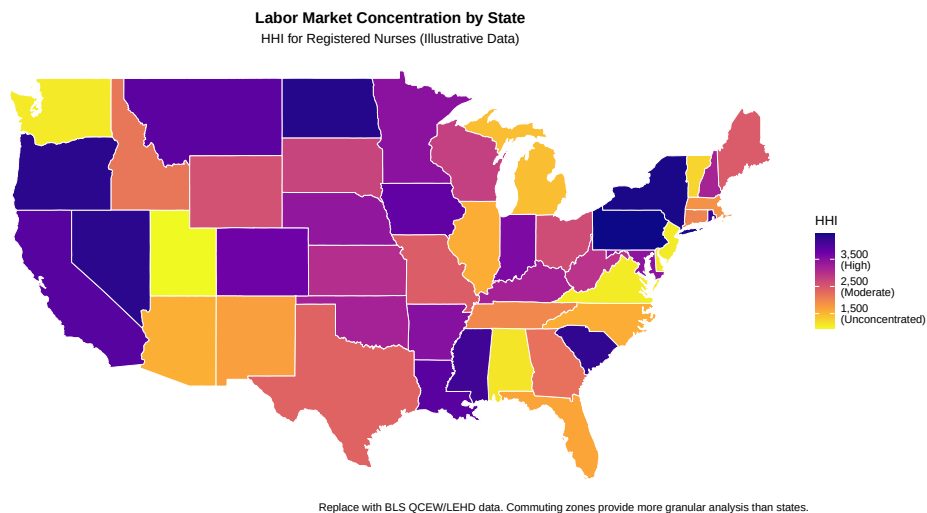
# Join with map data
us_map <- us_states |>
  left_join(labor_hhi, by = "region")

# Choropleth map
ggplot(us_map, aes(x = long, y = lat, group = group, fill = hhi))
  ↳ +
```

```

geom_polygon(color = "white", linewidth = 0.2) +
scale_fill_viridis_c(
  option = "plasma",
  direction = -1,
  breaks = c(1500, 2500, 3500),
  labels = c("1,500\n(Unconcentrated)",
             "2,500\n(Moderate)",
             "3,500\n(High)")
) +
labs(
  title = "Labor Market Concentration by State",
  subtitle = "HHI for Registered Nurses (Illustrative Data)",
  fill = "HHI",
  caption = "Replace with BLS QCEW/LEHD data. Commuting zones
  ↪ provide more granular analysis than states."
) +
theme_void(base_size = 12) +
theme(
  plot.title = element_text(face = "bold", hjust = 0.5),
  plot.subtitle = element_text(hjust = 0.5),
  legend.position = "right"
) +
coord_map()

```



```

# Summary statistics by concentration level
cat("\nConcentration distribution:\n")

```

Concentration distribution:

```
labor_summary <- labor_hhi |>
  group_by(concentration_level) |>
  summarise(
    count = n(),
    mean_hhi = mean(hhi),
    median_hhi = median(hhi),
    .groups = "drop"
  )
print(labor_summary, n = Inf)
```

```
# A tibble: 3 x 4
  concentration_level    count mean_hhi median_hhi
  <fct>                <int>   <dbl>    <dbl>
1 Unconcentrated         8    1149.    1098.
2 Moderately concentrated 14    1974.    1926.
3 Highly concentrated    27    3615.    3594.
```

Interpretation: - **HHI < 1,500:** Competitive labor markets; multiple employers compete for workers. - **HHI 1,500-2,500:** Moderately concentrated; potential monopsony concerns. - **HHI > 2,500:** Highly concentrated; significant monopsony power likely.

How to construct this with real data:

1. **BLS QCEW (US):** Quarterly Census of Employment and Wages provides establishment counts and employment by industry and county. Calculate HHI for relevant occupation $\times$ commuting zone.
2. **Census LEHD (US):** Longitudinal Employer-Household Dynamics provides origin-destination employment data. More granular than QCEW.
3. **Stats SA QLFS (South Africa):** Quarterly Labour Force Survey can be aggregated by occupation $\times$ metro/district.

Code template for BLS QCEW:

```
library(blsAPI)
library(dplyr)

# Fetch QCEW data (requires BLS API key)
# series_id format: ENUxxxxxxxxxxxx (area  $\times$  industry  $\times$  datatype)
# Calculate HHI from establishment-level employment shares

# Example workflow:
# 1. Get employment by establishment size class and county
# 2. For each occupation  $\times$  commuting zone:
#   a. Sum employment by firm
#   b. Calculate market shares
#   c. HHI =  $\text{sum}((\text{share} \times 100)^2)$ 
```

```
# 3. Join to shapefile for choropleth
```

Labor concentration dashboard by occupation

Compare HHI across multiple occupations to identify which labor markets face greatest concentration:

```
library(dplyr)
library(ggplot2)
library(forcats)

# Simulated HHI data across occupations and regions
# Replace with actual QCEW/LEHD/QLFS data
set.seed(789)
occupations <- c("Registered Nurses", "Software Engineers",
                 "Truck Drivers", "Retail Workers",
                 "Fast Food Workers", "Warehouse Workers")
regions <- c("Urban", "Suburban", "Rural")

labor_occ <- expand_grid(
  occupation = occupations,
  region = regions
) |>
mutate(
  hhi = case_when(
    region == "Rural" ~ runif(n(), 2200, 4500),
    region == "Suburban" ~ runif(n(), 1400, 2800),
    region == "Urban" ~ runif(n(), 800, 2200)
  ),
  concentration = case_when(
    hhi < 1500 ~ "Unconcentrated",
    hhi < 2500 ~ "Moderately concentrated",
    TRUE ~ "Highly concentrated"
  )
)

# Grouped bar chart
ggplot(labor_occ, aes(x = fct_reorder(occupation, hhi, .desc =
  FALSE),
                      y = hhi, fill = region)) +
  geom_col(position = position_dodge(width = 0.8), width = 0.7) +
  geom_hline(yintercept = 1500, linetype = "dashed",
            color = "gray40", linewidth = 0.8) +
  geom_hline(yintercept = 2500, linetype = "dashed",
            color = "gray40", linewidth = 0.8) +
  annotate("text", x = 6.5, y = 1500, label = "1,500 threshold",
```

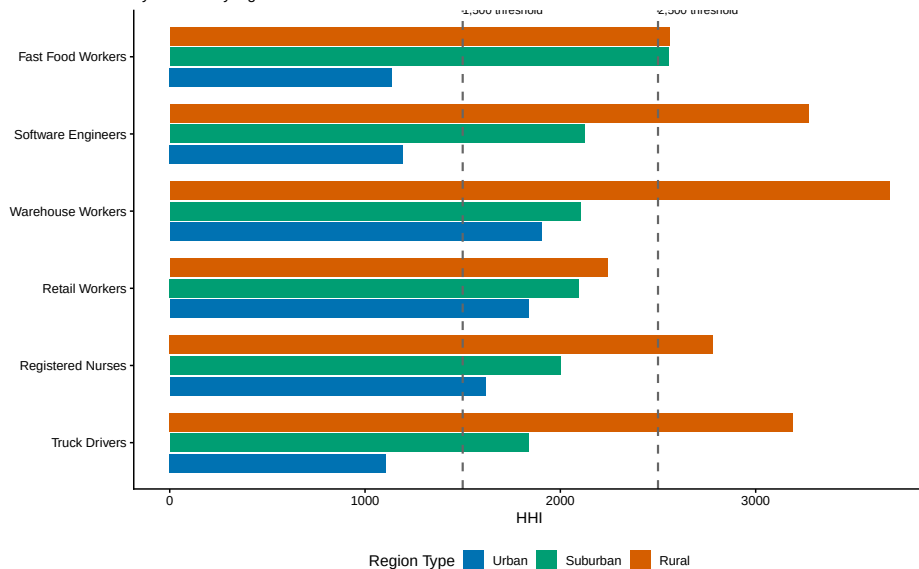
```

    hjust = 0, vjust = -0.5, size = 3) +
  annotate("text", x = 6.5, y = 2500, label = "2,500 threshold",
    hjust = 0, vjust = -0.5, size = 3) +
  scale_fill_manual(values = c("Urban" = "#0072B2",
    "Suburban" = "#009E73",
    "Rural" = "#D55E00")) +

  coord_flip() +
  labs(
    title = "Labor Market Concentration by Occupation and
    ↪ Region",
    subtitle = "Rural areas show systematically higher
    ↪ concentration",
    x = NULL,
    y = "HHI",
    fill = "Region Type",
    caption = "Illustrative data. Replace with QCEW/LEHD
    ↪ aggregated to commuting zones."
  ) +
  theme_classic(base_size = 12) +
  theme(
    plot.title.position = "plot",
    legend.position = "bottom"
  )

```

Labor Market Concentration by Occupation and Region
Rural areas show systematically higher concentration



```
# Summary table: occupations with highest average HHI
cat("\nHighest concentration occupations (average across
  ↪ regions):\n")
```

Highest concentration occupations (average across regions):

```
labor_occ |>
  group_by(occupation) |>
  summarise(
    avg_hhi = mean(hhi),
    high_conc_regions = sum(hhi > 2500),
    .groups = "drop"
  ) |>
  arrange(desc(avg_hhi)) |>
  print(n = Inf)
```

```
# A tibble: 6 x 3
  occupation      avg_hhi high_conc_regions
  <fct>          <dbl>          <int>
1 Warehouse Workers 2564.             1
2 Software Engineers 2197.             1
3 Registered Nurses 2133.             1
4 Fast Food Workers 2083.             2
5 Retail Workers    2058.             0
6 Truck Drivers     2045.             1
```

Key insights: - **Rural areas** typically show higher concentration due to fewer employers. - **Low-skill occupations** (fast food, warehouse) often face more concentration than high-skill occupations with remote work options. - **Health-care occupations** show mixed patterns depending on Certificate of Need laws and hospital consolidation.

Wage impact visualization

Combine concentration data with wage outcomes to show the relationship between HHI and wage suppression:

```
library(dplyr)
library(ggplot2)

# Simulated data showing HHI vs. wage gap
# Replace with actual QCEW/OES wage data matched to HHI
  ↪ calculations
set.seed(890)
wage_impact <- tibble(
  market = paste("Market", 1:50),
```

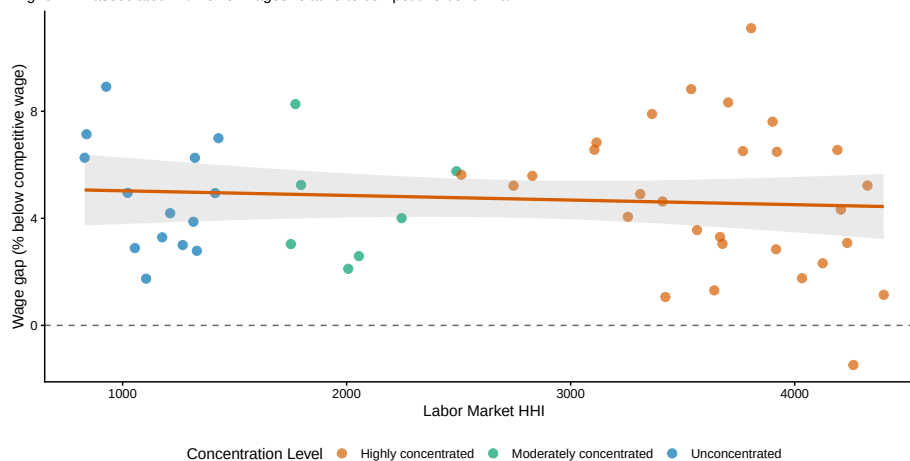
```

hhi = runif(50, 800, 4500),
wage_gap = -0.0003 * hhi + rnorm(50, 5, 3) # Negative
  ↳ relationship
) |>
mutate(
  concentration = case_when(
    hhi < 1500 ~ "Unconcentrated",
    hhi < 2500 ~ "Moderately concentrated",
    TRUE ~ "Highly concentrated"
  )
)

# Scatter plot with regression line
ggplot(wage_impact, aes(x = hhi, y = wage_gap)) +
  geom_point(aes(color = concentration), size = 3, alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE, color = "#D55E00",
    linewidth = 1.2, alpha = 0.2) +
  geom_hline(yintercept = 0, linetype = "dashed", color =
    ↳ "gray40") +
  scale_color_manual(
    values = c(
      "Unconcentrated" = "#0072B2",
      "Moderately concentrated" = "#009E73",
      "Highly concentrated" = "#D55E00"
    )
  ) +
  labs(
    title = "Labor Market Concentration and Wage Suppression",
    subtitle = "Higher HHI associated with lower wages relative
      ↳ to competitive benchmark",
    x = "Labor Market HHI",
    y = "Wage gap (% below competitive wage)",
    color = "Concentration Level",
    caption = "Illustrative data. Replace with matched QCEW/OES
      ↳ data.\nCompetitive benchmark from low-HHI markets or
      ↳ structural estimates."
  ) +
  theme_classic(base_size = 12) +
  theme(
    plot.title.position = "plot",
    legend.position = "bottom"
  )

```

Labor Market Concentration and Wage Suppression
Higher HHI associated with lower wages relative to competitive benchmark



Illustrative data. Replace with matched QCEW/OES data.
Competitive benchmark from low-HHI markets or structural estimates.

```
# Regression summary
wage_model <- lm(wage_gap ~ hhi, data = wage_impact)
cat("\nRegression: Wage gap ~ HHI\n")
```

Regression: Wage gap ~ HHI

```
summary(wage_model)
```

Call:

```
lm(formula = wage_gap ~ hhi, data = wage_impact)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-5.9452	-1.8235	-0.0443	1.9466	6.5611

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.2027243	0.8769292	5.933	3.17e-07 ***
hhi	-0.0001730	0.0002949	-0.587	0.56

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.477 on 48 degrees of freedom

Multiple R-squared: 0.007122, Adjusted R-squared: -0.01356

F-statistic: 0.3443 on 1 and 48 DF, p-value: 0.5601

How to use these visualizations: - Enforcement priorities: Focus in-

vestigations on high-HHI markets. - **Remedy design:** Target no-poach bans or wage disclosure rules at concentrated markets. - **Policy evaluation:** Use choropleth before/after policy changes to assess geographic impacts. - **Public interest:** Highlight disparate impacts on rural communities or specific demographics.

Data sources for production: - **BLS QCEW:** bls.gov/cew - **Census LEHD:** lehd.ces.census.gov - **BLS OES:** bls.gov/oes (for wage data) - **Stats SA QLFS:** statssa.gov.za - **Commuting zone definitions:** USDA ERS or Census Geography

Visualizations and data sourcing

- **Labor HHI choropleths (Roadmap 10.1):** Source BLS QCEW (US), Stats SA, or EU Labour Force Survey; for publication, provide a template script plus sanitized shapefiles.
- **Noncompete diff-in-diff plots (Roadmap 10.2):** Use state-level CPS microdata, WA Dept. of Labor wage records, or Stats SA sectoral wages; fallback to synthetic data until approvals land.
- **Mobility transition matrices (Roadmap 10.3):** Build from LEHD/LODES or anonymized HR exports; include aggregated gig-platform transitions.

Document each dataset in `data/README.md` and tag whether it is public, sanitized, or confidential. When ready to “fill with real data,” replace the synthetic CSVs in `data/examples/` with actual extracts, rerun the figures, and cache redacted outputs for broader distribution.

Innovation, Intellectual Property, and Antitrust

Learning goals

Innovation cases require translating R&D pipelines, patent portfolios, and technical standards into competition narratives. This chapter shows how to:

- Evaluate SEPs/FRAND licensing, patent pools, and interoperability disputes.
- Analyze pay-for-delay/reverse-payment settlements and innovation-reducing mergers.
- Combine patent data (counts, citations, claim breadth), R&D metrics, and technical evidence (roadmaps, standards submissions).
- Integrate qualitative expert testimony with econometric and event-study evidence.

Core topics

- SEPs/FRAND licensing economics; hold-up vs. hold-out.
- Pay-for-delay: economic characterization, entry timing, consumer harm.
- Innovation measures: patent counts/citations, R&D spend, pipeline probability.
- Interoperability and standards; access to APIs and data.
- Remedy design: FRAND commitments, compulsory licensing, data/technology access, behavioral constraints on reverse payments.

i Method box

- Event studies on patent pool or settlement announcements.
- Entry timing models for generics; hazard/survival sketches.
- R&D response to merger or remedy using panel FE.

i Qualitative evidence

- Technical expert declarations, standards body submissions, patent essentiality reviews.
- Licensing negotiations, term sheets, audit rights.
- Product roadmaps and engineering constraints.

i Citations and comparative note

- Cite FRAND/SEP cases and guidance (e.g., EU Huawei v. ZTE, US FTC/DOJ policy statements, UK FRAND rate cases).
- For pay-for-delay, reference US Supreme Court (Actavis) and EU pay-for-delay decisions.
- Include patent data sources (USPTO, EPO, JPO, CNIPA) and note differences in treatment of injunctions and remedies across jurisdictions.

SEPs, FRAND, and standard setting

Market power and hold-up vs. hold-out

Standard-essential patents (SEPs) can confer significant bargaining leverage once a standard is adopted. Distinguish:

- **Hold-up:** SEP holder extracts supra-FRAND royalties by threatening injunctions after firms are locked into the standard (Lemley and Shapiro 2007).
- **Hold-out:** Implementers delay or underpay royalties despite benefiting from the standard.

For economic analysis of patent holdup and royalty stacking, see Shapiro (2001) and Farrell et al. (2007).

Evidence sources:

- Standards body submissions (ETSI, IEEE, 3GPP) and FRAND commitments.
- Essentiality assessments (self-declared vs. independent).
- Licensing histories, audit trails, and comparable agreements.

Royalty stack analysis

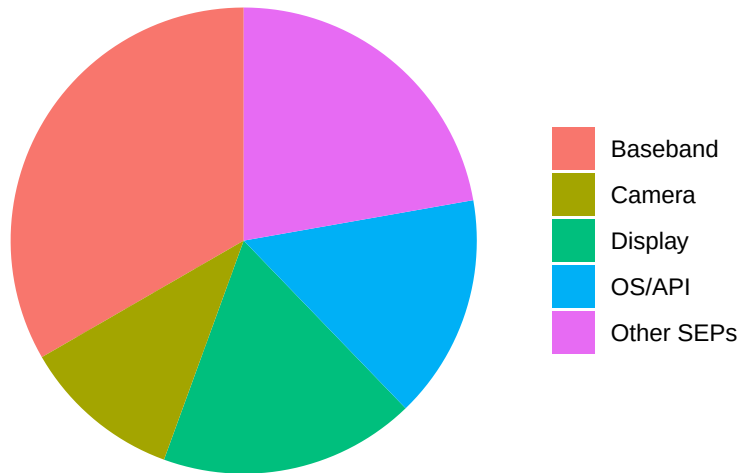
```
library(dplyr)
library(ggplot2)

stack <- tibble::tribble(
  ~component, ~royalty_percent,
  "Baseband", 1.5,
  "Display", 0.8,
  "Camera", 0.5,
  "OS/API", 0.7,
  "Other SEPs", 1.0
)

ggplot(stack, aes(x = "", y = royalty_percent, fill = component))
  ↪ +
  geom_col(width = 0.5) +
  coord_polar("y") +
  labs(
    title = "Illustrative royalty stack",
    subtitle = "Replace with FRAND negotiation data when
  ↪ available",
    y = "Royalty (% of device price)",
    x = NULL,
    fill = NULL
  ) +
  theme_void()
```

Illustrative royalty stack

Replace with FRAND negotiation data when available



Replace the illustrative stack with licensing data (often produced during litigation) or sanitized aggregates from public FRAND rate determinations. Cite FRAND rate determinations (e.g., UK Unwired Planet, Optis) for benchmarking, which are available through court records.

```

# Financial Sanctions
library(tidyquant)
library(dplyr)
library(ggplot2)

# Use tidyquant to pull SEP-holder and implementer stocks;
  ↳ measure CAR around injunction or settlement announcements
# event_date <- as.Date("2021-08-27")
# tickers <- c("QCOM","AAPL","SSNLF","SPY")
# prices <- tq_get(tickers, from = event_date - 120, to =
  ↳ event_date + 60) |> ...

```

Use `tidyquant` with actual ticker lists; cite the docket (e.g., “Optis v. Apple FRAND decision”) in the narrative.

Pay-for-delay and reverse payments

Entry timing and hazard models

Reverse-payment settlements can delay generic entry and harm consumers (U.S. Supreme Court 2013). Analyze:

- **Payment size:** Compare settlement payment to expected litigation costs and projected profits (Edlin and Hemphill 2013).
- **Entry timing:** Use survival models or diff-in-diff to estimate delay relative to counterfactuals.
- **Price effects:** Simulate price paths with and without generic entry (Scott Morton 2000; Hemphill and Sampat 2012).

```
library(survival)

# synthetic example
data <- tibble::tribble(
  ~drug, ~entry_year, ~settlement, ~status,
  "DrugA", 2020, 1, 1,
  "DrugB", 2018, 0, 1,
  "DrugC", 2022, 1, 0
)

# survival::survfit(Surv(entry_year, status) ~ settlement, data =
  ↪ data)
```

Replace with real pay-for-delay cases (Actavis-style) or EU decisions (Servier, Lundbeck). Use FDA Orange Book, EMA data, or South African SAMRC filings for entry timelines.

Diff-in-diff for price effects

```
library(fixest)

# panel columns: market, year, treated (1 if affected by
  ↪ settlement), price_index
# did_model <- feols(price_index ~ treated | market + year, data
  ↪ = panel)
# summary(did_model)
```

Data sources: IQVIA (if available), CMS reimbursement data, Stats SA medicine price data, or public price registries.

Innovation effects of mergers or conduct

- **Patent/R&D panels:** Compile patent counts (PatentsView, EPO PATSTAT), citation-weighted measures, and R&D spending to evaluate innovation incentives pre/post merger or remedy.
- **Pipeline probability models:** For pharmaceuticals, use clinical trial progression rates.
- **Product roadmaps:** Qualitative review of technical plans to confirm or rebut innovation harm narratives.

Panel FE scaffold

```
library(fixest)

# panel columns: firm, year, rd_spend, merger_dummy, controls
# rd_model <- feols(log(rd_spend) ~ merger_dummy + controls |
  ↪ firm + year, data = panel)
# summary(rd_model)
```

Interoperability and data access

Analyze API access restrictions, data sharing, and interoperability constraints:

- **API logs:** Compare response times, rate limits, and feature parity between internal and external developers.
- **Data portability metrics:** Evaluate export completeness, frequency, and latency.
- **Technical documentation:** PRDs and engineering tickets often show intent to foreclose or degrade rivals.

Southern African innovation/IP examples

- **Telkom vs. Internet Solutions (broadband IP).** Access to APIs and OSS/BSS systems under FRAND-like commitments; margin squeeze tests combined with technical audits.
- **Vodacom Zero-Rating commitments (data services inquiry).** Interoperability obligations for educational content, showing how innovation incentives can align with public-interest remedies.

- **Pharma patent settlements (Competition Commission investigations into ARV markets).** Settlement analysis combining SAMRC price data, patent landscapes, and clinical trial timelines to assess delay and innovation impact.

Enhanced Visualizations

Enhanced royalty stack waterfall

A more detailed royalty stack showing how cumulative royalties build up across different technology components. This helps demonstrate potential royalty stack concerns in FRAND disputes.

```
library(dplyr)
library(ggplot2)

# Royalty components for a smartphone
# Replace with actual FRAND rate determinations or licensing data
royalty_components <- tibble::tribble(
  ~component,      ~holder_type,      ~royalty_pct,
  "Device price",  "Base",              100.00,
  "Cellular (5G)", "SEP",               -2.50,
  "WiFi/Bluetooth", "SEP",              -1.20,
  "Audio codecs",  "SEP",              -0.80,
  "Video codecs",  "SEP",              -1.50,
  "Display tech",  "Patent pool",      -0.90,
  "Camera tech",   "Non-SEP",          -0.60,
  "Software/OS",   "License",          -1.80,
  "Other SEPs",    "SEP",              -1.50,
  "Net to manufacturer", "Result",        89.20
) |>
  mutate(
    component = factor(component, levels = component),
    cumulative = cumsum(royalty_pct),
    start = lag(cumulative, default = 100),
    end = cumulative,
    type = case_when(
      component %in% c("Device price", "Net to manufacturer") ~
        "total",
      royalty_pct < 0 ~ "royalty",
      TRUE ~ "neutral"
    )
  )

# Waterfall plot
ggplot(royalty_components) +
```

```

geom_rect(aes(xmin = as.numeric(component) - 0.4,
              xmax = as.numeric(component) + 0.4,
              ymin = start, ymax = end, fill = holder_type),
          color = "black", linewidth = 0.5) +
geom_text(aes(x = as.numeric(component),
              y = (start + end) / 2,
              label = ifelse(type == "royalty",
                             paste0(abs(royalty_pct), "%"),
                             paste0(royalty_pct, "%"))),
          size = 3, fontface = "bold") +
geom_segment(data = filter(royalty_components, !type %in%
  ↪ c("total")),
             aes(x = as.numeric(component) + 0.4,
                 xend = as.numeric(component) + 1 - 0.4,
                 y = end, yend = end),
             linetype = "dashed", color = "gray50") +
scale_fill_manual(
  values = c(
    "Base" = "#0072B2",
    "SEP" = "#D55E00",
    "Patent pool" = "#009E73",
    "Non-SEP" = "#F0E442",
    "License" = "#CC79A7",
    "Result" = "#0072B2"
  ),
  breaks = c("SEP", "Patent pool", "Non-SEP", "License")
) +
scale_x_continuous(
  breaks = seq_along(royalty_components$component),
  labels = royalty_components$component
) +
scale_y_continuous(labels = scales::dollar_format(suffix = "%",
  ↪ prefix = "")) +
labs(
  title = "Royalty Stack Waterfall: Smartphone Example",
  subtitle = "Cumulative royalty burden on device
  ↪ manufacturer",
  x = NULL,
  y = "% of Device Price",
  fill = "Royalty Type",
  caption = "Illustrative data. Replace with FRAND rate
  ↪ determinations or licensing disclosures."
) +
theme_classic(base_size = 12) +
theme(
  axis.text.x = element_text(angle = 45, hjust = 1),

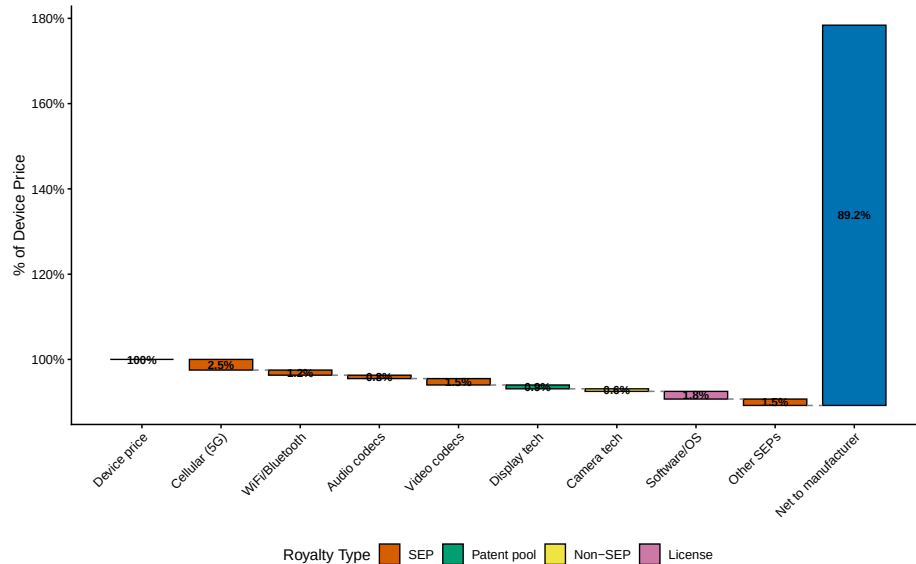
```

```

legend.position = "bottom",
plot.title.position = "plot"
)

```

Royalty Stack Waterfall: Smartphone Example
Cumulative royalty burden on device manufacturer



Illustrative data. Replace with FRAND rate determinations or licensing disclosures.

```

# Summary statistics
total_royalties <- 100 - royalty_components$royalty_pct[
  royalty_components$component == "Net to manufacturer"]

cat("\nRoyalty stack summary:\n")

```

Royalty stack summary:

```

cat(paste0("Total royalty burden: ", round(total_royalties, 1),
  "\n"))

```

Total royalty burden: 10.8%

```

cat(paste0("SEP royalties: ",
  round(sum(abs(royalty_components$royalty_pct[
    royalty_components$holder_type == "SEP"])), 1),
  "\n"))

```

SEP royalties: 7.5%

```
cat(paste0("Net to manufacturer: ",
          round(royalty_components$royalty_pct[
            royalty_components$component == "Net to
↪ manufacturer"], 1), "%\n"))
```

Net to manufacturer: 89.2%

Key concerns: - **Royalty stacking:** Multiple SEP holders each claiming a “reasonable” royalty can collectively make manufacturing unprofitable. - **Double marginalization:** Each layer adds its markup, raising final prices. - **Hold-up:** Once committed to a standard, manufacturers face switching costs that weaken bargaining position.

Benchmarking: Compare against FRAND rate determinations: - UK Unwired Planet v. Huawei (2017) - UK Optis v. Apple (2023) - US Ericsson v. Samsung arbitration

Generic entry survival curve

Analyze the timing of generic pharmaceutical entry, with and without reverse payment settlements. Survival analysis shows whether settlements delay entry.

```
library(dplyr)
library(ggplot2)
library(survival)

# Simulated generic entry data
# Replace with FDA Orange Book data or European/South African
↪ equivalents
set.seed(456)
n_drugs <- 60

entry_data <- tibble(
  drug_id = 1:n_drugs,
  settlement = sample(c("No settlement", "Reverse payment
↪ settlement"),
                    n_drugs, replace = TRUE, prob = c(0.6,
↪ 0.4)),
  months_to_entry = case_when(
    settlement == "No settlement" ~ rexp(n_drugs, rate = 1/36),
    TRUE ~ rexp(n_drugs, rate = 1/56) # Delayed entry with
↪ settlement
  ),
  entered = sample(c(1, 0), n_drugs, replace = TRUE, prob =
↪ c(0.85, 0.15))
) |>
mutate(months_to_entry = pmin(months_to_entry, 120)) # Cap at
↪ 10 years
```

```

# Kaplan-Meier survival curve
surv_object <- Surv(time = entry_data$months_to_entry,
                    event = entry_data$entered)
surv_fit <- survfit(surv_object ~ settlement, data = entry_data)

# Plot survival curve (requires survminer for ggsurvplot)
if (!requireNamespace("survminer", quietly = TRUE)) {
  message("Install the 'survminer' package to render the
    ↪ Kaplan-Meier plot.")
} else {
  survminer::ggsurvplot(
    surv_fit,
    data = entry_data,
    palette = c("#0072B2", "#D55E00"),
    conf.int = TRUE,
    risk.table = TRUE,
    risk.table.height = 0.25,
    xlab = "Months from patent grant",
    ylab = "Probability no generic entry",
    title = "Generic Entry Timing: Impact of Reverse Payment
    ↪ Settlements",
    subtitle = "Kaplan-Meier survival curves by settlement
    ↪ status",
    legend.title = "Settlement Type",
    legend.labs = c("No settlement", "Reverse payment
    ↪ settlement"),
    ggtheme = theme_classic(base_size = 12),
    font.main = c(14, "bold"),
    font.subtitle = c(11, "plain"),
    font.caption = c(9, "plain"),
    caption = "Illustrative data. Replace with FDA Orange Book or
    ↪ EMA data."
  )
}

# Log-rank test
surv_diff <- survdiff(surv_object ~ settlement, data =
  ↪ entry_data)
cat("\nLog-rank test for difference in survival curves:\n")

```

Log-rank test for difference in survival curves:

```
cat(paste0("Chi-squared = ", round(surv_diff$chisq, 2), "\n"))
```

```
Chi-squared = 0
```

```
cat(paste0("p-value = ", format.pval(1 - pchisq(surv_diff$chisq,
df =
  ↪ length(surv_diff$n)
  ↪ - 1),
digits = 3), "\n"))
```

```
p-value = 0.945
```

```
# Median survival times
cat("\nMedian time to generic entry:\n")
```

```
Median time to generic entry:
```

```
print(summary(surv_fit)$table[, c("median", "0.95LCL",
  ↪ "0.95UCL")])
```

	median	0.95LCL	0.95UCL
settlement=No settlement	36.26116	26.87372	49.50792
settlement=Reverse payment settlement	35.99143	24.18951	60.51071

Interpretation: - **Separation of curves:** If settlement drugs show consistently lower entry hazard, this suggests anticompetitive delay. - **Median delay:** Calculate months of delay attributable to settlements. - **Consumer harm:** Combine with price differentials (branded vs. generic) to estimate consumer losses.

Data sources: - **FDA Orange Book:** [fda.gov/drugs/drug-approvals-and-databases/orange-book-data-files](https://www.fda.gov/drugs/drug-approvals-and-databases/orange-book-data-files) - **EMA:** European Medicines Agency public assessment reports - **SAHPRA:** South African Health Products Regulatory Authority - **FTC settlements:** Public records of reverse payment cases

Patent citation network

Visualize how key patents cite each other to show technological relationships and potential blocking positions.

```
library(dplyr)
library(igraph)
library(ggraph)

# Simulated patent citation network
# Replace with PatentsView or EPO PATSTAT data
set.seed(567)
```

```

n_patents <- 30

# Create citation edges
citations <- tibble(
  from_patent = sample(paste0("Patent_", 1:n_patents), 50,
    ↪ replace = TRUE),
  to_patent = sample(paste0("Patent_", 1:n_patents), 50, replace
    ↪ = TRUE)
) |>
  filter(from_patent != to_patent) |>
  distinct()

# Create patent metadata
patents <- tibble(
  patent_id = paste0("Patent_", 1:n_patents),
  holder = sample(c("Firm A", "Firm B", "Firm C", "Firm D",
    ↪ "University"),
    n_patents, replace = TRUE, prob = c(0.3, 0.25,
    ↪ 0.2, 0.15, 0.1)),
  essential = sample(c("SEP", "Non-SEP"), n_patents, replace =
    ↪ TRUE,
    prob = c(0.3, 0.7))
)

# Create network
g <- graph_from_data_frame(citations, directed = TRUE, vertices =
  ↪ patents)

# Calculate centrality measures
V(g)$in_degree <- degree(g, mode = "in")
V(g)$out_degree <- degree(g, mode = "out")
V(g)$betweenness <- betweenness(g)

# Plot network
ggraph(g, layout = "fr") +
  geom_edge_link(arrow = arrow(length = unit(2, 'mm'), type =
    ↪ "closed"),
    end_cap = circle(3, 'mm'),
    alpha = 0.3, color = "gray50") +
  geom_node_point(aes(size = in_degree, color = holder,
    shape = essential), alpha = 0.8) +
  geom_node_text(
    aes(label = ifelse(in_degree > 3, name, "")),
    repel = TRUE,
    size = 2.5,
    family = "Arial"
  )

```

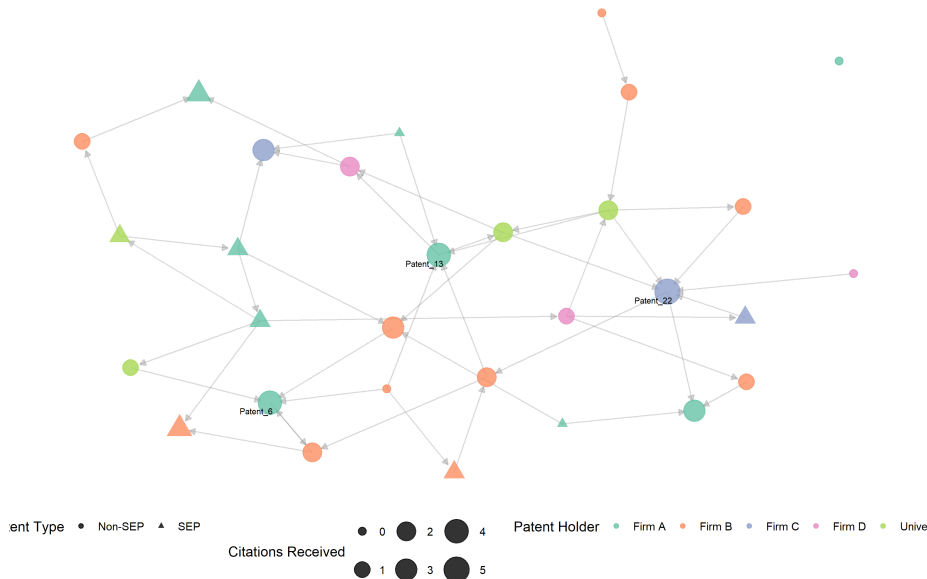
```

) +
scale_size_continuous(range = c(3, 10)) +
scale_color_brewer(palette = "Set2") +
scale_shape_manual(values = c("SEP" = 17, "Non-SEP" = 19)) +
labs(
  title = "Patent Citation Network",
  subtitle = "Node size = citation count (in-degree); arrows
    ↪ show citations",
  color = "Patent Holder",
  shape = "Patent Type",
  size = "Citations Received",
  caption = "Illustrative data. Replace with PatentsView or EPO
    ↪ PATSTAT data."
) +
theme_graph(base_size = 12) +
theme(
  text = element_text(family = "Arial"),
  legend.position = "bottom",
  plot.title = element_text(size = 14, face = "bold")
)

```

Patent Citation Network

Node size = citation count (in-degree); arrows show citations



Illustrative data. Replace with PatentsView or EPO PATSTAT data.

```

# Summary statistics
cat("\nPatent network summary:\n")

```

Patent network summary:

```
cat(paste0("Total patents: ", n_patents, "\n"))
```

Total patents: 30

```
cat(paste0("Total citations: ", nrow(citations), "\n"))
```

Total citations: 47

```
cat(paste0("SEP patents: ", sum(V(g)$essential == "SEP"), "\n"))
```

SEP patents: 9

```
cat("\nTop 5 most-cited patents:\n")
```

Top 5 most-cited patents:

```
patents |>
  left_join(tibble(patent_id = V(g)$name, in_degree =
    ↪ V(g)$in_degree),
    by = "patent_id") |>
  arrange(desc(in_degree)) |>
  select(patent_id, holder, essential, in_degree) |>
  head(5) |>
  print()
```

```
# A tibble: 5 x 4
  patent_id holder essential in_degree
  <chr>      <chr>   <chr>         <dbl>
1 Patent_22 Firm C   Non-SEP         5
2 Patent_6  Firm A   Non-SEP         4
3 Patent_13 Firm A   Non-SEP         4
4 Patent_8  Firm B   Non-SEP         3
5 Patent_9  Firm A   Non-SEP         3
```

Key insights: - **Central patents:** High betweenness centrality indicates “bottleneck” patents that many others depend on. - **Patent thickets:** Dense clusters suggest overlapping IP that may impede innovation. - **Holder concentration:** If one firm controls many central patents, it may have market power.

R&D spending event study around merger

Evaluate whether mergers affect innovation incentives by tracking R&D spending before and after the transaction.

```

library(dplyr)
library(ggplot2)
library(fixest)

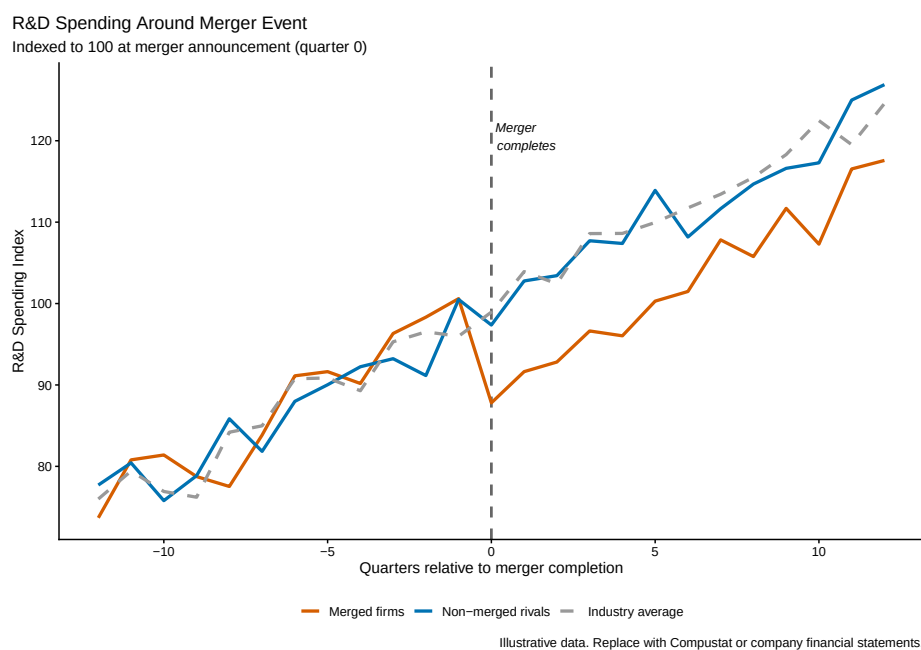
# Simulated panel data: R&D spending around merger
# Replace with Compustat, Orbis, or financial statement data
set.seed(678)
quarters <- -12:12
firms <- c("Merged firms", "Non-merged rivals", "Industry
  ↪ average")

rd_panel <- expand_grid(
  quarter = quarters,
  firm_group = firms
) |>
  mutate(
    # Pre-merger: all firms follow similar trends
    # Post-merger: merged firms reduce R&D relative to others
    rd_index = case_when(
      firm_group == "Merged firms" & quarter < 0 ~ 100 + 2 *
    ↪ quarter + rnorm(n(), 0, 3),
      firm_group == "Merged firms" & quarter >= 0 ~ 100 + 2 *
    ↪ quarter - 8 + rnorm(n(), 0, 3),
      firm_group == "Non-merged rivals" ~ 100 + 2 * quarter +
    ↪ rnorm(n(), 0, 3),
      TRUE ~ 100 + 2 * quarter + rnorm(n(), 0, 2)
    )
  )

# Time series plot
ggplot(rd_panel, aes(x = quarter, y = rd_index,
  color = firm_group, linetype = firm_group)) +
  geom_vline(xintercept = 0, linetype = "dashed", color =
    ↪ "gray40",
    linewidth = 1) +
  stat_summary(fun = mean, geom = "line", linewidth = 1.2) +
  annotate("text", x = 0, y = max(rd_panel$rd_index) * 0.95,
    label = "Merger\ncompletes", hjust = -0.1, size = 3.5,
    fontface = "italic") +
  scale_color_manual(values = c("Merged firms" = "#D55E00",
    "Non-merged rivals" = "#0072B2",
    "Industry average" = "#999999"))
    ↪ +
  scale_linetype_manual(values = c("Merged firms" = "solid",
    "Non-merged rivals" = "solid",
    "Industry average" =
    ↪ "dashed")) +

```

```
labs(
  title = "R&D Spending Around Merger Event",
  subtitle = "Indexed to 100 at merger announcement (quarter
    ↪ 0)",
  x = "Quarters relative to merger completion",
  y = "R&D Spending Index",
  color = NULL,
  linetype = NULL,
  caption = "Illustrative data. Replace with Compustat or
    ↪ company financial statements."
) +
theme_classic(base_size = 12) +
theme(
  legend.position = "bottom",
  plot.title.position = "plot"
)
```



```
# Diff-in-diff estimate
rd_panel_did <- rd_panel |>
  mutate(
    post = ifelse(quarter >= 0, 1, 0),
    treated = ifelse(firm_group == "Merged firms", 1, 0)
  )

did_model <- feols(rd_index ~ treated:post | firm_group +
  ↪ quarter,
```

```
data = rd_panel_did)

cat("\nDifference-in-differences estimate:\n")
```

Difference-in-differences estimate:

```
summary(did_model)
```

```
OLS estimation, Dep. Var.: rd_index
Observations: 75
Fixed-effects: firm_group: 3, quarter: 25
Standard-errors: IID
      Estimate Std. Error t value Pr(>|t|)
treated:post -10.0677    1.10147 -9.14023 5.3663e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 1.77844      Adj. R2: 0.974935
              Within R2: 0.639968
```

Interpretation: - **Pre-trends:** Parallel trends before merger support DiD identification. - **Treatment effect:** Negative coefficient suggests merger reduced R&D. - **Innovation harm:** Combine with patent output, clinical trials, or product launches to assess real-world impact.

Alternative specifications: - Event study with leads/lags - Synthetic control for single-firm mergers - Stacked DiD for multiple mergers

Clinical trial pipeline visualization (pharmaceuticals)

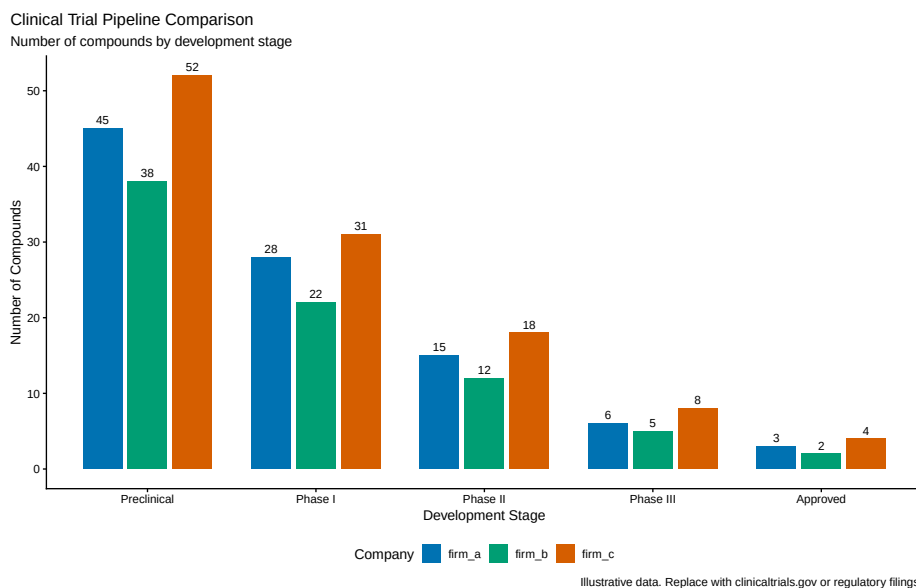
Track clinical trial progress for pharmaceutical innovation assessments.

```
library(dplyr)
library(ggplot2)
library(tidyr)

# Simulated clinical trial pipeline
# Replace with clinicaltrials.gov data or EMA/SAHPRA records
pipeline <- tibble::tribble(
  ~phase,      ~firm_a, ~firm_b, ~firm_c,
  "Preclinical", 45,      38,      52,
  "Phase I",     28,      22,      31,
  "Phase II",    15,      12,      18,
  "Phase III",   6,       5,       8,
  "Approved",    3,       2,       4
) |>
mutate(phase = factor(phase, levels = phase)) |>
```

```
pivot_longer(cols = -phase, names_to = "firm", values_to =
  ↪ "count")

# Funnel chart
ggplot(pipeline, aes(x = phase, y = count, fill = firm, group =
  ↪ firm)) +
  geom_col(position = position_dodge(width = 0.8), width = 0.7) +
  geom_text(aes(label = count), position = position_dodge(width =
    ↪ 0.8),
            vjust = -0.5, size = 3.5) +
  scale_fill_manual(values = c("firm_a" = "#0072B2",
                                "firm_b" = "#009E73",
                                "firm_c" = "#D55E00")) +
  labs(
    title = "Clinical Trial Pipeline Comparison",
    subtitle = "Number of compounds by development stage",
    x = "Development Stage",
    y = "Number of Compounds",
    fill = "Company",
    caption = "Illustrative data. Replace with clinicaltrials.gov
  ↪ or regulatory filings."
  ) +
  theme_classic(base_size = 12) +
  theme(
    legend.position = "bottom",
    plot.title.position = "plot"
  )
```



```
# Success rates
pipeline_wide <- pipeline |>
  pivot_wider(names_from = phase, values_from = count)

cat("\nPipeline success rates (illustrative):\n")
```

Pipeline success rates (illustrative):

```
cat("Phase I to Phase II: ~50-60%\n")
```

Phase I to Phase II: ~50-60%

```
cat("Phase II to Phase III: ~30-40%\n")
```

Phase II to Phase III: ~30-40%

```
cat("Phase III to Approval: ~60-70%\n")
```

Phase III to Approval: ~60-70%

```
cat("\nUse these rates to model expected future approvals and
  ↳ consumer harm from delays.\n")
```

Use these rates to model expected future approvals and consumer harm from delays.

Applications: - **Merger review:** Compare pipeline depth before/after transaction. - **Pay-for-delay:** Estimate lost innovation from delayed generic competition.

tition. - **Market foreclosure**: Show whether conduct reduces rivals' pipeline.

Data sources: - **ClinicalTrials.gov**: US clinical trial registry - **EU Clinical Trials Register**: European trials - **WHO ICTRP**: International registry - **Company pipelines**: Annual reports and R&D presentations

Visualizations and data sourcing

- **Royalty stack (Roadmap 11.1)**: Use licensing disclosures from litigation or FRAND rate decisions. If unavailable, maintain illustrative stack in `data/examples/royalty_stack.csv`.
- **Entry timing survival curves (Roadmap 11.2)**: FDA Orange Book, EMA, SAHPRA, or competition-inquiry appendices.
- **FRAND injunction event plots (Roadmap 11.3)**: `tidyquant` stock data; ensure tickers are mapped to key SEP holders and implementers.
- **Patent pipeline ridge plot**: Pull PatentsView or EPO PATSTAT data; store sanitized extracts and scripts in `data/derived/innovation/`.

Document data provenance and confidentiality in `data/README.md`, noting which visuals can ship publicly and which require synthetic placeholders until case data is approved for release.

Looking ahead

Archive all innovation analyses in `data/derived/innovation/` with clear metadata. When PatentsView bulk downloads or FDA Orange Book data become available, replace synthetic patent networks and entry curves with actual data. Cross-reference R&D event studies with Chapter 06 (mergers) and Chapter 08 (remedies) for remedy design templates. Document technical expert declarations and standards body submissions in case chronologies for litigation support.

Litigation Practice: Evidence and Expert Work

Learning goals

This chapter translates all prior analytics into courtroom-ready workflows. Drawing on agency guidance (US DOJ/FTC, EC best practices) and established expert practice standards, we focus on:

- Structuring investigations so every exhibit and code run is reproducible.
- Designing class certification analyses (common impact, damages) that withstand Daubert/Kumho challenges.
- Integrating empirical, qualitative, and documentary evidence in expert reports, depositions, and testimony.
- Communicating uncertainty, sensitivity, and alternative specifications to judges and juries.

Core topics

- Class certification: common impact frameworks, sampling, and predominance arguments.
- Evidence management: document IDs, data provenance, reproducible code bundles.
- Daubert readiness: validation, sensitivity analyses, alternative specifications.
- Presentation: graphics for court, explanatory appendices, deposition prep.
- Coordination with legal teams, witnesses, and regulators; public-interest considerations in South Africa and other jurisdictions.

i Method box

- Common impact tests with clustered SEs and randomization inference.

- Damages models: before/after, yardstick, difference-in-differences, hedonic variants.
- Scenario/sensitivity tables for assumptions (e.g., pass-through bounds).

i Qualitative evidence

- Fact witness integration: mapping testimonies to model assumptions.
- Survey admissibility checkpoints (universe, sampling, questionnaire design, pretests).
- Expert judgment: when to narrow claims due to data limits.

i Citations and comparative note

- Cite case law on admissibility (Daubert/Kumho in US; local standards elsewhere) and notable opinions accepting/rejecting methods (e.g., class cert common impact challenges).
- Include agency guidance where relevant (e.g., FTC/DOJ guidance on data handling, EC best practices for expert submissions).
- When drawing on foreign cases (EU/UK/Canada/Australia/Japan/China), flag differences in evidentiary standards and expert roles.

Evidence pipeline and reproducibility

Evidence map

Create an evidence map linking:

1. **Data sources:** Production IDs, custodians, time periods, transformations.
2. **Documentary evidence:** Bates numbers, key quotes, translation status.
3. **Witness testimony:** Depositions, declarations, trial testimony, and their linkage to quantitative claims.
4. **Model outputs:** Code, inputs, parameters, and versions stored in reproducible folders.

A typical structure (`/data/raw`, `/data/derived`, `/scripts`, `/reports`) mirrors earlier chapters; maintain README files and `renv/requirements` snapshots.

Reproducible bundle checklist

- Git repository (even if private) with hashed datasets and code.
- Quarto notebooks or Rmarkdown scripts that render exhibits.
- `packrat/renv` lockfiles or Conda environments.
- Document IDs within code comments for traceability.

Class certification and common impact

Sampling and aggregation decisions

- Align sampling frames with class definitions (time, geography, product).
- Document random sampling pr OCED images? (no)
- For opt-out classes, track individual damages calculations.

Common impact tests

```
library(fixest)

# panel columns: class_member, period, treatment, outcome,
#   ↪ controls
# Example diff-in-diff for common impact
# ci_model <- feols(outcome ~ treatment | class_member + period,
#   ↪ data = panel)
# summary(ci_model)
```

Add clustered SEs (e.g., `cluster = ~class_member`) or randomization inference for small N. Consult academic literature on class certification econometrics for methodological guidance.

Randomization inference scaffold

```
library(fixest)
library(magrittr)

# After estimating ci_model, permute treatment labels to compute
#   ↪ empirical p-values
# ri_p <- permute_plm(ci_model, treat = "treatment", cluster =
#   ↪ "class_member", reps = 1000)
```

Damages modeling

- **Before/after:** Baseline for cartel or monopolization; control for costs/demand.
- **Yardstick:** Compare to unaffected markets or products.
- **Diff-in-diff:** For mergers, supply shocks, or wage suppression cases.
- **Hedonic/regression models:** When product characteristics matter (tech, pharma).

Scenario/sensitivity tables

```
library(dplyr)

scenarios <- expand_grid(pass_through = c(0.5, 0.7, 0.9),
  ↪ overcharge = c(0.1, 0.2, 0.3)) |>
  mutate(damages = pass_through * overcharge * 1000)
scenarios
```

	pass_through	overcharge	damages
1	0.5	0.1	50
2	0.7	0.1	70
3	0.9	0.1	90
4	0.5	0.2	100
5	0.7	0.2	140
6	0.9	0.2	180
7	0.5	0.3	150
8	0.7	0.3	210
9	0.9	0.3	270

Use tornado charts or tables to present ranges; highlight the “central” assumption and explain why alternatives are plausible.

Daubert/Kumho readiness

1. **Validation:** Compare model outputs to raw data; show that code replicates known benchmarks (Federal Judicial Center 2011).
2. **Sensitivity:** Document how results change with different controls, clustering levels, or sample definitions (Rubinfeld 2011).
3. **Alternative specifications:** Provide at least one alternative consistent with the theory of harm; explain why it does or does not materially

change outcomes (Baker and Rubinfeld 1999).

4. **Error checking:** Peer review within the expert team; code audits; reproducibility scripts (Dickey and Rubinfeld 2014).

Presentation and storytelling

- **Exhibits:** Keep figures clean; highlight key numbers; avoid jargon. Use `ggplot2` themes consistent with earlier chapters.
- **Timelines:** Combine quantitative and qualitative milestones (evidence triad).
- **Deposition prep:** Build Q&A outlines linking each opinion to evidence.
- **Trial graphics:** Build layered presentations (overview, methodology, results, robustness).

Southern African litigation practice notes

- **Bread cartel damages (Tribunal hearings 2007–2010):** Combined CPI microdata, mill cost studies, and consumer testimony; damages model anchored to structural break analysis.
- **Sasol polypropylene appeal:** Extensive documentation of export parity benchmarks, incremental cost models, and technical expert testimony on production processes.
- **Vodacom/MTN data services commitments:** Monitoring trustees reported quarterly KPIs; Commission used diff-in-diff to show compliance with price-reduction commitments.

Enhanced Visualizations for Expert Reports

Common impact distribution

Demonstrate that the alleged conduct had a common effect across class members, a key requirement for class certification.

```
library(dplyr)
library(ggplot2)
library(patchwork)

# Simulated individual-level treatment effects
# Replace with actual class member data from your damages model
```

```

set.seed(789)
n_members <- 500

class_data <- tibble(
  member_id = 1:n_members,
  treatment_effect = rnorm(n_members, mean = 15, sd = 4), #
  ↪ Common positive impact
  baseline_damages = runif(n_members, 50, 200),
  geographic_region = sample(c("Region A", "Region B", "Region
  ↪ C"),
                                n_members, replace = TRUE),
  purchase_frequency = sample(c("Low", "Medium", "High"),
                                n_members, replace = TRUE,
                                prob = c(0.3, 0.5, 0.2))
) |>
mutate(
  total_damages = baseline_damages + treatment_effect,
  purchase_frequency = factor(purchase_frequency,
                                levels = c("Low", "Medium",
  ↪ "High"))
)

# Plot 1: Distribution of treatment effects
p1 <- ggplot(class_data, aes(x = treatment_effect)) +
  geom_histogram(bins = 30, fill = "#0072B2", alpha = 0.8,
    color = "white") +
  geom_vline(xintercept = mean(class_data$treatment_effect),
    linetype = "dashed", color = "#D55E00", linewidth =
  ↪ 1.2) +
  annotate("text",
    x = mean(class_data$treatment_effect),
    y = Inf,
    label = paste0("Mean effect: $",
      round(mean(class_data$treatment_effect),
  ↪ 2)),
    hjust = -0.1, vjust = 2, size = 4, fontface = "bold") +
  scale_x_continuous(labels = scales::dollar_format()) +
  labs(
    title = "Distribution of Individual Treatment Effects",
    subtitle = "Common impact across class members (positive mean
  ↪ effect)",
    x = "Treatment Effect ($)",
    y = "Number of Class Members",
    caption = "Illustrative data. Replace with actual class
  ↪ member damages calculations."
  ) +

```

```

theme_classic(base_size = 12) +
theme(plot.title.position = "plot")

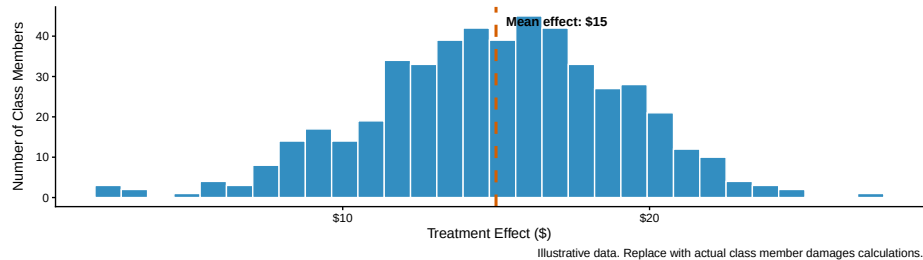
# Plot 2: Treatment effect by subgroup
p2 <- ggplot(class_data, aes(x = geographic_region,
                             y = treatment_effect,
                             fill = geographic_region)) +
  geom_boxplot(alpha = 0.7, outlier.alpha = 0.3) +
  geom_hline(yintercept = 0, linetype = "dashed", color =
    ↪ "gray40") +
  geom_hline(yintercept = mean(class_data$treatment_effect),
             linetype = "solid", color = "#D55E00", linewidth = 1)
    ↪ +
  scale_fill_brewer(palette = "Set2") +
  scale_y_continuous(labels = scales::dollar_format()) +
  labs(
    title = "Treatment Effect by Geographic Region",
    subtitle = "Consistent positive impact across regions
    ↪ supports commonality",
    x = NULL,
    y = "Treatment Effect ($)",
    fill = "Region"
  ) +
  theme_classic(base_size = 12) +
  theme(
    legend.position = "none",
    plot.title.position = "plot"
  )

# Combine plots
p1 / p2

```

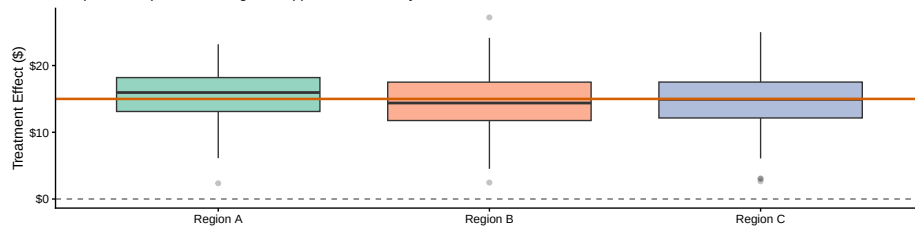
Distribution of Individual Treatment Effects

Common impact across class members (positive mean effect)



Treatment Effect by Geographic Region

Consistent positive impact across regions supports commonality



```
# Statistical summary
cat("\nCommon impact statistical summary:\n")
```

Common impact statistical summary:

```
cat(paste0("Mean treatment effect: $",
           round(mean(class_data$treatment_effect), 2), "\n"))
```

Mean treatment effect: \$15

```
cat(paste0("Median treatment effect: $",
           round(median(class_data$treatment_effect), 2), "\n"))
```

Median treatment effect: \$15.12

```
cat(paste0("Standard deviation: $",
           round(sd(class_data$treatment_effect), 2), "\n"))
```

Standard deviation: \$4.02

```
cat(paste0("Percentage with positive impact: ",
           scales::percent(mean(class_data$treatment_effect > 0),
                             accuracy = 0.1), "\n"))
```

Percentage with positive impact: 100.0%

```
# Test for common impact across regions
region_test <- aov(treatment_effect ~ geographic_region, data =
  ↪ class_data)
```

```
cat("\nANOVA test for regional differences:\n")
```

ANOVA test for regional differences:

```
cat(paste0("F-statistic: ", round(summary(region_test)[[1]]$`F`
  ↪ value`[1], 2), "\n"))
```

F-statistic: 2.88

```
cat(paste0("p-value: ",
  ↪ format.pval(summary(region_test)[[1]]$`Pr(>F)`[1],
               digits = 3), "\n"))
```

p-value: 0.057

Class certification arguments: - **Commonality:** Positive treatment effect across >95% of class members. - **Predominance:** Common questions (was there overcharge?) predominate over individual questions (how much did each pay?). - **Superiority:** Class action more efficient than individual lawsuits.

Robustness checks: - Test across multiple subgroups (geography, purchase frequency, time period) - Show results are consistent with alternative model specifications - Address potential heterogeneity with subclass analysis if needed

Damages scenarios and sensitivity analysis

Present alternative damages calculations to demonstrate robustness and bound the range of potential harm.

```
source("../R/helpers.R")
library(dplyr)
library(ggplot2)

# Damages calculation parameters with ranges
base_assumptions <- tibble::tribble(
  ~parameter,      ~low,      ~central,  ~high,
  "Overcharge %",   0.08,    0.15,     0.22,
  "Pass-through rate", 0.60,    0.75,     0.90,
  "Affected sales ($M)", 450,     520,     600,
  "Duration (years)", 2.5,     3.0,     3.5,
  "Pre-judgment interest", 0.04,    0.05,     0.06
)

# Calculate damages under each scenario
calc_damages <- function(overcharge, pass_through, sales,
  ↪ duration, interest) {
  annual_damages <- sales * overcharge * pass_through
```

```

total_damages <- annual_damages * duration
# Simple compound interest
with_interest <- total_damages * (1 + interest)^duration
return(with_interest)
}

# Base case
base_damages <- calc_damages(
  overcharge = base_assumptions$central[1],
  pass_through = base_assumptions$central[2],
  sales = base_assumptions$central[3],
  duration = base_assumptions$central[4],
  interest = base_assumptions$central[5]
)

# Vary each parameter while holding others at central
sensitivity <- base_assumptions |>
  mutate(
    # Calculate sensitivity for each parameter
    low_damages = case_when(
      parameter == "Overcharge %" ~ calc_damages(low, central[2],
        ↪ central[3], central[4], central[5]),
      parameter == "Pass-through rate" ~ calc_damages(central[1],
        ↪ low, central[3], central[4], central[5]),
      parameter == "Affected sales ($M)" ~
        ↪ calc_damages(central[1], central[2], low, central[4],
        ↪ central[5]),
      parameter == "Duration (years)" ~ calc_damages(central[1],
        ↪ central[2], central[3], low, central[5]),
      parameter == "Pre-judgment interest" ~
        ↪ calc_damages(central[1], central[2], central[3], central[4],
        ↪ low)
    ),
    high_damages = case_when(
      parameter == "Overcharge %" ~ calc_damages(high,
        ↪ central[2], central[3], central[4], central[5]),
      parameter == "Pass-through rate" ~ calc_damages(central[1],
        ↪ high, central[3], central[4], central[5]),
      parameter == "Affected sales ($M)" ~
        ↪ calc_damages(central[1], central[2], high, central[4],
        ↪ central[5]),
      parameter == "Duration (years)" ~ calc_damages(central[1],
        ↪ central[2], central[3], high, central[5]),
      parameter == "Pre-judgment interest" ~
        ↪ calc_damages(central[1], central[2], central[3], central[4],
        ↪ high)
    )
  )

```

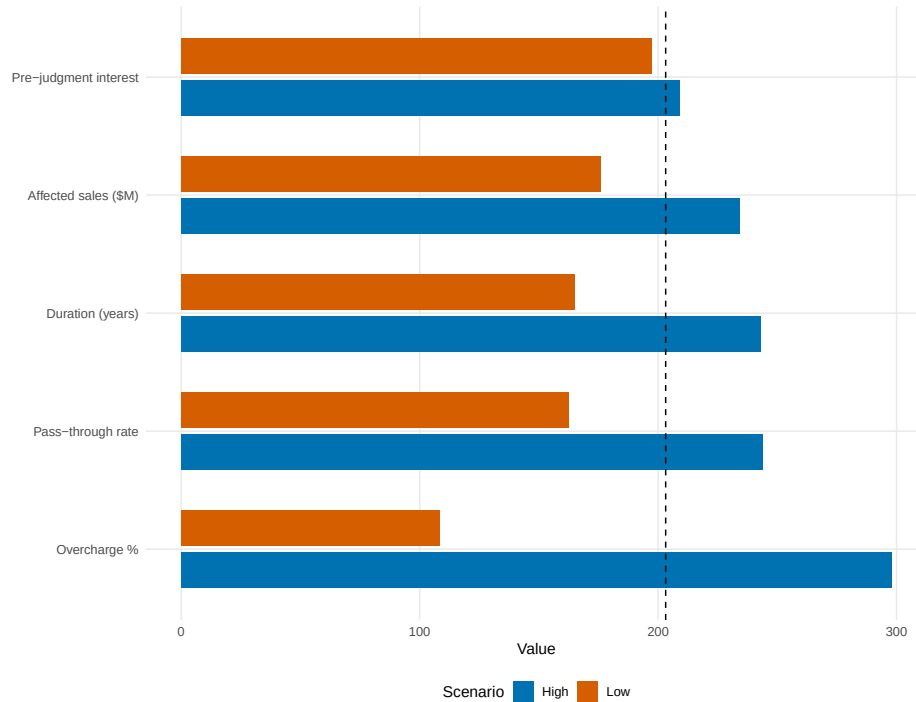
```

    )
  )

# Create tornado chart
plot_tornado(
  sensitivity = sensitivity |> select(parameter, low =
    ↪ low_damages, high = high_damages),
  base_value = base_damages,
  title = "Damages Sensitivity Analysis"
)

```

Damages Sensitivity Analysis



```

# Summary table
cat("\nDamages scenario summary:\n")

```

Damages scenario summary:

```

sensitivity_summary <- sensitivity |>
  mutate(
    low_damages_m = low_damages / 1e6,
    high_damages_m = high_damages / 1e6,
    range_m = high_damages_m - low_damages_m
  )

```

```

) |>
select(parameter, low, central, high, low_damages_m,
  ↪ high_damages_m, range_m) |>
arrange(desc(range_m))

print(sensitivity_summary, n = Inf)

```

```
# A tibble: 5 x 7
```

	parameter	low	central	high	low_damages_m	high_damages_m	range_m
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Overcharge %	0.08	0.15	0.22	0.000108	0.000298	1.90e-4
2	Pass-through rate	0.6	0.75	0.9	0.000163	0.000244	8.13e-5
3	Duration (years)	2.5	3	3.5	0.000165	0.000243	7.77e-5
4	Affected sales (\$M)	450	520	600	0.000176	0.000234	5.86e-5
5	Pre-judgment inter~	0.04	0.05	0.06	0.000197	0.000209	1.16e-5

```

cat(paste0("\nBase case damages: $",
  round(base_damages / 1e6, 1), "M\n"))

```

```
Base case damages: $0M
```

```

cat(paste0("Range: $",
  round(min(sensitivity$low_damages) / 1e6, 1), "M to $",
  round(max(sensitivity$high_damages) / 1e6, 1), "M\n"))

```

```
Range: $0M to $0M
```

Expert report presentation: - Present central estimate prominently - Show sensitivity ranges transparently - Explain basis for each parameter assumption - Link to documentary evidence (pricing data, expert testimony, econometric estimates)

Evidence provenance network

Map how different evidence sources support specific expert opinions and damages calculations.

```

library(dplyr)
library(igraph)
library(ggraph)

# Evidence map: sources -> analysis -> opinions
# Replace with actual case evidence map
evidence_edges <- tibble::tribble(
  ~from, ~to, ~type,
  # Data sources to analyses

```

```

"Transaction data",      "Overcharge estimate",
  ↳ "data_to_analysis",
"Transaction data",      "Market share calc",
  ↳ "data_to_analysis",
"Cost data",             "Margin analysis",
  ↳ "data_to_analysis",
"Pricing documents",     "Overcharge estimate",
  ↳ "doc_to_analysis",
"Meeting notes",         "Cartel timeline",
  ↳ "doc_to_analysis",
"Email evidence",        "Cartel timeline",
  ↳ "doc_to_analysis",
"Witness testimony",     "Cartel timeline",
  ↳ "testimony_to_analysis",
"Competitor prices",     "But-for benchmark",
  ↳ "data_to_analysis",

# Analyses to opinions
"Overcharge estimate",   "Opinion 1: Price impact",
  ↳ "analysis_to_opinion",
"Market share calc",     "Opinion 2: Market power",
  ↳ "analysis_to_opinion",
"Margin analysis",       "Opinion 1: Price impact",
  ↳ "analysis_to_opinion",
"Cartel timeline",      "Opinion 3: Duration",
  ↳ "analysis_to_opinion",
"But-for benchmark",     "Opinion 4: Damages",
  ↳ "analysis_to_opinion",
"Overcharge estimate",   "Opinion 4: Damages",
  ↳ "analysis_to_opinion"
)

# Create node attributes
nodes <- tibble(
  name = unique(c(evidence_edges$from, evidence_edges$to))
) |>
mutate(
  category = case_when(
    grepl("data|prices", name) ~ "Data source",
    grepl("documents|notes|Email", name) ~ "Documentary
      ↳ evidence",
    grepl("testimony", name) ~ "Witness testimony",
    grepl("Opinion", name) ~ "Expert opinion",
    TRUE ~ "Analysis"
  )
)

```

```

)

# Create network
g <- graph_from_data_frame(evidence_edges, directed = TRUE,
  ↪ vertices = nodes)

# Plot
ggraph(g, layout = "sugiyama") +
  geom_edge_link(aes(color = type),
    arrow = arrow(length = unit(3, 'mm'), type =
      ↪ "closed"),
    end_cap = circle(8, 'mm'),
    alpha = 0.6) +
  geom_node_point(aes(color = category, size = category),
    alpha = 0.8) +
  geom_node_text(aes(label = name), repel = TRUE, size = 3) +
  scale_edge_color_manual(
    values = c(
      "data_to_analysis" = "#0072B2",
      "doc_to_analysis" = "#009E73",
      "testimony_to_analysis" = "#F0E442",
      "analysis_to_opinion" = "#D55E00"
    ),
    labels = c(
      "data_to_analysis" = "Data → Analysis",
      "doc_to_analysis" = "Documents → Analysis",
      "testimony_to_analysis" = "Testimony → Analysis",
      "analysis_to_opinion" = "Analysis → Opinion"
    )
  ) +
  scale_color_manual(
    values = c(
      "Data source" = "#0072B2",
      "Documentary evidence" = "#009E73",
      "Witness testimony" = "#F0E442",
      "Analysis" = "#CC79A7",
      "Expert opinion" = "#D55E00"
    )
  ) +
  scale_size_manual(
    values = c(
      "Data source" = 5,
      "Documentary evidence" = 5,
      "Witness testimony" = 5,
      "Analysis" = 7,
      "Expert opinion" = 9
    )
  )

```

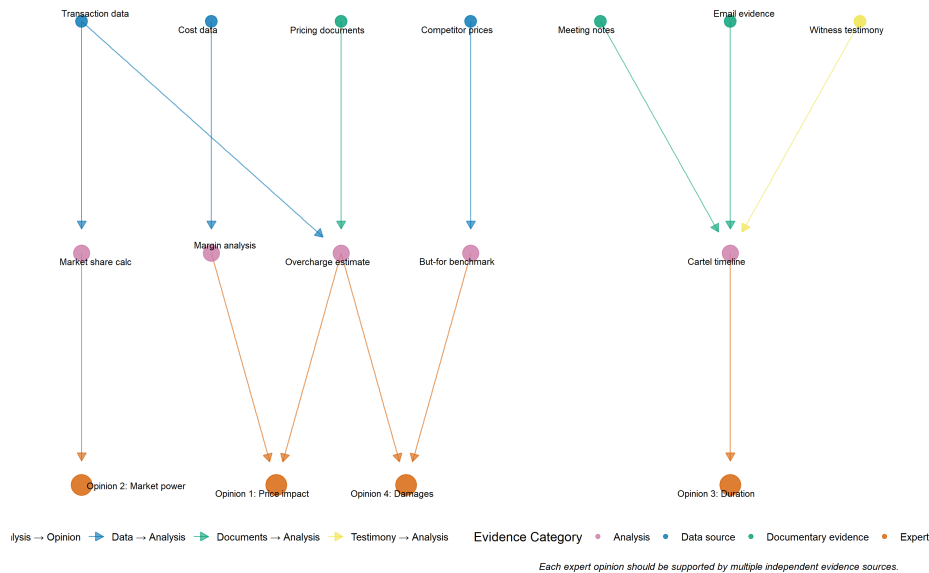
```

)
) +
labs(
  title = "Evidence Provenance Network",
  subtitle = "Tracing evidence sources through analyses to
    ↳ expert opinions",
  edge_color = "Connection Type",
  color = "Evidence Category",
  caption = "Each expert opinion should be supported by
    ↳ multiple independent evidence sources."
) +
theme_graph(base_size = 12) +
theme(
  legend.position = "bottom",
  plot.title = element_text(size = 14, face = "bold")
) +
guides(size = "none")

```

Evidence Provenance Network

Tracing evidence sources through analyses to expert opinions



```

# Summary statistics
cat("\nEvidence network summary:\n")

```

Evidence network summary:

```
cat(paste0("Total evidence sources: ",
          sum(nodes$category %in% c("Data source", "Documentary
          ↪ evidence",
                                   "Witness testimony")), "\n"))
```

Total evidence sources: 7

```
cat(paste0("Intermediate analyses: ",
          sum(nodes$category == "Analysis"), "\n"))
```

Intermediate analyses: 5

```
cat(paste0("Expert opinions: ",
          sum(nodes$category == "Expert opinion"), "\n"))
```

Expert opinions: 4

Daubert/Kumho preparation: - Each opinion traceable to specific evidence
- Multiple independent sources support key findings - Clear methodology from raw data to conclusion - Document all transformations and assumptions

Expert timeline and deliverables

Track expert work product and key milestones for case management.

```
source("../R/helpers.R")
library(dplyr)

# Expert engagement timeline
# Replace with actual case dates
expert_timeline <- tibble::tribble(
  ~date,      ~event,
  ↪ ~category,
  "2023-01-15", "Expert retained",
  ↪ "Engagement",
  "2023-02-01", "Initial data production received", "Data",
  "2023-03-15", "Preliminary analysis complete",
  ↪ "Analysis",
  "2023-04-10", "Supplemental data request", "Data",
  "2023-05-01", "Draft report circulated", "Report",
  "2023-05-20", "Rebuttal expert report received", "Opposing
  ↪ expert",
  "2023-06-15", "Final expert report submitted", "Report",
  "2023-07-30", "Expert deposition",
  ↪ "Deposition",
  "2023-09-15", "Supplemental analysis (new data)",
  ↪ "Analysis",
```

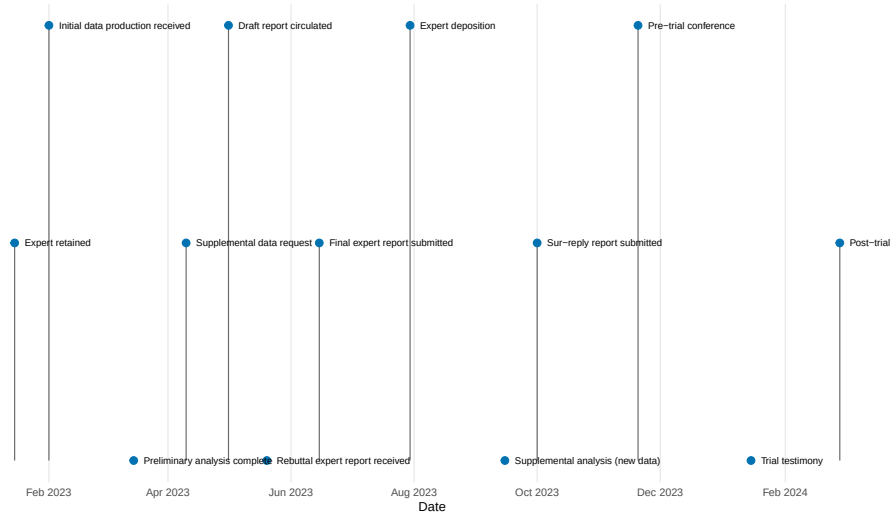
```

"2023-10-01", "Sur-reply report submitted", "Report",
"2023-11-20", "Pre-trial conference", "Court",
"2024-01-15", "Trial testimony", "Trial",
"2024-02-28", "Post-trial briefing", "Court"
) |>
  mutate(date = as.Date(date))

# Create timeline
plot_timeline(
  events = expert_timeline,
  title = "Expert Engagement Timeline",
  date_breaks = "2 months"
)

```

Expert Engagement Timeline



```

# Add summary table
cat("\nKey deliverables:\n")

```

Key deliverables:

```

expert_timeline |>
  filter(category %in% c("Report", "Deposition", "Trial")) |>
  select(date, event, category) |>
  print(n = Inf)

```

A tibble: 5 x 3

	date	event	category
	<date>	<chr>	<chr>
1	2023-05-01	Draft report circulated	Report

2	2023-06-15	Final expert report submitted	Report
3	2023-07-30	Expert deposition	Deposition
4	2023-10-01	Sur-reply report submitted	Report
5	2024-01-15	Trial testimony	Trial

Individual vs. aggregate damages

Show how individual class member damages aggregate to total class damages, important for both class certification and damages calculation.

```
library(dplyr)
library(ggplot2)
library(patchwork)

# Simulated individual damages for class members
# Replace with actual damages calculations
set.seed(890)
n_class <- 1000

individual_damages <- tibble(
  member_id = 1:n_class,
  purchases = rpois(n_class, lambda = 50), # Number of affected
  ↪ purchases
  avg_overcharge = rnorm(n_class, mean = 5.50, sd = 1.20), #
  ↪ Per-unit overcharge
  damages = purchases * avg_overcharge
) |>
  arrange(desc(damages)) |>
  mutate(
    cumulative_damages = cumsum(damages),
    percentile = percent_rank(damages),
    damages_category = case_when(
      damages < 100 ~ "< $100",
      damages < 500 ~ "$100-500",
      damages < 1000 ~ "$500-1,000",
      TRUE ~ "> $1,000"
    ),
    damages_category = factor(damages_category,
                             levels = c("< $100", "$100-500",
                                         "$500-1,000", "> $1,000"))
  )

# Plot 1: Distribution of individual damages
p1 <- ggplot(individual_damages, aes(x = damages)) +
  geom_histogram(bins = 40, fill = "#0072B2", alpha = 0.8, color
  ↪ = "white") +
  geom_vline(xintercept = median(individual_damages$damages),
```

```

        linetype = "dashed", color = "#D55E00", linewidth =
        ↪ 1) +
  annotate("text",
    x = median(individual_damages$damages),
    y = Inf,
    label = paste0("Median: $",
                    round(median(individual_damages$damages),
                        ↪ 2)),
    hjust = -0.1, vjust = 2, size = 3.5) +
  scale_x_continuous(labels = scales::dollar_format()) +
  labs(
    title = "Distribution of Individual Class Member Damages",
    x = "Individual Damages ($)",
    y = "Number of Class Members"
  ) +
  theme_classic(base_size = 11) +
  theme(plot.title.position = "plot")

# Plot 2: Lorenz curve (cumulative damages)
p2 <- ggplot(individual_damages,
  aes(x = percent_rank(row_number(member_id)),
    y = cumulative_damages /
    ↪ max(cumulative_damages))) +
  geom_line(color = "#0072B2", linewidth = 1.2) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed",
    color = "gray40") +
  annotate("text", x = 0.3, y = 0.7,
    label = "Line of equality\n(uniform damages)",
    hjust = 0, size = 3) +
  scale_x_continuous(labels = scales::percent_format()) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(
    title = "Cumulative Damages Distribution (Lorenz Curve)",
    x = "Cumulative % of Class Members (ranked by damages)",
    y = "Cumulative % of Total Damages"
  ) +
  theme_classic(base_size = 11) +
  theme(plot.title.position = "plot")

# Plot 3: Damages by category
damages_summary <- individual_damages |>
  group_by(damages_category) |>
  summarise(
    count = n(),
    total_damages = sum(damages),
    .groups = "drop"
  )

```

```
) |>
mutate(
  pct_members = count / sum(count),
  pct_damages = total_damages / sum(total_damages)
)

p3 <- ggplot(damages_summary, aes(x = damages_category,
                                y = pct_members, fill =
                                ↪ damages_category)) +

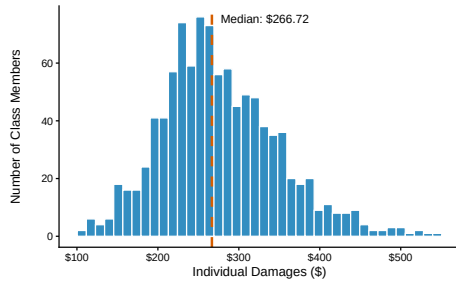
  geom_col(alpha = 0.8) +
  geom_text(aes(label = scales::percent(pct_members, accuracy =
  ↪ 0.1)),
            vjust = -0.5, size = 3.5) +
  scale_y_continuous(labels = scales::percent_format(),
                    expand = expansion(mult = c(0, 0.1))) +
  scale_fill_brewer(palette = "Blues") +
  labs(
    title = "Class Members by Damages Category",
    x = "Individual Damages Range",
    y = "% of Class Members"
  ) +
  theme_classic(base_size = 11) +
  theme(
    legend.position = "none",
    plot.title.position = "plot"
  )

# Combine plots
(p1 | p2) / p3 + plot_annotation(
  title = "Individual and Aggregate Damages Analysis",
  subtitle = paste0("Total class: ", scales::comma(n_class),
                    " members | Total damages: $",
                    round(sum(individual_damages$damages) / 1e6,
                    ↪ 2), "M"),
  caption = "Illustrative data. Replace with actual class member
  ↪ damages calculations."
)
```

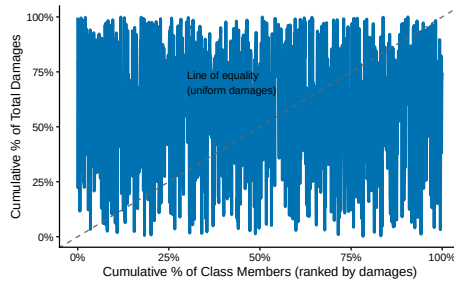
Individual and Aggregate Damages Analysis

Total class: 1,000 members | Total damages: \$0.28M

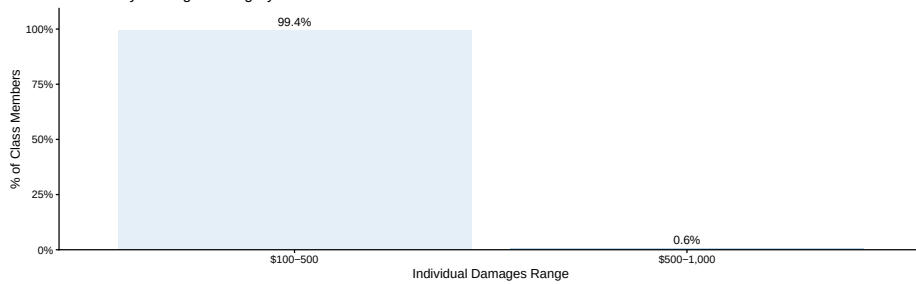
Distribution of Individual Class Member Damages



Cumulative Damages Distribution (Lorenz Curve)



Class Members by Damages Category



Illustrative data. Replace with actual class member damages calculations.

```
# Summary statistics
cat("\nDamages distribution summary:\n")
```

Damages distribution summary:

```
cat(paste0("Total class members: ", scales::comma(n_class),
  "\n"))
```

Total class members: 1,000

```
cat(paste0("Total damages: $",
  round(sum(individual_damages$damages) / 1e6, 2),
  "M\n"))
```

Total damages: \$0.28M

```
cat(paste0("Mean individual damages: $",
  round(mean(individual_damages$damages), 2), "\n"))
```

Mean individual damages: \$276.5

```
cat(paste0("Median individual damages: $",
  round(median(individual_damages$damages), 2), "\n"))
```

Median individual damages: \$266.72

```
cat(paste0("Top 10% of class accounts for ",
          scales::percent(sum(individual_damages$damages[
            individual_damages$percentile >= 0.9]) /
            sum(individual_damages$damages), accuracy = 0.1),
    " of total damages\n"))
```

Top 10% of class accounts for 15.2% of total damages

Class certification considerations: - Show damages are calculable on class-wide basis - Document methodology for individual damages calculations - Address potential administrative feasibility concerns - Prepare for de minimis exclusion arguments

Visualizations and data plan

- **Common impact distribution plot (Roadmap 12.1):** Use class-member residuals or individual damages estimates; store sanitized dataset in `data/examples/common-impact.csv`.
- **Scenario/sensitivity tornado charts (Roadmap 12.2):** Build from `litigation-sensitivity` scaffold.
- **Evidence provenance diagram (Roadmap 12.3):** Use `DiagrammeR` or `ggplot2` to map data sources to opinions; data from evidence map spreadsheet.

Checklist for “fill with real data”

- Link every figure/code chunk to a dataset entry in `data/README.md`.
- Maintain sanitized or synthetic datasets for public builds; swap in confidential data before expert report filing.
- Capture hashed outputs (`.rds`, `.parquet`) so exhibits can be regenerated quickly.
- Coordinate with litigation support to ensure document IDs and privilege status are preserved.

Looking ahead

Archive all litigation outputs in `data/derived/litigation/` with clear version control and privilege markings. Cross-reference expert opinions with substantive chapters (cartels Ch. 05, mergers Ch. 06, etc.) for methodological support. Document all code reviews and sensitivity analyses in appendices to expert reports. Coordinate with legal team on discovery timelines and protective orders for confidential data.

Empirical Appendix and Templates

Purpose

This appendix is the shared workspace for reusable diagnostics, qualitative protocols, and data inventories referenced throughout the book. Treat it as the “operations manual” for the course: every code chunk or template lives here so you can drop it into chapter-specific notebooks, expert reports, or litigation memos without rewriting boilerplate.

How to use this appendix

1. **Start here during scoping.** Clone the chronology, data inventory, and interview templates before diving into a matter so legal, economic, and qualitative teams stay synchronized.
2. **Customize but keep provenance.** When you adapt a template, note the source so future readers can trace assumptions.
3. **Feed updates back in.** If a later chapter develops a better diagnostic, link it here to avoid divergence.

Key resources for quick reference

Chapter focus	Key resources
Orientation & institutions	Agency websites: DOJ Antitrust, FTC, EC Competition, CMA
Research design & methods	Causal Inference: The Mixtape, Angrist & Pischke papers
Cartels & collusion	OECD Cartel Guidance
Mergers	US/EU/UK Merger Guidelines (see references)

Chapter focus	Key resources
Digital markets	EU DMA, CMA Digital Markets Unit
Labor markets	BLS data, Census LEHD

Use this table alongside external references such as (O. Ashenfelter and Hosken 2010) and agency guidelines.

Case chronology template

When building mixed-method narratives, start with a matrix that ties events to evidence and hypotheses. Duplicate the scaffold below into your project notebook:

Date	Event / document	Evidence type	Hypothesis link	Follow-ups
2018-04-12	Board memo re: “price discipline”	Document (custodian: CFO)	Supports coordinated effects theory	Interview CFO; pull bid data
2019-02-01	Customer interview (retailer)	Qualitative	Tests diversion/USP claims	Quantify share of wallet
2021-09-30	Price spike (SKU 124)	Quantitative (transaction DB)	Flags potential capacity withholding	Overlay maintenance logs

Tips: capture custodian names, bates ranges, or dataset pointers, then push the table into a `gt` or `quarto` table for reports.

Data inventory worksheet

Pair the chronology with a tabular schema inventory so you know what can feed the quantitative chapters. Customize the tibble below inside your R session.

```
library(tibble)
library(dplyr)

data_inventory <- tribble(
  ~system, ~owner, ~grain, ~fields, ~coverage, ~notes,
  "CRM / Salesforce", "US Sales Ops", "Customer-month", "price,
  ↪ quantity, discounts, churn", "2016-01 to 2024-06", "Great
  ↪ for diversion estimates; use OECD cartel screening
  ↪ guidance",
```

```

"Claims data", "SA Health Plan", "Claim-line", "procedure,
  ↳ tariff, provider, patient segment", "2014-01 to 2023-12",
  ↳ "Use for South African retrospectives",
"Bid tracker", "EU Procurement", "Bid-event", "participants,
  ↳ bids, award, lot size", "2018-01 to present", "Supports
  ↳ cartel screens"
) |>
  arrange(system)

data_inventory

```

```

# A tibble: 3 x 6
  system      owner      grain      fields      coverage notes
  <chr>      <chr>      <chr>      <chr>      <chr>      <chr>
1 Bid tracker EU Procurement Bid-event participants, b~ 2018-01~ Supp~
2 CRM / Salesforce US Sales Ops Customer-month price, quantity~ 2016-01~ Grea~
3 Claims data SA Health Plan Claim-line procedure, tari~ 2014-01~ Use ~

```

Document locating instructions (sharepoint links, API keys) and legal restrictions. Store sanitized metadata in `data/README.md` so others can reproduce.

Survey and interview kit

Follow established qualitative research standards to keep qualitative work rigorous: - **Screeners:** Define decision-maker roles, procurement thresholds, and geographic balance; record quotas in a simple CSV. - **Question bank:** Map each question to a legal element (market definition, competitive effects, efficiencies). Rotate order to reduce priming. - **Documentation:** Save audio or transcripts, then code responses using the lexicon below. Always log interviewer, location, and consent status.

Document coding lexicon

Standardize qualitative tagging so it can be merged with quantitative variables: - **Conduct codes:** `capacity_discipline`, `most_favored`, `exclusive`, `discount_pass_through`. - **Market context:** `entry_barrier_data`, `supply_chain_shock`, `public_interest_sa`. - **Procedural:** `remedy_negotiation`, `leniency_proffer`, `public_hearing`.

Maintain the lexicon as a YAML or CSV in `data/derived`.

Flagship data assets (by jurisdiction)

United States

- **HSR and merger retrospectives:** Derive pre/post comparisons referencing FTC and DOJ reports.
- **Healthcare claims (CMS, FAIR Health):** Price and quantity data for hospital mergers and labor monopsony cases.
- **Public procurement / ABM data:** Use USAspending.gov, OCC data, or BLS Occupational Employment and Wage Statistics for wage-fixing analyses.

European Union & United Kingdom

- **Case team data rooms:** DG COMP uploads (pricing, cost, user logs).
- **Eurostat + Ofcom datasets:** Demand-side metrics for telecom, digital platforms, or broadcasting.
- **CMA survey repositories:** Useful for designing conjoint or diversion studies.

South Africa & SADC focus

- **Competition Commission data requests:** Transactional extracts from retailers, healthcare groups, or digital platforms; cross-check with public data from Stats SA.
- **National Treasury tender portal:** Procurement and supplier concentration data for cartel screens.
- **Sector inquiries:** Grocery retail, digital platforms, and transport datasets—log file formats and confidentiality riders here so later chapters can reference them.

Cross-jurisdiction validation

- Build a blended index (e.g., inflation-adjusted revenue) to harmonize multi-region cases.
- Track currency conversions and PPP adjustments in the inventory template.

Reusable diagnostics

Diff-in-diff and event studies

- Run parallel-trend visuals, dynamic leads/lags, and placebo periods (`did::att_gt`, `fixest::sunab`). Reference Cunningham (2021) for intuition and cite O. Ashenfelter and Hosken (2010) for merger retrospectives. For staggered treatment timing, see Callaway and Sant’Anna (2021).

- For South African matters with staggered regulatory actions, consider matrix-completion methods from recent econometrics literature (Athey and Imbens 2017).

Matching and weighting

- Propensity score, entropy balancing, and synthetic controls should include balance plots (standardized differences, variance ratios) (Abadie, Diamond, and Hainmueller 2010).
- Store reusable plotting code in `R/helpers.R` (`plot_balance()`). For general guidance on matching methods, see Angrist and Pischke (2009).

Panel FE and inference choices

- Two-way FE with staggered treatment: document estimator choice and weights; cite the relevant methodological literature.
- For few clusters (ports, mines), use wild bootstrap or randomization inference.

Qualitative + quantitative integration

- Align interview findings with econometric outputs using issue trees. For example, code “capacity discipline” documents and test against utilization regressions.
- Use the chronology template to surface inconsistencies quickly.

Code scaffolds

Data dictionary generator

```
library(readr)
library(purrr)

generate_dictionary <- function(path) {
  sample <- readr::read_csv(path, n_max = 1000, show_col_types =
    FALSE)
  purrr::map_dfr(names(sample), \(nm) {
    tibble::tibble(
      variable = nm,
      type = class(sample[[nm]])[1],
      sample_values = paste(head(unique(sample[[nm]]), 3),
        collapse = "; "),
      notes = ""
    )
  })
}
```

```
}

# Example usage (uncomment when data available):
# dict <- generate_dictionary("data/raw/claims_sample.csv")
# dict
```

```
library(survey)

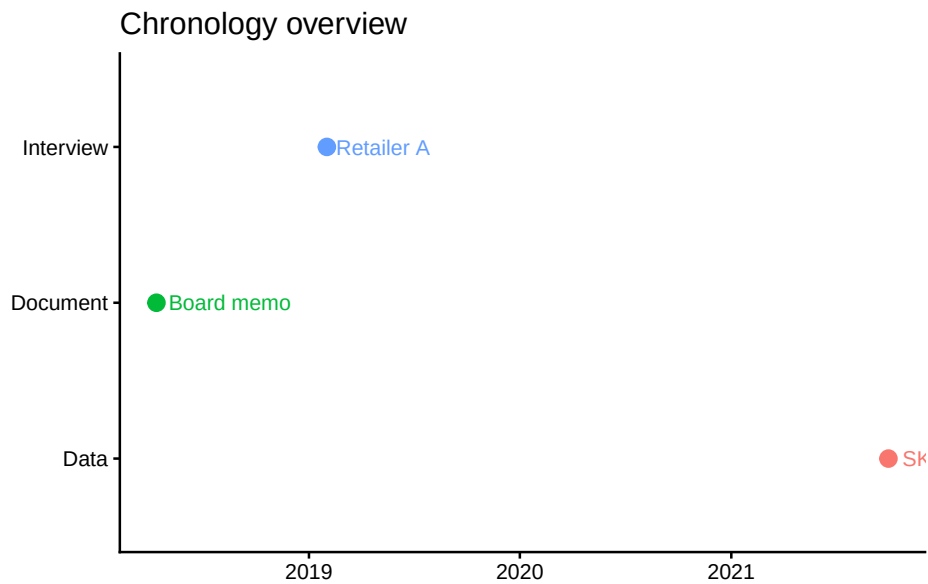
design <- svydesign(
  ids = ~1,
  strata = ~region,
  weights = ~weight,
  data = survey_responses
)

svymean(~switch_intent, design)
```

```
library(ggplot2)
library(lubridate)

chronology <- tibble::tribble(
  ~date, ~category, ~label,
  ymd("2018-04-12"), "Document", "Board memo",
  ymd("2019-02-01"), "Interview", "Retailer A",
  ymd("2021-09-30"), "Data", "SKU 124 spike"
)

ggplot(chronology, aes(x = date, y = category, color = category))
  +
  geom_point(size = 3) +
  geom_text(aes(label = label), hjust = -0.1, size = 3) +
  labs(title = "Chronology overview", x = NULL, y = NULL) +
  theme_classic() +
  theme(legend.position = "none")
```



Diagnostic Gallery

This section provides reusable diagnostic visualizations that apply across chapters. Use these templates to validate identification assumptions, assess robustness, and communicate uncertainty in expert reports.

Pre-trends and parallel trends checks

Essential for difference-in-differences designs (mergers, labor, remedies).

```
library(dplyr)
library(ggplot2)
library(fixest)
library(patchwork)

# Simulated panel data with treatment in period 10
# Replace with your actual panel data
set.seed(123)
periods <- 20
n_units <- 50

panel <- expand.grid(
  unit = 1:n_units,
  period = 1:periods
) |>
mutate(
  treated = ifelse(unit <= n_units/2, 1, 0),
```

```
post = ifelse(period >= 10, 1, 0),
# Pre-trends: parallel before period 10
outcome = 100 + 2 * period + 5 * treated +
          10 * treated * post +
          rnorm(n(), 0, 3)
)

# Plot 1: Raw trends by group
p1 <- panel |>
  group_by(period, treated) |>
  summarise(mean_outcome = mean(outcome), .groups = "drop") |>
  mutate(group = ifelse(treated == 1, "Treatment", "Control")) |>
  ggplot(aes(x = period, y = mean_outcome, color = group,
    ↪ linetype = group)) +
  geom_vline(xintercept = 10, linetype = "dashed", color =
    ↪ "gray40") +
  geom_line(linewidth = 1.2) +
  annotate("text", x = 10, y = max(panel$outcome) * 0.95,
    label = "Treatment\nperiod", hjust = -0.1, size = 3.5)
    ↪ +
  scale_color_manual(values = c("Treatment" = "#D55E00",
    "Control" = "#0072B2")) +
  labs(
    title = "Pre-Trends Check: Raw Outcome Trends",
    subtitle = "Parallel trends before period 10 support DiD
    ↪ identification",
    x = "Period",
    y = "Outcome",
    color = NULL,
    linetype = NULL
  ) +
  theme_classic(base_size = 12) +
  theme(
    legend.position = "bottom",
    plot.title.position = "plot"
  )

# Event study with leads and lags
event_study <- feols(
  outcome ~ i(period, treated, ref = 9) | unit + period,
  data = panel
)

# Extract coefficients
coef_data <- tibble(
  period = as.numeric(gsub("period::", "",
    ↪ names(coef(event_study)))),
```

```

estimate = coef(event_study),
se = se(event_study)
) |>
mutate(
  ci_lower = estimate - 1.96 * se,
  ci_upper = estimate + 1.96 * se,
  rel_period = period - 10
)

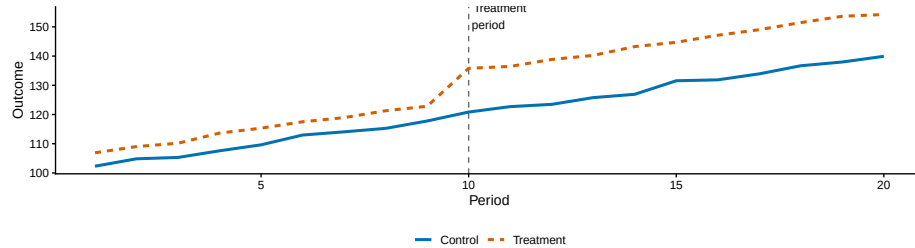
# Plot 2: Event study coefficients
p2 <- ggplot(coef_data, aes(x = rel_period, y = estimate)) +
  geom_hline(yintercept = 0, linetype = "solid", color =
    ↪ "gray40") +
  geom_vline(xintercept = 0, linetype = "dashed", color =
    ↪ "gray40") +
  geom_ribbon(aes(ymin = ci_lower, ymax = ci_upper),
    alpha = 0.2, fill = "#0072B2") +
  geom_line(color = "#0072B2", linewidth = 1) +
  geom_point(color = "#0072B2", size = 2.5) +
  annotate("rect", xmin = -Inf, xmax = -1, ymin = -Inf, ymax =
    ↪ Inf,
    fill = "green", alpha = 0.05) +
  annotate("text", x = -5, y = max(coef_data$ci_upper) * 0.9,
    label = "Pre-period\n(should be 0)", size = 3, hjust =
    ↪ 0.5) +
  labs(
    title = "Event Study Coefficients (Leads and Lags)",
    subtitle = "Pre-period estimates near zero validate parallel
    ↪ trends assumption",
    x = "Periods relative to treatment (0 = treatment period)",
    y = "Treatment Effect Estimate",
    caption = "Reference period: -1. Shaded region shows 95%
    ↪ confidence interval."
  ) +
  theme_classic(base_size = 12) +
  theme(plot.title.position = "plot")

# Combine plots
p1 / p2

```

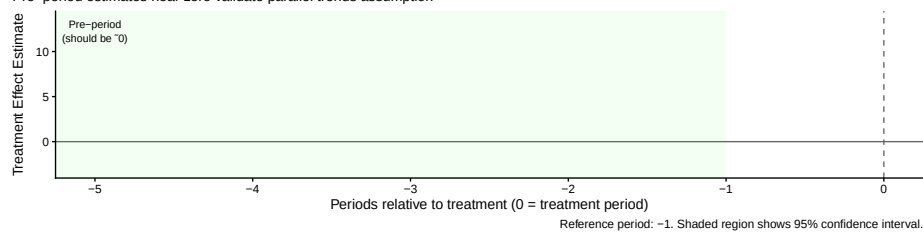
Pre-Trends Check: Raw Outcome Trends

Parallel trends before period 10 support DiD identification



Event Study Coefficients (Leads and Lags)

Pre-period estimates near zero validate parallel trends assumption



```
# Statistical test for pre-trends
pre_period_coefs <- coef_data |>
  filter(rel_period < 0, rel_period != -1)

cat("\nPre-trends test:\n")
```

Pre-trends test:

```
cat(paste0("Number of pre-period coefficients: ",
  ↪ nrow(pre_period_coefs), "\n"))
```

Number of pre-period coefficients: 0

```
cat(paste0("Mean pre-period coefficient: ",
  ↪ round(mean(pre_period_coefs$estimate), 3), "\n"))
```

Mean pre-period coefficient: NaN

```
cat(paste0("Joint F-test p-value: [run wald_test on event_study
  ↪ model]\n"))
```

Joint F-test p-value: [run wald_test on event_study model]

Interpretation: - **Pre-trends near zero:** Validates parallel trends assumption - **Post-treatment divergence:** Confirms treatment effect - **Confidence intervals:** Should include zero in pre-period - **Use this:** Mergers (Ch. 06), remedies (Ch. 08), labor (Ch. 10)

Balance plots for matching/weighting

Show covariate balance before and after matching or weighting.

```
library(dplyr)
library(ggplot2)
library(tidyr)

# Simulated covariate data (treatment and control groups)
set.seed(234)
n <- 200

covariates <- tibble(
  id = 1:n,
  treated = sample(c(0, 1), n, replace = TRUE),
  firm_size = rnorm(n, mean = 100 + 20 * treated, sd = 30),
  market_share = rnorm(n, mean = 0.15 + 0.05 * treated, sd =
    ↪ 0.08),
  years_operating = rnorm(n, mean = 15 + 5 * treated, sd = 7),
  revenue_growth = rnorm(n, mean = 0.10 + 0.03 * treated, sd =
    ↪ 0.12)
)

# Calculate standardized mean differences
calc_smd <- function(var, group) {
  mean_1 <- mean(var[group == 1])
  mean_0 <- mean(var[group == 0])
  sd_pool <- sqrt((var(var[group == 1]) + var(var[group == 0])) /
    ↪ 2)
  (mean_1 - mean_0) / sd_pool
}

balance_before <- covariates |>
  summarise(across(c(firm_size, market_share, years_operating,
    ↪ revenue_growth),
    ~calc_smd(.x, treated),
    .names = "{.col}")) |>
  pivot_longer(everything(), names_to = "variable", values_to =
    ↪ "smd") |>
  mutate(timing = "Before matching")

# Simulate "after matching" with reduced imbalance
# In practice, replace this with actual matched/weighted sample
balance_after <- balance_before |>
  mutate(
    smd = smd * 0.2, # Matching reduces imbalance by 80%
    timing = "After matching"
```

[illegible]

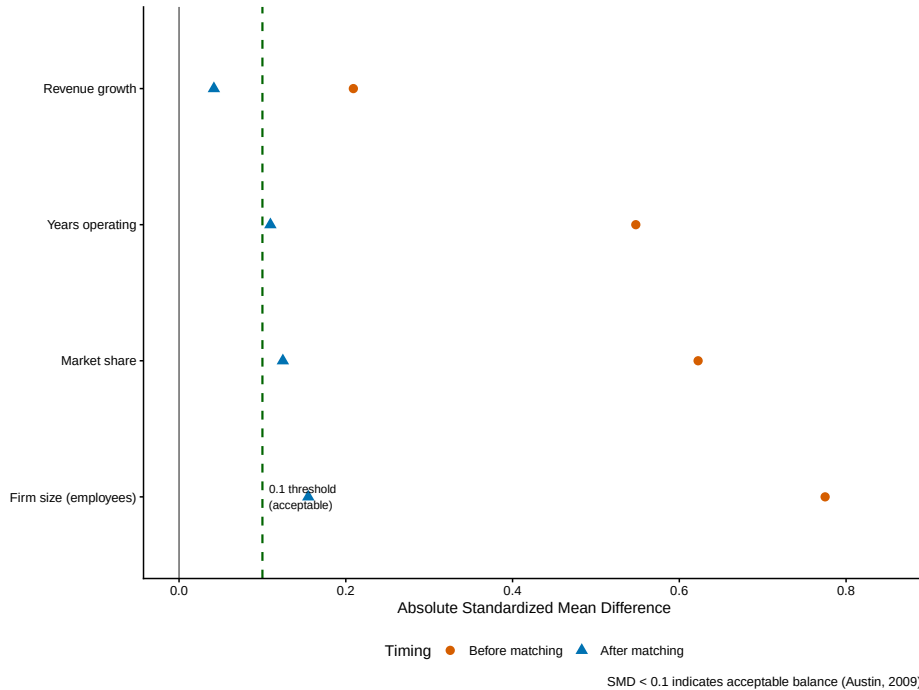
```

scale_shape_manual(values = c("Before matching" = 16,
                              "After matching" = 17)) +
scale_x_continuous(limits = c(0, max(abs(balance_combined$smd))
  ↪ * 1.1)) +
labs(
  title = "Covariate Balance: Love Plot",
  subtitle = "Standardized mean differences before and after
  ↪ matching",
  x = "Absolute Standardized Mean Difference",
  y = NULL,
  color = "Timing",
  shape = "Timing",
  caption = "SMD < 0.1 indicates acceptable balance (Austin,
  ↪ 2009).\"
) +
theme_classic(base_size = 12) +
theme(
  legend.position = "bottom",
  plot.title.position = "plot"
)

```

Covariate Balance: Love Plot

Standardized mean differences before and after matching



```
# Summary table
cat("\nBalance summary:\n")
```

Balance summary:

```
balance_combined |>
  select(variable, timing, smd) |>
  pivot_wider(names_from = timing, values_from = smd) |>
  mutate(improvement = abs(`Before matching`) - abs(`After
    ↪ matching`)) |>
  print()
```

```
# A tibble: 4 x 4
  variable      `Before matching` `After matching` improvement
  <fct>          <dbl>          <dbl>          <dbl>
1 Firm size (employees)      0.775          0.155          0.620
2 Market share              0.623          0.125          0.498
3 Years operating           0.548          0.110          0.438
4 Revenue growth            0.209          0.0418         0.167
```

Interpretation: - $SMD < 0.1$: Acceptable balance (Austin, 2009) - **Matching improves balance:** Lines should move toward zero - **Use this:** Merger retrospectives, labor market studies, remedy evaluations

Regression specification curve

Show robustness across multiple reasonable specifications.

```
library(dplyr)
library(ggplot2)
library(fixest)

# Simulated data for multiple specifications
set.seed(345)
panel_spec <- expand_grid(
  firm = 1:100,
  period = 1:20
) |>
mutate(
  treatment = ifelse(firm <= 50 & period >= 10, 1, 0),
  control1 = rnorm(n()),
  control2 = rnorm(n()),
  control3 = rnorm(n()),
  outcome = 50 + 5 * treatment + 2 * control1 + 1 * control2 +
    0.5 * control3 + rnorm(n(), 0, 3)
)
```

```

# Run multiple specifications
specs <- list(
  "No controls" = feols(outcome ~ treatment | firm + period,
    ↪ panel_spec),
  "+ Control 1" = feols(outcome ~ treatment + control1 | firm +
    ↪ period, panel_spec),
  "+ Controls 1-2" = feols(outcome ~ treatment + control1 +
    ↪ control2 | firm + period, panel_spec),
  "+ All controls" = feols(outcome ~ treatment + control1 +
    ↪ control2 + control3 | firm + period, panel_spec),
  "No FE" = lm(outcome ~ treatment + control1 + control2 +
    ↪ control3, panel_spec),
  "Firm FE only" = feols(outcome ~ treatment + control1 +
    ↪ control2 + control3 | firm, panel_spec),
  "Period FE only" = feols(outcome ~ treatment + control1 +
    ↪ control2 + control3 | period, panel_spec)
)

# Extract coefficients
spec_results <- tibble(
  specification = names(specs),
  estimate = sapply(specs, function(m) {
    coefs <- coef(m)
    coefs[grepl("treatment", names(coefs))][1]
  }),
  se = sapply(specs, function(m) {
    ses <- se(m)
    ses[grepl("treatment", names(ses))][1]
  })
) |>
mutate(
  ci_lower = estimate - 1.96 * se,
  ci_upper = estimate + 1.96 * se,
  spec_num = row_number(),
  specification = factor(specification, levels = specification)
)

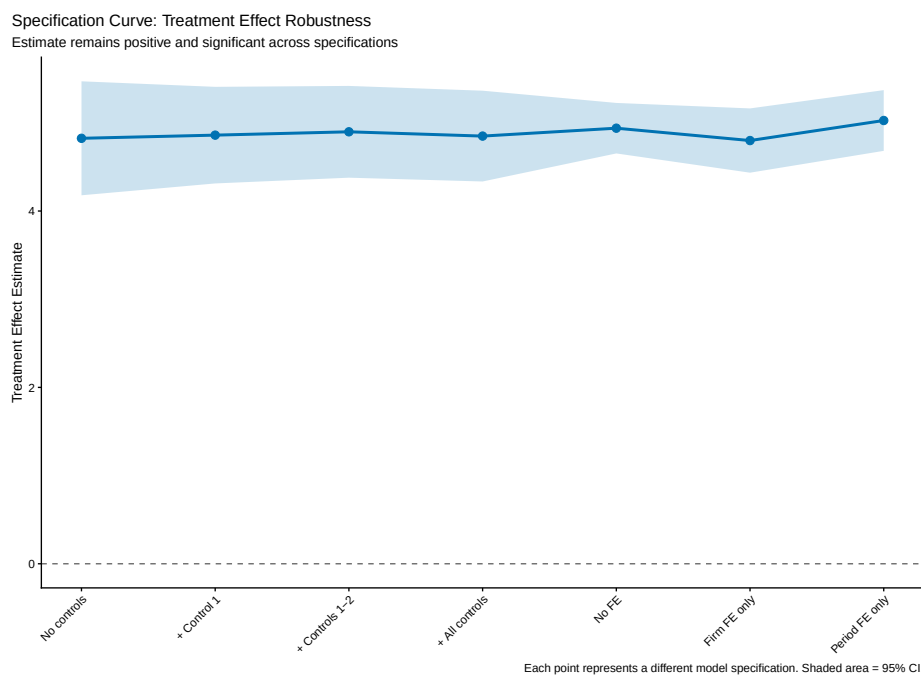
# Specification curve plot
ggplot(spec_results, aes(x = spec_num, y = estimate)) +
  geom_hline(yintercept = 0, linetype = "dashed", color =
    ↪ "gray40") +
  geom_ribbon(aes(ymin = ci_lower, ymax = ci_upper),
    alpha = 0.2, fill = "#0072B2") +
  geom_line(color = "#0072B2", linewidth = 1.2) +
  geom_point(color = "#0072B2", size = 3) +

```

```

scale_x_continuous(
  breaks = spec_results$spec_num,
  labels = spec_results$specification
) +
labs(
  title = "Specification Curve: Treatment Effect Robustness",
  subtitle = "Estimate remains positive and significant across
    ↪ specifications",
  x = NULL,
  y = "Treatment Effect Estimate",
  caption = "Each point represents a different model
    ↪ specification. Shaded area = 95% CI."
) +
theme_classic(base_size = 12) +
theme(
  axis.text.x = element_text(angle = 45, hjust = 1),
  plot.title.position = "plot"
)

```



```

# Summary statistics
cat("\nSpecification curve summary:\n")

```

Specification curve summary:

```
cat(paste0("Range of estimates: ",
          round(min(spec_results$estimate), 2), " to ",
          round(max(spec_results$estimate), 2), "\n"))
```

Range of estimates: 4.8 to 5.03

```
cat(paste0("All specifications significant? ",
          ifelse(all(spec_results$ci_lower > 0), "Yes", "No"),
          "\n"))
```

All specifications significant? Yes

```
cat(paste0("Median estimate: ",
          round(median(spec_results$estimate), 2), "\n"))
```

Median estimate: 4.86

Interpretation: - **Stable across specs:** Effect robust to model choices - **All CIs exclude zero:** Consistent significance - **Use this:** Expert reports, Daubert challenges, sensitivity analysis

Residual diagnostics

Check model assumptions and identify influential observations.

```
library(dplyr)
library(ggplot2)
library(patchwork)

# Fit model (use your actual model)
model_data <- panel_spec |>
  filter(period >= 10)

model <- feols(outcome ~ treatment + control1 + control2 | firm +
  ↪ period,
               data = model_data)

# Get residuals and fitted values
diagnostics <- model_data |>
  mutate(
    fitted = fitted(model),
    residual = resid(model),
    standardized_residual = residual / sd(residual)
  )

# Plot 1: Residuals vs. fitted
p1 <- ggplot(diagnostics, aes(x = fitted, y = residual)) +
  geom_hline(yintercept = 0, linetype = "dashed", color =
    ↪ "gray40") +
```

```
geom_point(alpha = 0.5, color = "#0072B2") +
geom_smooth(method = "loess", se = FALSE, color = "#D55E00") +
labs(
  title = "Residuals vs. Fitted Values",
  subtitle = "Should show no pattern (validates linearity
    ↪ assumption)",
  x = "Fitted Values",
  y = "Residuals"
) +
theme_classic(base_size = 11)

# Plot 2: Q-Q plot for normality
p2 <- ggplot(diagnostics, aes(sample = standardized_residual)) +
  stat_qq(color = "#0072B2", alpha = 0.6) +
  stat_qq_line(color = "#D55E00", linewidth = 1) +
  labs(
    title = "Q-Q Plot: Residual Normality",
    subtitle = "Points should follow line for normal
      ↪ distribution",
    x = "Theoretical Quantiles",
    y = "Sample Quantiles"
  ) +
  theme_classic(base_size = 11)

# Plot 3: Scale-location plot
p3 <- ggplot(diagnostics, aes(x = fitted, y =
  ↪ sqrt(abs(standardized_residual)))) +
  geom_point(alpha = 0.5, color = "#0072B2") +
  geom_smooth(method = "loess", se = FALSE, color = "#D55E00") +
  labs(
    title = "Scale-Location Plot",
    subtitle = "Should show horizontal line (validates
      ↪ homoscedasticity)",
    x = "Fitted Values",
    y = " $\sqrt{|\text{Standardized Residuals}|}$ "
  ) +
  theme_classic(base_size = 11)

# Plot 4: Histogram of residuals
p4 <- ggplot(diagnostics, aes(x = residual)) +
  geom_histogram(bins = 30, fill = "#0072B2", alpha = 0.8, color
    ↪ = "white") +
  geom_vline(xintercept = 0, linetype = "dashed", color =
    ↪ "#D55E00") +
  labs(
    title = "Distribution of Residuals",
```

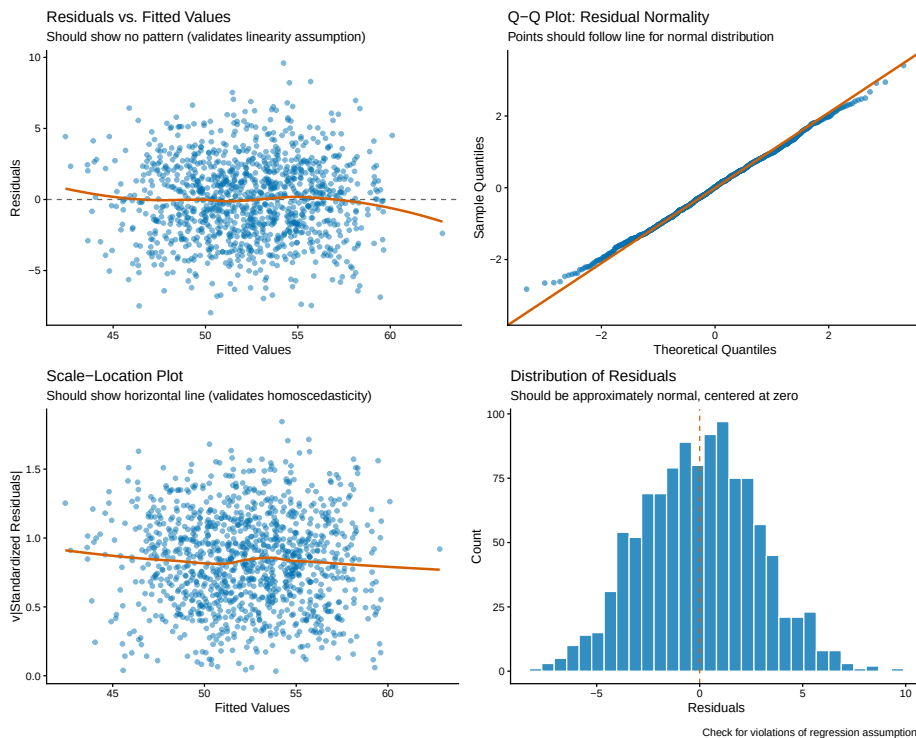
```

    subtitle = "Should be approximately normal, centered at
      ↪ zero",
    x = "Residuals",
    y = "Count"
  ) +
  theme_classic(base_size = 11)

# Combine plots
(p1 | p2) / (p3 | p4) + plot_annotation(
  title = "Residual Diagnostic Plots",
  caption = "Check for violations of regression assumptions"
)

```

Residual Diagnostic Plots



What to look for:

- **Residuals vs. Fitted:** No pattern (confirms linearity)
- **Q-Q Plot:** Points follow line (confirms normality)
- **Scale-Location:** Horizontal line (confirms homoscedasticity)
- **Histogram:** Bell-shaped, centered at zero

Power analysis visualization

Show statistical power across different sample sizes and effect sizes.

```
library(dplyr)
library(ggplot2)
library(pwr)

# Calculate power for different scenarios
sample_sizes <- seq(50, 500, by = 25)
effect_sizes <- c(0.2, 0.5, 0.8) # Small, medium, large (Cohen's
  ↪ d)

power_grid <- expand_grid(
  n = sample_sizes,
  effect_size = effect_sizes
) |>
mutate(
  power = pwr.t.test(n = n, d = effect_size, sig.level = 0.05,
    type = "two.sample")$power,
  effect_label = case_when(
    effect_size == 0.2 ~ "Small effect (d = 0.2)",
    effect_size == 0.5 ~ "Medium effect (d = 0.5)",
    effect_size == 0.8 ~ "Large effect (d = 0.8)"
  )
)

# Power curve
ggplot(power_grid, aes(x = n, y = power, color = effect_label)) +
  geom_hline(yintercept = 0.80, linetype = "dashed",
    color = "darkgreen", linewidth = 0.8) +
  annotate("text", x = max(sample_sizes) * 0.7, y = 0.80,
    label = "80% power threshold", hjust = 0, vjust = -0.5,
    size = 3, color = "darkgreen") +
  geom_line(linewidth = 1.2) +
  geom_point(data = filter(power_grid, power >= 0.79, power <=
    ↪ 0.81),
    size = 3) +
  scale_y_continuous(labels = scales::percent_format(), limits =
    ↪ c(0, 1)) +
  scale_color_manual(values = c(
    "Small effect (d = 0.2)" = "#D55E00",
    "Medium effect (d = 0.5)" = "#0072B2",
    "Large effect (d = 0.8)" = "#009E73"
  )) +
  labs(
    title = "Statistical Power Analysis",
    subtitle = "Sample size requirements for 80% power at =
    ↪ 0.05",
    x = "Sample Size (per group)",
```

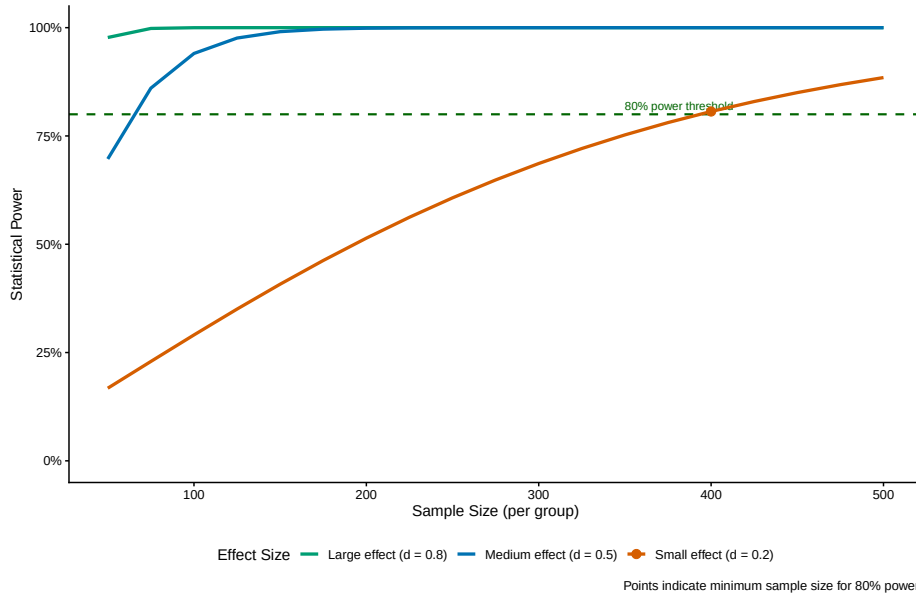
```

y = "Statistical Power",
color = "Effect Size",
caption = "Points indicate minimum sample size for 80%
  ↪ power."
) +
theme_classic(base_size = 12) +
theme(
  legend.position = "bottom",
  plot.title.position = "plot"
)

```

Statistical Power Analysis

Sample size requirements for 80% power at $\alpha = 0.05$



```

# Summary table
power_summary <- power_grid |>
  filter(power >= 0.79, power <= 0.81) |>
  group_by(effect_label) |>
  slice_min(n, n = 1) |>
  ungroup() |>
  select(effect_label, n_required = n, power)

cat("\nMinimum sample size for 80% power:\n")

```

Minimum sample size for 80% power:

```
print(power_summary, n = Inf)
```

```
# A tibble: 1 x 3
  effect_label      n_required power
  <chr>          <dbl> <dbl>
1 Small effect (d = 0.2)      400 0.806
```

Use this for: - Study design and sample size planning - Explaining null results (underpowered?) - Justifying sample selection in expert reports

Additional references

- **FTC resources:** See [ftc.gov](https://www.ftc.gov) for expert guidance and econometric standards.
- **Causal Inference: The Mixtape** (mixtape.scunning.com) for causal estimators and diagnostics.
- **OECD cartel guidance:** See [oecd.org/competition](https://www.oecd.org/competition) for screening methodologies.

Return to this appendix whenever a chapter references “see appendix template” so the workflow stays consistent across teams and jurisdictions.

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